Pure-k_T Perturbative TMD Extraction: Formalism Development Notes

k_T Perturbative Extraction Project

Version 1.7 (physical validation update) – October 1, 2026

Abstract

These development notes construct and test a transverse-momentum representation of the Collins/CSS2 leading-power TMD operator. At fixed perturbative order the polynomial distribution algebra, matching dictionary, and representation identities are exact on their stated domains. Canonical cumulative scale setting retains its origin contact term; momentum-disk boundary conversions and ordered flavor-matrix logarithmic reconstruction are derived through three loops.

An explicit unexpanded construction is now supplied. The CS kernel is first RG-resummed from a spectral canonical scale with a smooth, declared infrared completion and no ultraviolet cap. Its endpoint evolution is a well-defined operator on a common logarithmically weighted domain, while complete canonical legs are bounded. Resolvent polynomials of the momentum-space Laplacian provide convergent transform-free evaluation, including contact-complete perturbative kernels and finite bins. An all-order logarithmic degree argument establishes N^{3}LL evolution with four-loop cusp and running, three-loop noncusp and CS boundary inputs; physical three-loop hard and matching boundaries would provide the prime once imported and validated. A separate uniform remainder statement has explicit weak-coupling hypotheses. This construction uses the CSS2 canonical RG solution rather than literal exponentiation of a fixed-scale three-loop polynomial.

Version 1.6 added a constrained shared-resolvent evaluator, a radial deterministic forward map and input-gradient tests, positivity/branching compatibility results, and process-level kinematic support and matching. It retains the CSS2 operator rather than modifying it to obtain probability weights. The uncapped canonical symbol has a proved obstruction to a positive finite-measure branching interpretation. An auxiliary positive recoil proposal can be combined with an exact signed correction. Its first lower-order same-symbol comparison to continuum Hankel quadrature and scalar-input closure exposed residual conditioning, discretization and matching tasks.

The present update adds sourced N^{3}LL evolution with strict-NLO hard/OPE boundaries, endpoint-preserving Chebyshev evaluation, non-Gaussian continuum and three-scale physical-PDF comparisons, contacts, input derivatives, and a conditional finite-table ultraviolet bound. A restricted neural closure and qualified proton–proton Drell–Yan calculation connect the operator to inverse use. These accepted results supplement the preserved development studies.

The result remains a model-agnostic, fixed-flavor perturbative prescription. Physical NNLO/N^{3}LO boundary completion, full-spectrum matching, massive thresholds, and a global joint-(x, k_T) extraction remain separate tasks. DNNs supply possible nonperturbative inputs, not perturbative accuracy. The methods paper precedes the identical-data comparison project.

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1 Status and purpose

This note is a method-development record for fitting TMDs directly in k_T while preserving their Collins/CSS2 definition and perturbative accuracy connections. It supplies one admitted all-order-in-logarithms construction, in Sec. 23; it is not a certificate of a completed numerical high-order extraction. Its purpose is not to report a fitted phenomenological extraction, but to define the prescription that such an extraction must follow if it is to remain in k_T space while carrying the same perturbative accuracy labels as the corresponding Collins/CSS2 calculation. The note therefore fixes:

1. the parent scheme, $\mathcal{K}_{\mathrm{CSS2}}$, and its relation to Collins/CSS2;
2. the k_T-space distribution basis and convolution algebra;
3. the perturbative evolution, matching, and fixed-order expansion checks needed for $N^k\mathrm{LL}$ and $N^k\mathrm{LL}'$ claims;
4. a process-level W-term template that is not restricted to fixed-target Drell-Yan;

5. a model-agnostic nonperturbative module interface;
6. regulated normalization and transverse-moment definitions;
7. the $W + Y$ full-spectrum matching prescription and a perturbative uncertainty decomposition;
8. a claim ledger separating an exact fixed-order representation, a genuinely resummed $N^3\mathrm{LL}'$ prescription, and a completed numerical extraction.

What changes in v1.7. This is an incremental update of v1.6: the existing derivations, section ordering, and historical tests are retained. Section 39 adds the accepted physical canonical realization, practical evaluator, current error budgets, finite-PDF-table UV assumption, and qualified inverse demonstration. The label is $\mathrm{N^3LL}$ evolution with strict-NLO hard/OPE boundaries, not an implemented $\mathrm{N^3LL}'$ spectrum. The immediate project is the complete methods paper; the identical-data joint-(x, k_T) versus b_T extraction follows as a separate project. Dated status annotations below distinguish old development records from current acceptance without erasing either.

What changed in v1.6. At the v1.6 stage the central target was a reproducible momentum-space fit, rather than a publication-readiness statement. Sections 24–29 separate the fixed CSS2 operator from a signed numerical evaluator, from a positive auxiliary branching proposal, and from the measured process support. They retain the v1.5 operator/tower results with their stated hypotheses. Bernstein remains a constructive existence argument; a constrained shared-shift radial evaluator is developed and tested as a practical candidate. The new positivity obstructions, exact-support formula, rational error theorem and split-covariance identities are new derivations in this revision. The companion measurements are lower-order method tests, explicitly separated from physical high-order validation.

What changed in v1.5. The substantive v1.4 corrections are retained: contact preservation, partial multiplier domains, finite canonical-boundary translations, and the distinction between fixed-order checks and resummation. Section 23 now resolves the admitted-operator and logarithmic-accuracy problem for one explicit spectral-canonical realization. It does not assert completion of the alternative cumulative-profile production route. The absence of validated physical three-loop payloads still prevents a numerical $\mathrm{N^3LL}'$ prediction from being claimed. The completion ledger in Sec. 38 specifies the scope.

The starting phenomenological anchor is the momentum-space neural-network extraction of Fernando and Keller [38]. Their method remains in k_T space, learns a transverse structure kernel from E288/E605 Drell-Yan data, and reconstructs a normalized profile $s(x, k; Q)$ with a differentiable quadrature layer. For the present project, that work is best viewed as an inverse-convolution and nonparametric-extraction prototype. It is not, by itself, an $N^k\mathrm{LL}$ TMD extraction because it does not impose an explicit Sudakov kernel, fixed-order matching, or Y term.

The logical scope is deliberately narrower than a new factorization proof. We import the operator definitions, soft-factor subtraction, Glauber cancellation assumptions, and leading-power factorization statement from Collins/CSS2. What is established here is a representation theorem for that singular operator, a perturbative coefficient dictionary, and a set of implementation checks. The construction does not prove that an arbitrary fitted function is a valid TMD, that a chosen nonperturbative module is unique, or that power corrections are negligible in a specified data set. Those statements require separate operator, phenomenological, and data-level analyses.

Working decision. The project target is not "a neural network instead of resummation." The target is a renormalized k_T-space TMD framework whose perturbative content is equivalent to a chosen N^kLL or N^kLL$'$ Collins/CSS calculation. The neural network supplies only nonperturbative boundary information.

2 Prior art and project gap

Before deriving new kernels, we separate what is already established from what is specific to this project. The short conclusion is:

Direct transverse-momentum-space resummation exists. High-order perturbative TMD ingredients exist. Neural-network TMD fits exist. Direct-k_T inverse reconstruction exists. The project seeks a universal TMDPDF extraction framework that keeps the fitted representation in k_T space while implementing Collins/CSS-equivalent N^kLL or N^kLL$'$ evolution, matching, and $W + Y$ consistency.

2.1 The standard baseline: Collins/CSS and modern high-accuracy TMD fits

The dominant precision framework for small-q_T Drell-Yan is the Collins-Soper-Sterman logic: factorize at low transverse momentum, Fourier transform to b_T space, resum large logarithms through Collins-Soper and RG evolution, and match back to fixed-order collinear factorization outside the strict TMD region [4, 5, 6]. The recent TMD literature provides a broad review of the associated operator definitions, process dependence, evolution, and phenomenological status [7].

This baseline has already reached very high perturbative accuracy in phenomenological extractions. Examples include unpolarized Drell-Yan fits at N^4LL accuracy that extract TMDPDFs and the Collins-Soper kernel [28], and combined Drell-Yan plus SIDIS analyses at N^4LL that also include TMD fragmentation functions [29]. Neural-network parametrizations of nonperturbative TMD components have also already been demonstrated at high perturbative accuracy in the standard impact-parameter-space framework [30]. A 2026 conference analysis further reports a simultaneous Drell–Yan plus SIDIS neural parametrization at N^3LL [45]; it is additional evidence that the neural representation should be treated as a modular long-distance parametrization rather than as the definition of perturbative accuracy.

2.2 Direct q_T-space and momentum-space resummation

The idea of resumming directly in transverse momentum is not new. Ellis and Veseli constructed a q_T-space resummation for electroweak boson transverse-momentum spectra and showed agreement with the leading, next-to-leading, and next-to-next-to-leading towers of the usual b-space result [8]. Frixione, Nason, and Ridolfi analyzed analytic issues in both b-space and q_T-space resummation and obtained a closed next-to-leading-logarithmic q_T-space expression, while identifying limitations of its analytic structure [9]. Kulesza and Stirling further explored explicit transverse-momentum-space Sudakov resummation for electroweak boson production and its matching behavior [10, 11].

Modern direct-space approaches made this line much more powerful. Kang, Lee, and Vaidya introduced a hybrid construction in which the virtuality scale is chosen in momentum space while the rapidity scale remains b_T dependent, obtaining NNLL spectra and a fast semi-analytic inversion [44]. This is an important intermediate construction, but it still retains a b_T-space rapidity scale and a precomputed inverse Fourier step. Monni, Re, and Torrielli formulated a momentum-space resummation for Higgs transverse momentum at NNLL+NNLO [12]. Bizon, Monni, Re, Rottoli,

and Torrielli then generalized momentum-space resummation for transverse observables up to N^3LL+NNLO, with a formula that reduces to the standard impact-parameter-space result in known cases [13]. Ebert and Tackmann developed a related distribution-space RG method in which the singular plus distributions in q_T are treated directly as the objects of evolution [14]. More recently, Simonelli proposed an analytic transverse-momentum-space continuation of TMDPDFs and TMD-factorized cross sections to NLL accuracy [15].

Cerutti and Simonelli subsequently showed that different infrared prescriptions in the standard CSS implementation can fit low-energy Drell–Yan data comparably while producing sizable differences at intermediate transverse momentum and in high-energy predictions [46]. This does not imply that a k_T representation removes nonperturbative ambiguity; it sharpens the motivation for keeping the perturbative kernel, its continuation, and the fitted long-distance module separately auditable in the measured transverse variable.

2.3 Momentum-space TMD ingredients and high-order perturbative inputs

There is also substantial prior work on perturbative TMD ingredients that can be used as input to a direct-k_T construction. In the rapidity-renormalization-group formulation, rapidity-renormalized TMD soft and beam functions were computed at two loops, providing NNLL$'$ ingredients for transverse-momentum-differential predictions [19]. TMDPDF matching coefficients are now known to very high order: Ebert, Mistlberger, and Vita computed quark and gluon TMDPDFs at N^3LO [20], and Luo, Yang, Zhu, and Zhu computed unpolarized quark and gluon TMDPDF and TMDFF matching coefficients through N^3LO [22]. The rapidity anomalous dimension, equivalently the perturbative Collins-Soper kernel in common schemes, is also known at very high order [23, 24].

For our purposes, the key lesson from this literature is that the perturbative ingredients should not be refit by the neural network. They should enter as hard, evolution, matching, and subtraction kernels. The nonperturbative model should only cover the long-distance or small-k_T boundary information left unresolved by perturbation theory.

2.4 Large-k_T consistency, integral constraints, and hadron-structure-oriented fits

A central issue for a pure-k_T extraction is the transition between the nonperturbative core and the perturbative large-k_T tail. Gonzalez-Hernandez, Rainaldi, and Rogers stressed that TMD parametrizations must be made consistent with their large-transverse-momentum behavior in collinear factorization [26]. The hadron-structure-oriented approach implements related constraints in practical Drell-Yan and Z^0 phenomenology, emphasizing preservation of the TMD-PDF integral relationship and controlled comparisons among nonperturbative transverse structures [27]. Ebert, Michel, Stewart, and Sun likewise showed that momentum-space TMD integral functionals provide a systematic way to disentangle long and short distances, replacing naive unregulated integral constraints by regulated, perturbatively controlled objects [25].

2.5 Neighboring momentum-space phenomenology

The Parton Branching approach is an important neighboring framework because it evolves TMD parton densities in momentum space and has been used to describe Drell-Yan transverse momentum spectra across small and large k_T regions [32]. However, PB TMDs and CSS TMDs are not the same formal objects. A direct comparison paper emphasizes that PB TMDs evolve with an explicit single scale in momentum space, whereas CSS TMDs have coupled renormalization and rapidity-scale

evolution [31]. For this project, PB is prior art and a useful comparison target, but not a replacement for a Collins-equivalent double-scale k_T-space TMD factorization framework.

Fernando and Keller provide the closest direct predecessor on the extraction side: they perform a physics-informed neural inverse reconstruction directly in momentum space, using fixed-target Drell-Yan data and a normalized transverse profile whose auto-convolution reproduces the measured spectra [38]. Their construction is precisely why the present project starts from a direct-k_T inverse problem. The perturbative upgrade is to replace their tree-level/profile-only forward model by a distribution-valued $N^k\mathrm{LL}_{k_T}$ forward operator.

2.6 Gap statement

The prior art implies a useful division of labor, summarized in Table 1. The project should not claim to invent momentum-space resummation, high-order TMD matching, or neural-network TMD fitting. The defensible claim is the synthesis of these ingredients into a TMD-level extraction architecture. A literature review through August 2026 did not identify a published construction that simultaneously supplies a CSS2-parented TMD-level k_T representation, double-scale convolutional evolution, an explicit $N^3\mathrm{LL}'$ ingredient certificate, and a model-independent extraction interface with no numerical b_T transform. This is a bounded literature statement, not a proof of priority. In particular, direct distribution- space RG, high-accuracy direct-space cross-section resummation, and recent NLL analytic momentum-space TMD resummation are essential predecessors [14, 13, 15, 46].

Table 1: Prior art and the remaining project gap.

Prior-art line	What already exists	What remains for this project
CSS/Collins TMD factorization	Double-scale TMD evolution, small-q_T factorization, OPE matching, and $W+Y$ logic in a b_T-space representation [5, 6].	Represent the same perturbative content by k_T-space convolution kernels and fit the boundary object directly in k_T.
Direct q_T resummation	Cross-section-level resummation directly in q_T or momentum space, including hybrid and modern high-accuracy direct-space implementations [8, 44, 12, 13].	Promote the direct-space logic from color-singlet spectra to universal TMDPDF evolution and extraction.
Distribution-space RG	Plus-distribution evolution kernels and distributional scale setting for transverse-momentum spectra [14].	Turn the distribution algebra into a practical TMDPDF Green's function $\mathcal{U}_i(\boldsymbol{k};\mu_f,\zeta_f;\mu_i,\zeta_i)$.
High-order TMD ingredients	Soft/beam functions, matching coefficients, and rapidity anomalous dimensions through high perturbative orders [19, 20, 22, 23].	Convert and organize these inputs consistently in the chosen k_T-space scheme.

Prior-art line	What already exists	What remains for this project
Modern TMD fits	N^3LL and N^4LL phenomenological extractions, including neural nonperturbative parametrizations [28, 29, 30, 45].	Keep the fitted representation in k_T and make the neural model a boundary core rather than the full TMD.
Direct-k_T inverse reconstruction	Fernando-Keller learn a normalized k_T profile from fixed-target Drell-Yan data without a b_T transform [38].	Add perturbative evolution, large-k_T matching, regulated integral constraints, and Y-term consistency.
Parton Branching	Momentum-space TMD evolution and Drell-Yan phenomenology with an explicit single evolution scale [32, 31].	Build a Collins/CSS-equivalent two-scale (μ, ζ) momentum-space TMD framework rather than a PB-to-CSS conversion.

The target object is therefore the simultaneous implementation of

$$\frac{\partial}{\partial \ln \sqrt{\zeta}} F_{i/h}(x, \boldsymbol{k}; \mu, \zeta) = -\left[\mathcal{D}_i(\mu) \otimes_T F_{i/h}(x; \mu, \zeta)\right](\boldsymbol{k}), \tag{1}$$

$$F_{i/h}(x, \boldsymbol{k}; \mu_0, \zeta_0) = \sum_j \mathcal{C}_{i \leftarrow j}(\boldsymbol{k}; \mu_0, \zeta_0) \otimes_x f_{j/h}(\mu_0) + F^{\text{core}}_{i/h}(x, \boldsymbol{k}; \theta) - F^{\text{sub}}_{i/h}(x, \boldsymbol{k}; \theta), \tag{2}$$

$$\frac{\mathrm{d}\sigma}{\mathrm{d}Q^2\, \mathrm{d}y\, \mathrm{d}^2\boldsymbol{q}} = \sigma_0 \left[W^{N^k\mathrm{LL}}_{k_T}(Q, y, \boldsymbol{q}) + Y_{k_T}(Q, y, \boldsymbol{q})\right] + \mathcal{O}\left(q_T^2/Q^2\right), \tag{3}$$

with all singular kernels treated as transverse-momentum distributions and all nonperturbative information supplied by a fitted boundary core.

Working decision. Claim discipline: we do not claim the first momentum-space resummation, the first momentum-space TMD calculation, or the first neural-network TMD fit. We claim a TMD-level, Collins-equivalent, pure-k_T extraction framework that synthesizes those lines of work.

3 Core thesis

The Collins/CSS advantage is not fundamentally the coordinate b_T itself [4, 5, 6]. The advantage is that Fourier transformation converts transverse convolutions into products, making Collins-Soper and renormalization-group evolution local in transverse separation. A pure k_T formulation must keep the same perturbative content while accepting a different mathematical structure:

$$\text{local multiplication in } b_T \quad \Longleftrightarrow \quad \text{convolution operators in } k_T. \tag{4}$$

The working thesis is therefore

$N^k\mathrm{LL}_{k_T} \equiv N^k\mathrm{LL}_{b_T}$ perturbative content represented by k_T-space distributions.

That means the same hard functions, anomalous dimensions, matching coefficients, and $W + Y$ subtractions as the corresponding Collins/CSS calculation.

4 Working conventions and retained operator choices

The following choices define the paper-level prescription. They are conservative by design: we first represent Collins/CSS2 directly in k_T space and require explicit conversion factors before introducing any child scheme. These choices are broad enough for a formal paper and narrow enough for a later implementation repository to test.

Table 2: Paper-ready working choices.

Issue	Working choice
Reference scheme	Use Collins/CSS2 as the parent scheme. Define the initial pure-k_T implementation, denoted $\mathcal{K}_{\mathrm{CSS2}}$, as the momentum-space representation of the Collins/CSS2 TMDs, hard factors, Collins-Soper kernel, and matching coefficients. This follows the scheme-translation logic of Collins and Rogers [39].
Custom scheme	Do not introduce a nontrivial finite custom conversion at the first step. First build $\mathcal{K}_{\mathrm{CSS2}}$ and verify exact equivalence to CSS2 at fixed perturbative order. A more adapted child scheme $\mathcal{K}$ may be introduced later only with explicit finite kernels $\mathcal{Z}^{\mathcal{K}\leftarrow\mathcal{K}_{\mathrm{CSS2}}}$ and inverse kernels.
Distribution basis	Keep both the conventional two-dimensional plus distributions $\mathcal{L}_n$ and the analytic generator $\mathcal{G}_\eta$. Use $\mathcal{L}_n$ for comparison with singular coefficients and bin integrals; use $\mathcal{G}_\eta$ for convolution algebra and exponentiation.
Perturbative tail	Define the universal TMD large-k_T tail from TMD matching coefficients, $\mathcal{C}_{i\leftarrow j}^{\mathcal{K}_{\mathrm{CSS2}}} \otimes_x f_j$. Do not use a cross-section generator to define the TMD tail.
Large-q_T matching	Define the cross-section tail through the usual Y term, $Y = \sigma_{\mathrm{FO}} - W_{\mathrm{FO\,exp.}}$. Use DYTurbo, CuTe-MCFM/MCFM, or equivalent fixed-order/resummed tools as validation and benchmarking references, not as definitions of the universal TMDPDF [40, 41, 42].
Normalization	Use regulated cumulants and weighted moments. Do not impose an unregulated full-k_T integral as a fundamental identity. The same definitions must work from fixed-target kinematics to collider kinematics.
Uncertainty	In the formalism stage, track perturbative uncertainties only: scale/path choices, matching truncation, scheme-conversion truncation, and distributional numerical errors. Leave neural/TMD model, experimental, and replica uncertainties for the later fitting stage.

Issue	Working choice
Flavor	Use a full flavor-vector formalism from day one. Fixed-target Drell-Yan can be the first data application, but the formalism must also accommodate collider Drell-Yan, SIDIS, e^+e^-, and gluon-sensitive color-singlet channels.

Working decision. The first concrete scheme is $\mathcal{K}_{\mathrm{CSS2}}$: a pure-$k_T$ representation of Collins/CSS2. This is deliberately conservative. It lets us prove equivalence to the standard formalism before making any finite, numerically adapted scheme redefinitions.

5 Conventions

Transverse vectors are bold: $\boldsymbol{k}$, $\boldsymbol{q}$, $\boldsymbol{p}$, $\boldsymbol{b}$. Their magnitudes are denoted by $k_T = |\boldsymbol{k}|$, $q_T = |\boldsymbol{q}|$, and so on. The longitudinal convolution is

$$(C \otimes_x f)(x) \equiv \int_x^1 \frac{\mathrm{d}z}{z}\, C(z)\, f\left(\frac{x}{z}\right). \tag{5}$$

For ordinary transverse functions the convolution is

$$(A \otimes_T B)(\boldsymbol{q}) \equiv \int \mathrm{d}^2\boldsymbol{k}\, A(\boldsymbol{k})\, B(\boldsymbol{q}-\boldsymbol{k}). \tag{6}$$

For singular kernels Eq. (6) is shorthand for the distributional definition below and is not an ordinary Lebesgue integral.

We fix the Fourier pair

$$\widetilde{F}(x,\boldsymbol{b};\mu,\zeta) = \int \mathrm{d}^2\boldsymbol{k}\, e^{i\boldsymbol{b}\cdot\boldsymbol{k}} F(x,\boldsymbol{k};\mu,\zeta), \tag{7}$$

$$F(x,\boldsymbol{k};\mu,\zeta) = \int \frac{\mathrm{d}^2\boldsymbol{b}}{(2\pi)^2}\, e^{-i\boldsymbol{b}\cdot\boldsymbol{k}} \widetilde{F}(x,\boldsymbol{b};\mu,\zeta). \tag{8}$$

The intended production representation remains in k_T space. Fourier-space expressions are used to define the parent scheme, import known perturbative coefficients, and prove representation identities; they need not be evaluated numerically during an extraction.

5.1 Functional domain and meaning of convolution

We use Schwartz test functions $\varphi \in \mathcal{S}(\mathbb{R}^2)$ and their tempered-distribution dual $\mathcal{S}'(\mathbb{R}^2)$ [1]. The Fourier pair is

$$\widetilde{A}(\boldsymbol{b}) = \int \mathrm{d}^2\boldsymbol{k}\, e^{i\boldsymbol{b}\cdot\boldsymbol{k}} A(\boldsymbol{k}),$$

$$A(\boldsymbol{k}) = \int \frac{\mathrm{d}^2\boldsymbol{b}}{(2\pi)^2}\, e^{-i\boldsymbol{b}\cdot\boldsymbol{k}} \widetilde{A}(\boldsymbol{b}). \tag{9}$$

The integrals mean distributional duality when they are not ordinary integrals. Arbitrary tempered distributions cannot be multiplied or convolved.

The genuinely closed perturbative algebra used below is

$$\mathscr{P}_\mu = \mathbb{C}[L_b], \qquad \mathscr{A}_\mu^{\mathrm{pol}} = \mathcal{F}_{b\to k}^{-1}\mathscr{P}_\mu, \qquad L_b = \ln(\mu^2 b_T^2/b_0^2). \tag{10}$$

Every polynomial in L_b is locally integrable at $b_T = 0$ and has at most polynomial growth at infinity; the same is true of its finite products. Thus

$$A \otimes_T B := \mathcal{F}_{b\to k}^{-1}[\widetilde{A}\,\widetilde{B}] \tag{11}$$

defines an associative, commutative product on $\mathscr{A}_\mu^{\mathrm{pol}}$ with identity $\delta^{(2)}$. The formal series algebra $\mathscr{A}_\mu^{\mathrm{pol}}[[a_s]]$ is also well defined coefficient by coefficient. This product agrees with ordinary transverse convolution where that integral exists; otherwise it specifies its renormalized extension and is not an assertion that the unregularized integral converges.

For evolution factors and nonperturbative boundaries outside this polynomial algebra, we write $\mathscr{M}_\mu$ for a *partial multiplier domain*: each multiplier and each product actually used must be a locally integrable function defining a tempered distribution, or must have a separately specified distributional extension. The product must be checked, not assumed. This domain is not asserted to be one globally closed algebra. For example, the two factors $e^{-3L_b/4}$ are individually locally integrable in two transverse dimensions, whereas their product $e^{-3L_b/2}$ is not locally integrable at the origin. A closed algebra of ordinary multipliers cannot contain every generator with $0 < \mathrm{Re}\,\eta < 1$. The earlier unrestricted closure assumption is therefore not used.

Sharp bins are not Schwartz test functions. For the radial logarithmic distributions in this note their actions are defined by the common cumulative prescription below, or by smooth approximations constant near the origin and converging at a regular bin edge. Arbitrary singular measurements require their own domain analysis.

5.2 Evolution and sign conventions

The explicit fundamental-representation formulas use $T_F = 1/2$, $C_A = N_c$, and $C_F = (N_c^2-1)/(2N_c)$. Thus a finite $q \leftarrow g$ coefficient written as $2z(1-z)$ is $4T_F z(1-z)$ with the trace normalization restored. Quark/gluon quadratic-Casimir rescaling must not be extrapolated to the four-loop cusp, which contains quartic color invariants [33].

We use

$$a_s \equiv \frac{\alpha_s}{4\pi}, \qquad \frac{\mathrm{d}a_s}{\mathrm{d}\ln\mu} = -2\sum_{n\ge 0}\beta_n a_s^{n+2}. \tag{12}$$

Our Collins–Soper convention is

$$\frac{\partial}{\partial \ln\sqrt{\zeta}}\widetilde{F}_i = -D_i\widetilde{F}_i, \qquad D_i = -\widetilde{K}_i^{\mathrm{Collins}}, \tag{13}$$

so that

$$\frac{\mathrm{d}D_i}{\mathrm{d}\ln\mu} = \gamma_K^i = 2\Gamma^i_{\mathrm{cusp}}, \qquad \frac{\partial\gamma_F^i}{\partial\ln\sqrt{\zeta}} = -2\Gamma^i_{\mathrm{cusp}}. \tag{14}$$

These equations fix the signs and factors of two used in every subsequent kernel formula. They are the $D = -\widetilde{K}$ rewriting of the CSS2 equations in Ref. [39].

For Drell–Yan we impose $\zeta_A\zeta_B = Q^4$. The symmetric point is $\zeta_A = \zeta_B = Q^2$, but correlated rapidity-scale profiles may be used for uncertainty studies as long as the same convention is maintained.

5.3 Proof and verification ledger

The note uses five logically different levels of support:

Status	Meaning in this note
Imported parent theorem	CSS2 factorization, operator definitions, the small-b_T OPE, and the coupled Collins–Soper/RG equations are hypotheses taken from the cited parent literature. They are not reproved here.
Exact representation identity	An equality in $\mathcal{S}'(\mathbb{R}^2)$ following from the Fourier transform, injectivity, and the multiplier-domain assumptions in Eqs. (10)–(11). It applies at fixed symbolic scales, or when the complete scale-dependent parent multiplier is inverse transformed as one object. The distributional profile construction of Sec. 11 is a separate momentum-space resummation prescription and is not asserted to be the exact inverse transform of an arbitrary b_T-space profile.
Perturbative theorem/corollary	An equality order by order in a_s after importing the same renormalized CSS2 anomalous dimensions, matching coefficients, hard functions, and a scale prescription consistent with the claimed logarithmic counting. It inherits the factorization and power corrections of the parent theorem.
Symbolic or numerical regression check	A check that the code reproduces a stated algebraic identity, coefficient map, bin action, or semigroup relation. Passing such a check is not an independent proof of TMD factorization, of the physical coefficient input, or of logarithmic minimization by a profile.
Phenomenological claim	A statement requiring physical coefficient payloads, a specified nonperturbative module, data, cuts, power-correction treatment, and uncertainty analysis. It is not implied by the formal representation alone.

The dependency structure is summarized explicitly in Table 4.

Table 4: Theorem and verification dependency ledger.

Object or statement	Status	Source of support
CSS2 leading-power factorization, TMD definition, small-b_T OPE, and two-scale evolution	Imported theorem	Collins/CSS2 and scheme-translation literature, primarily Refs. [6, 39].
$\mathcal{K}_{\mathrm{CSS2}}$ representation of a fixed-scale CSS2 singular operator	Proved here	Fourier convolution theorem and injectivity on the declared admissible multiplier domain; Theorem 6.1.
Collins–Soper/RG path independence	Derived here from imported consistency equations	Proposition 9.1; no fit or numerical assumption enters.

Admissible generator composition and polynomial convolution algebra	Proved here	Fourier images of $\mathcal{G}_\eta$ and $\mathfrak{L}_n$; Theorem 10.1.
Canonical and profile scale setting for distributions	Imported one-scale construction plus a conditional extension	The one-scale, two-dimensional cumulative prescription is imported from Ref. [14] and is restated in Definition 11.1. The simultaneous (μ, ζ) profile is defined weakly here with origin contacts. Its mathematical domain and finite-order inheritance are proved; coupled RG and final-state rapidity constraints are additional physical requirements, not a proof of all-order resummation.
One-, two-, and three-loop kernel formulas	Perturbative derivation	CSS2 RG equations plus the cited loop coefficients; companion scripts check algebra and projections but do not independently calculate the QCD inputs.
$N^k\mathrm{LL}'$ accuracy inheritance	Conditional corollary	The fixed-truncation representation theorem, the declared ingredient convention, and an admissible, logarithmically accurate resummed solution. A finite-order distributional two-scale prescription alone does not establish the latter; Sec. 22.4 identifies the obstruction; Sec. 23 resolves it for the spectral-canonical class.
Nonperturbative module and extraction	Prescription/open implementation	Must satisfy the stated overlap, evolution, large-k_T, and regulated-integral conditions; no existence or uniqueness theorem is claimed.

6 Reference scheme: KCSS2 to Collins/CSS2

The initial scheme is not a new operator definition. It is the Collins/CSS2 TMD scheme represented directly in transverse momentum. The factorization theorem, Wilson-line directions, soft subtraction, and renormalized TMD definitions are those of the parent CSS2 formalism [6, 39]. The present construction addresses only their representation and perturbative implementation.

Define

$$\mathcal{F}^{-1}_{b\to k}[\widetilde{A}](\boldsymbol{k}) \equiv \int \frac{\mathrm{d}^2\boldsymbol{b}}{(2\pi)^2} e^{-i\boldsymbol{b}\cdot\boldsymbol{k}}\widetilde{A}(\boldsymbol{b}), \qquad \mathcal{F}_{k\to b}[A](\boldsymbol{b}) \equiv \int \mathrm{d}^2\boldsymbol{k}\, e^{i\boldsymbol{b}\cdot\boldsymbol{k}} A(\boldsymbol{k}). \tag{15}$$

For every parent object in the admissible domain, define

$$F^{[\mathcal{K}_{\mathrm{CSS2}}]}_{i/h}(x, \boldsymbol{k}; \mu, \zeta) \equiv \mathcal{F}^{-1}_{b\to k}\left[\widetilde{F}^{[\mathrm{CSS2}]}_{i/h}(x, \boldsymbol{b}; \mu, \zeta)\right], \tag{16}$$

$$\mathcal{D}^{[\mathcal{K}_{\mathrm{CSS2}}]}_{i}(\boldsymbol{k}; \mu) \equiv \mathcal{F}^{-1}_{b\to k}\left[D^{[\mathrm{CSS2}]}_{i}(\boldsymbol{b}; \mu)\right], \tag{17}$$

$$\mathcal{C}^{[\mathcal{K}_{\mathrm{CSS2}}]}_{i\leftarrow j}(z, \boldsymbol{k}; \mu, \zeta) \equiv \mathcal{F}^{-1}_{b\to k}\left[\widetilde{C}^{[\mathrm{CSS2}]}_{i\leftarrow j}(z, \boldsymbol{b}; \mu, \zeta)\right]. \tag{18}$$

The equality is distributional. The sign convention for D_i is fixed by Eq. (13).

Theorem 6.1 (Representation of the CSS2 singular operator)**.** *Let the two parent CSS2 TMD factors have Fourier images in $\mathscr{M}_\mu$, and assume that their product is an admissible multiplier. Then the CSS2 singular term*

$$W_{AB}^{[\mathrm{CSS2}]}(Q,y,\boldsymbol{q}) = \sum_{ij} H_{ij}^{[\mathrm{CSS2}]}(Q,\mu)\mathcal{F}_{b\to k}^{-1}\Big[\widetilde{F}_{i/A}^{[\mathrm{CSS2}]}(x_A,\boldsymbol{b};\mu,\zeta_A)\widetilde{F}_{j/B}^{[\mathrm{CSS2}]}(x_B,\boldsymbol{b};\mu,\zeta_B)\Big](\boldsymbol{q}) \tag{19}$$

is exactly equal in $\mathcal{S}'(\mathbb{R}^2)$ to

$$W_{AB}^{[\mathcal{K}_{\mathrm{CSS2}}]}(Q,y,\boldsymbol{q}) = \sum_{ij} H_{ij}^{[\mathrm{CSS2}]}(Q,\mu)\left[F_{i/A}^{[\mathcal{K}_{\mathrm{CSS2}}]} \otimes_T F_{j/B}^{[\mathcal{K}_{\mathrm{CSS2}}]}\right](\boldsymbol{q}). \tag{20}$$

Proof. Apply Eq. (11) to the two inverse Fourier images in Eqs. (16). Their convolution has Fourier image equal to the product of the two parent TMDs. Fourier injectivity on $\mathcal{S}'(\mathbb{R}^2)$ then gives Eq. (20). The hard function is independent of the transverse representation and is unchanged. □

Remark 6.2. Theorem 6.1 is a representation theorem, not a new proof of leading-power TMD factorization. It applies to any nonperturbative boundary only if that boundary and its Fourier image belong to the stated domain. In particular, an arbitrary DNN output is not automatically an admissible TMD distribution; admissibility must be imposed by the extraction architecture.

6.1 Optional child schemes and finite conversion kernels

A later numerically adapted scheme may be defined by finite, process-independent kernels,

$$F_{i/h}^{[\mathcal{K}]}(x,\boldsymbol{k};\mu,\zeta) = \sum_j \left[\mathcal{Z}_{i\leftarrow j}^{\mathcal{K}\leftarrow\mathcal{K}_{\mathrm{CSS2}}} \otimes_x \otimes_T F_{j/h}^{[\mathcal{K}_{\mathrm{CSS2}}]}\right](x,\boldsymbol{k};\mu,\zeta), \tag{21}$$

with

$$\mathcal{Z}_{i\leftarrow j}^{\mathcal{K}\leftarrow\mathcal{K}_{\mathrm{CSS2}}}(z,\boldsymbol{k}) = \delta_{ij}\,\delta(1-z)\,\delta^{(2)}(\boldsymbol{k}) + \mathcal{O}(\alpha_s)\,. \tag{22}$$

The inverse relation is

$$F_{i/h}^{[\mathcal{K}_{\mathrm{CSS2}}]} = \sum_j \left[\left(\mathcal{Z}^{\mathcal{K}\leftarrow\mathcal{K}_{\mathrm{CSS2}}}\right)^{-1}_{i\leftarrow j} \otimes_x \otimes_T F_{j/h}^{[\mathcal{K}]}\right]. \tag{23}$$

If $\mathcal{Z}$ has nontrivial transverse-momentum dependence, the physical W term written in terms of $F^{[\mathcal{K}]}$ contains these inverse kernels explicitly:

$$W_{AB} = \sum_{ij} H_{ij}^{[\mathrm{CSS2}]}\left[\left(\mathcal{Z}^{-1} \otimes_x \otimes_T F_A^{[\mathcal{K}]}\right)_i \otimes_T \left(\mathcal{Z}^{-1} \otimes_x \otimes_T F_B^{[\mathcal{K}]}\right)_j\right](\boldsymbol{q}). \tag{24}$$

Therefore a nonlocal transverse conversion kernel cannot simply be hidden inside a b-independent hard function. A finite scheme change is acceptable only if the complete cross section, including any accompanying hard-factor redefinition and both inverse leg kernels, is invariant to the claimed perturbative order. Equation (21) by itself is not a proof of scheme equivalence. This is why the first implementation sets $\mathcal{Z} = \mathbf{1}$ and works in $\mathcal{K}_{\mathrm{CSS2}}$. Only after fixed-order and RG checks pass should a nontrivial child scheme be introduced.

Working decision. Separate three notions that are often conflated: the TMD operator scheme, the transverse distribution basis, and the numerical implementation. The first implementation keeps the Collins/CSS2 operator scheme, derives the one-loop k_T-space matching and Collins-Soper kernels, validates the bin and flavor-vector checks, and now adds the first one-loop evolution Green function.

6.2 First conversion targets

The $\mathcal{K}_{\rm CSS2} \leftrightarrow$ CSS2 bridge reduces the first derivations to a finite list:

1. validate the mapped one-loop CSS2 quark-channel coefficients $\widetilde{C}_{q\leftarrow q}^{[\rm CSS2]}$ and $\widetilde{C}_{q\leftarrow g}^{[\rm CSS2]}$ in the $\mathcal{L}_n$ and $\mathcal{G}_\eta$ bases;
2. map the perturbative Collins-Soper kernel $D_q^{[\rm CSS2]}(b;\mu)$ into $\mathcal{D}_q^{[\mathcal{K}_{\rm CSS2}]}(k;\mu)$;
3. verify that Eq. (20), expanded through $\mathcal{O}(\alpha_s)$, reproduces the known Drell-Yan small-q_T singular spectrum;
4. add the Y term only after the singular W operator passes the CSS2 equivalence tests.

7 Momentum-space TMD factorization for Drell–Yan

For a color-singlet Drell–Yan final state with invariant mass Q, rapidity y, and $q_T \ll Q$, the Born-projected leading-power momentum fractions are

$$x_A = \frac{Q}{\sqrt{s}} e^{y}, \qquad x_B = \frac{Q}{\sqrt{s}} e^{-y}. \tag{25}$$

Beyond leading power or in a fully differential fiducial measurement, the precise recoil map is part of the measurement operator and need not coincide with Eq. (25) event by event.

We take σ_0 to contain the common electromagnetic/color normalization. An un-hatted channel hard factor includes its Born charge. Whenever e_q^2 is written outside the sum explicitly, the reduced factor is $\widehat{H}_q$, with $H_{q\bar{q}} = e_q^2 \widehat{H}_q$ and $\widehat{H}_q^{(0)} = 1$. This avoids counting the charge twice in the generic and explicit one-loop formulas.

The leading-power singular term is

$$\begin{aligned} W_{AB}(Q,y,\boldsymbol{q};\mu,\zeta_A,\zeta_B) &= \sum_{ij} H_{ij}(Q,\mu) \int \mathrm{d}^2\boldsymbol{k}_A\, \mathrm{d}^2\boldsymbol{k}_B\, \delta^{(2)}(\boldsymbol{q}-\boldsymbol{k}_A-\boldsymbol{k}_B) \\ &\quad \times F_{i/A}^{\rm DY}(x_A,\boldsymbol{k}_A;\mu,\zeta_A) F_{j/B}^{\rm DY}(x_B,\boldsymbol{k}_B;\mu,\zeta_B) \qquad (26) \\ &= \sum_{ij} H_{ij} \left[F_{i/A}^{\rm DY}(x_A;\mu,\zeta_A) \otimes_T F_{j/B}^{\rm DY}(x_B;\mu,\zeta_B) \right](\boldsymbol{q}). \qquad (27) \end{aligned}$$

The full spectrum is organized as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}Q^2\,\mathrm{d}y\,\mathrm{d}^2\boldsymbol{q}} = \sigma_0(Q,s,y)\,[W_{AB}^{\rm resum} + Y_{AB}] + \delta\sigma_{\rm pow}, \tag{28}$$

with

$$Y_{AB} = \left[\frac{1}{\sigma_0}\frac{\mathrm{d}\sigma}{\mathrm{d}Q^2\,\mathrm{d}y\,\mathrm{d}^2\boldsymbol{q}}\right]_{\rm FO} - \left[W_{AB,\rm pert}^{\rm LP}\right]_{\text{expanded to the same FO}}. \tag{29}$$

The subtraction uses the perturbative leading-power operator, not a fitted nonperturbative model. The term $\delta\sigma_{\rm pow}$ contains only corrections not already present in the full fixed-order remainder Y or an explicit recoil improvement; it must not add those power terms a second time. For sufficiently inclusive recoil-insensitive observables these commonly begin at relative order q_T^2/Q^2, but fiducial leptonic cuts and recoil-sensitive measurement maps can generate linear q_T/Q corrections that require their own resummation or exact kinematic treatment [47]. We therefore do not assign a universal power to $\delta\sigma_{\rm pow}$ in Eq. (28). Equation (29) is representation independent.

Working decision. A restricted fixed-target study may initially use only W in a stated small-q_T region. A full-spectrum or collider-level prediction requires the matched remainder and a separate assessment of fiducial and dynamical power corrections.

8 Generic process-level W operator

Drell–Yan is the first validation channel, not the definition of the framework. For a process F whose leading-power factorization contains two transverse-momentum-carrying legs, introduce a process-specific measurement convolution $\otimes_{T,\mathcal{M}_F}$. It contains the momentum-conservation map, Jacobians, fragmentation rescalings, angular projectors, and fiducial measurement function appropriate to F. The schematic singular operator is

$$W_F^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{q};Q,\{x\},\mu)=\sum_{ab}H_{ab\to F}(Q,\mu)\left[\mathbb{F}_{a/A}^{\mathcal{K}_{\rm CSS2}}\otimes_{T,\mathcal{M}_F}\mathbb{F}_{b/B}^{\mathcal{K}_{\rm CSS2}}\right](\boldsymbol{q}). \tag{30}$$

For unpolarized Drell–Yan with the simple recoil definition, $\otimes_{T,\mathcal{M}_F}$ reduces to $\otimes_T$ in Eq. (6).

The following equations display only their transverse-convolution skeletons:

$$W_{\rm DY}=\sum_q e_q^2\widehat{H}_{q\bar{q}}^{\rm DY}\left[F_{q/A}\otimes_T F_{\bar{q}/B}+F_{\bar{q}/A}\otimes_T F_{q/B}\right], \tag{31}$$

$$W_{\rm SIDIS}\sim\sum_{ij}H_{ij}^{\rm SIDIS}\left[F_{i/A}\otimes_{T,\mathcal{M}_{\rm SIDIS}}D_{h/j}\right], \tag{32}$$

$$W_{e^+e^-}\sim\sum_{ij}H_{ij}^{e^+e^-}\left[D_{h_1/i}\otimes_{T,\mathcal{M}_{e^+e^-}}D_{h_2/j}\right], \tag{33}$$

$$W_{gg\to F}\sim H_{gg\to F}\,\mathcal{P}_F^{\alpha\beta\rho\sigma}\left[F_{g/A}^{\alpha\beta}\otimes_{T,\mathcal{M}_F}F_{g/B}^{\rho\sigma}\right]+\cdots. \tag{34}$$

The symbols $\sim$ emphasize that SIDIS and fragmentation formulas require their process-dependent z factors and measurement maps, while gluon channels can require nontrivial transverse tensor structures, including linearly polarized gluon TMDs. Those details do not invalidate the scalar subalgebra, but tensor harmonics require additional tensor distributions and transform/projector rules; changing only the hard and measurement factors is not generally sufficient. The recent availability of high-order polarized and tensor matching coefficients makes such extensions possible, but they are not proven merely by the unpolarized scalar examples.

Collecting the partonic channels into

$$\mathbf{F}_h=(F_{g/h},F_{u/h},F_{\bar{u}/h},F_{d/h},F_{\bar{d}/h},\ldots)^T, \tag{35}$$

allows $\mathcal{C}$, DGLAP evolution, and any finite scheme conversion to act as flavor matrices. Flavor simplifications are fit assumptions, not properties of $\mathcal{K}_{\rm CSS2}$.

9 Evolution as convolutional dynamics

The CSS2 evolution equations become, in k_T space,

$$\frac{\partial}{\partial\ln\sqrt{\zeta}}F_i(x,\boldsymbol{k};\mu,\zeta)=-\left[\mathcal{D}_i(\mu)\otimes_T F_i(x;\mu,\zeta)\right](\boldsymbol{k}), \tag{36}$$

$$\frac{\partial}{\partial\ln\mu}F_i(x,\boldsymbol{k};\mu,\zeta)=\gamma_F^i(\mu,\zeta)F_i(x,\boldsymbol{k};\mu,\zeta), \tag{37}$$

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\mathcal{D}_i(\boldsymbol{k};\mu)=2\Gamma^i_{\rm cusp}(\alpha_s(\mu))\delta^{(2)}(\boldsymbol{k}). \tag{38}$$

The first equation is nonlocal in transverse momentum, whereas the second is local because γ_F^i is independent of k_T.

Proposition 9.1 (Integrability of the two-scale evolution)**.** *Within a fixed perturbative truncation and the domain $\mathscr{A}_\mu$, the μ and ζ evolutions commute if*

$$\frac{\mathrm{d}\mathcal{D}_i}{\mathrm{d}\ln\mu} = 2\Gamma_i\delta^{(2)}, \qquad \frac{\partial\gamma_F^i}{\partial\ln\sqrt{\zeta}} = -2\Gamma_i. \tag{39}$$

Consequently the endpoint kernel is path independent up to terms beyond the common perturbative truncation.

Proof. Apply the two derivatives to F_i in opposite orders. The commutator is the convolution with $-\mathrm{d}\mathcal{D}_i/\mathrm{d}\ln\mu - \partial\gamma_F^i/\partial\ln\sqrt{\zeta}\,\delta^{(2)}$, which vanishes by Eq. (39). At finite order the statement requires all anomalous dimensions and running-coupling terms to be truncated consistently. □

At fixed μ, the solution is

$$F_i(x,\boldsymbol{k};\mu,\zeta_f) = \left[\mathrm{Exp}_{\otimes_T}\left\{-\ln\sqrt{\frac{\zeta_f}{\zeta_i}}\,\mathcal{D}_i(\mu)\right\}\otimes_T F_i(x;\mu,\zeta_i)\right](\boldsymbol{k}), \tag{40}$$

where

$$\mathrm{Exp}_{\otimes_T}\{A\} \equiv \delta^{(2)} + A + \frac{1}{2!}A\otimes_T A + \frac{1}{3!}A\otimes_T A\otimes_T A + \cdots. \tag{41}$$

Equation (40) is exact when the corresponding Fourier multiplier exponential defines an element of the admissible domain. In fixed-order derivations it is a formal distribution series in a_s; equality then means equality coefficient by coefficient through the retained order. This distinction is essential when profile choices or downward evolution would produce a non-tempered exponential.

Working decision. The implementation object is a radial distributional Green function $\mathcal{U}_i$ satisfying $F_i(\mu_f,\zeta_f) = \mathcal{U}_i\otimes_T F_i(\mu_i,\zeta_i)$. Path independence is an RG integrability property, not an empirical numerical assumption.

10 Distribution basis for pure-k_T resummation

Perturbative transverse spectra contain $\delta^{(2)}(\boldsymbol{k})$ and two-dimensional plus distributions. For $\varphi\in\mathcal{S}(\mathbb{R}^2)$, define

$$\langle\mathcal{L}_n(\mu),\varphi\rangle \equiv \int\frac{\mathrm{d}^2\boldsymbol{k}}{\pi k_T^2}\ln^n\!\left(\frac{\mu^2}{k_T^2}\right)\left[\varphi(\boldsymbol{k}) - \varphi(\boldsymbol{0})\,\theta(\mu-k_T)\right], \qquad n\geq 0. \tag{42}$$

Equivalently,

$$\mathcal{L}_n(\boldsymbol{k},\mu) = \frac{1}{\pi\mu^2}\left[\frac{\mu^2}{k_T^2}\ln^n\!\left(\frac{\mu^2}{k_T^2}\right)\right]_+, \tag{43}$$

with the subtraction scale fixed to μ. Equation (42) fixes the sign, the contact terms, and the zero-bin cumulants. Pointwise values at $k_T=0$ are undefined and physically irrelevant.

Let

$$L_b \equiv \ln\!\left(\frac{\mu^2 b_T^2 e^{2\gamma_E}}{4}\right). \tag{44}$$

For $0 < \operatorname{Re}\eta < 1$, define the analytic generator by

$$\mathcal{F}_{k\to b}[\mathcal{G}_\eta](\boldsymbol{b}) = e^{-\eta L_b}. \tag{45}$$

Its ordinary representative away from the origin is

$$\mathcal{G}_\eta(\boldsymbol{k};\mu) = \frac{1}{\pi\mu^2}\left(\frac{\mu^2}{k_T^2}\right)^{1-\eta} \frac{e^{-2\gamma_E\eta}\Gamma(1-\eta)}{\Gamma(\eta)}. \tag{46}$$

The family has a distributional analytic continuation to a neighborhood of $\eta = 0$.

Theorem 10.1 (Admissible generator composition)**.** *For η, η' in a common domain where the multipliers and their product are admissible,*

$$\mathcal{G}_\eta \otimes_T \mathcal{G}_{\eta'} = \mathcal{G}_{\eta+\eta'}. \tag{47}$$

The same identity holds coefficient by coefficient after analytic continuation about $\eta = \eta' = 0$.

Proof. For ordinary transverse integrals an explicit common strip is $\operatorname{Re}\eta > 0$, $\operatorname{Re}\eta' > 0$, $\operatorname{Re}(\eta+\eta') < 1$. In that strip Feynman parametrization gives

$$\begin{aligned}&\int \mathrm{d}^2\boldsymbol{p}\,(p_T^2)^{\eta-1}[(\boldsymbol{k}-\boldsymbol{p})^2]^{\eta'-1}\\ &\quad= \pi(k_T^2)^{\eta+\eta'-1}\frac{\Gamma(\eta)\Gamma(\eta')\Gamma(1-\eta-\eta')}{\Gamma(1-\eta)\Gamma(1-\eta')\Gamma(\eta+\eta')}.\end{aligned} \tag{48}$$

The normalizations in Eq. (46) reduce this to $\mathcal{G}_{\eta+\eta'}$. Equivalently the Fourier multipliers multiply to $e^{-(\eta+\eta')L_b}$ and Fourier injectivity gives the result. Differentiating the holomorphic multiplier family near zero proves the coefficientwise identity. Continuation through its poles is not an unrestricted group law for ordinary functions; any such continuation requires its own prescription. □

Define the derivative basis

$$\mathfrak{L}_n(\boldsymbol{k};\mu) \equiv (-1)^{n+1}\,\frac{\partial^{n+1}}{\partial\eta^{n+1}}\mathcal{G}_\eta(\boldsymbol{k};\mu)\bigg|_{\eta=0}, \qquad \mathcal{F}_{k\to b}[\mathfrak{L}_n] = L_b^{n+1}. \tag{49}$$

The conventional-basis expansion is

$$\mathcal{G}_\eta = h(\eta)\left[\delta^{(2)} + \eta\mathcal{L}_0 - \eta^2\mathcal{L}_1 + \frac{\eta^3}{2}\mathcal{L}_2 + \mathcal{O}\left(\eta^4\right)\right], \tag{50}$$

with

$$h(\eta) = e^{-2\gamma_E\eta}\frac{\Gamma(1-\eta)}{\Gamma(1+\eta)} = 1 + \frac{2\zeta_3}{3}\eta^3 + \mathcal{O}\left(\eta^5\right). \tag{51}$$

Hence

$$\mathfrak{L}_0 = -\mathcal{L}_0, \tag{52}$$

$$\mathfrak{L}_1 = -2\mathcal{L}_1, \tag{53}$$

$$\mathfrak{L}_2 = -3\mathcal{L}_2 - 4\zeta_3\delta^{(2)}, \tag{54}$$

and

$$\mathcal{F}_{k\to b}[\mathcal{L}_0] = -L_b, \tag{55}$$

$$\mathcal{F}_{k\to b}[\mathcal{L}_1] = -\frac{1}{2}L_b^2, \tag{56}$$

$$\mathcal{F}_{k\to b}[\mathcal{L}_2] = -\frac{1}{3}L_b^3 - \frac{4}{3}\zeta_3. \tag{57}$$

The ζ values in these equations are basis-conversion constants, not new dynamical coefficients. The signs follow from using $\ln(\mu^2/k_T^2)$ in Eq. (43).

Remark 10.2. The generator construction establishes an exact algebra for the perturbative distribution subspace. It does not imply that the inverse Fourier transform of every all-order Sudakov ansatz is a positive function or even an ordinary function. Physical predictions are obtained from its action on bins or smooth measurement functions.

11 Distributional canonical and profile scales

Changing a scale after evaluating a differential plus distribution is not, without a prescription, the same operation as changing that scale in its cumulant and then differentiating. We adopt the cumulative scale-setting approach of Ref. [14], with the origin term made explicit. This section defines one admissible scale-setting construction; it does not exclude other momentum-space resummation methods with separately proved accuracy.

Definition 11.1 (Weak profile operator including origin contact terms)**.** Put $t = k_T^2$ and, for $\varphi \in \mathcal{S}(\mathbb{R}^2)$, define its angular average

$$\psi_\varphi(t) = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{d}\theta\, \varphi(\sqrt{t}\,\hat{e}_\theta). \tag{58}$$

For a radial family $A(\boldsymbol{k};\lambda)$ with a specified cumulative $\Sigma_A(K;\lambda)$, set

$$C_p(t) = \Sigma_A(\sqrt{t};\lambda_p(\sqrt{t})). \tag{59}$$

Assume that C_p is locally integrable on $[0,\infty)$ and of at most polynomial growth. Its origin need not have a finite limit: finite powers of $\ln t$ are allowed. The profiled distribution is defined by

$$\boxed{\langle \mathcal{S}_p A, \varphi\rangle = -\int_0^\infty \mathrm{d}t\, C_p(t)\psi'_\varphi(t).} \tag{60}$$

This is the radial lift of $\mathrm{d}[\theta(t)C_p(t)]/\mathrm{d}t$, not just the ordinary derivative for $t > 0$. It defines a continuous tempered-distribution action under the stated integrability and growth bounds. It is a restricted construction for families with function-valued cumulants, not a definition for arbitrary elements of $\mathcal{S}'(\mathbb{R}^2)$, such as unspecified derivatives of contact distributions.

Proposition 11.2 (Cumulative consistency, including a zero bin)**.** *At a positive continuity point K of the cumulative, the common bin extension gives*

$$\begin{aligned}\int_{|\boldsymbol{k}|\le K} \mathrm{d}^2\boldsymbol{k}\, \mathcal{S}_p A(\boldsymbol{k}) &= C_p(K^2),\\ \int_{K_a<|\boldsymbol{k}|\le K_b} \mathrm{d}^2\boldsymbol{k}\, \mathcal{S}_p A(\boldsymbol{k}) &= C_p(K_b^2) - C_p(K_a^2), \quad K_a > 0.\end{aligned} \tag{61}$$

In particular, the zero-bin action does not *subtract $C_p(0^+)$. When that limit is finite and $C_p - C_p(0^+) \in W^{1,1}_{\mathrm{loc}}([0,\infty))$,*

$$\mathcal{S}_p A = C_p(0^+)\delta^{(2)}(\boldsymbol{k}) + \frac{1}{\pi}C_p'(k_T^2) \quad (k_T > 0 \text{ for the second term}). \tag{62}$$

For logarithmically divergent $C_p(0^+)$ only the weak formula is used.

Proof. Approximate the disk indicator by a radial test equal to one near $t = 0$ whose derivative tends to $-\delta(t - K^2)$. Equation (60) then yields $C_p(K^2)$. Taking a difference gives the annulus formula. For a finite origin limit, integrate by parts in the weak definition; the endpoint contribution is $C_p(0^+)\psi_\varphi(0)$, which is exactly the contact term. For $C_p(t) = c$ the formula gives $c\delta^{(2)}$, whereas subtracting the origin cumulative would incorrectly give zero. Logarithmic divergences cause no problem in the weak integral because ψ_φ' is bounded near zero. □

Proposition 11.3 (Finite-order inheritance under weak profiling)**.** *Suppose the two profiled cumulants have a common perturbative expansion through a_s^N. For their remainder $R_N(t)$ assume, on every compact interval including the origin, $|R_N(t)| \le a_s^{N+1} r(t)$ with r locally integrable. For Schwartz tests also require $|R_N(t)| \le a_s^{N+1} C(1+t)^m$ at sufficiently large t, with C, m independent of a_s in the stated expansion regime; compact tests need no such additional bound. Then their weak profile actions agree through a_s^N. Annular or zero-bin actions agree to that order whenever the respective endpoint remainder obeys the same order estimate.*

Proof. The action difference is

$$\langle \mathcal{S}_p A_N - \mathcal{S}_p B_N, \varphi\rangle = -\int_0^\infty \mathrm{d}t\, R_N(t)\psi_\varphi'(t). \tag{63}$$

The stated dominated bound gives $\mathcal{O}\left(a_s^{N+1}\right)$ on a compact support, including any origin contribution. Polynomial growth and Schwartz decay extend the result to the stated noncompact domain. Bins follow from the endpoint identities. Bounds only on intervals bounded away from zero would not justify this conclusion for tests or bins containing the origin. □

For reference the finite-order hypothesis may be written as

$$\Sigma_{A_N}(K;\lambda_p(K)) - \Sigma_{B_N}(K;\lambda_p(K)) = \mathcal{O}\left(a_s^{N+1}\right), \tag{64}$$

but its uniformity/integrability assumptions are essential. This proposition is not a theorem about all-order logarithmic accuracy.

For the conventional logarithmic distributions,

$$C_{n,p}(t) = -\frac{1}{n+1}\ln^{n+1}\frac{\mu_p^2(\sqrt{t})}{t}, \tag{65}$$

so their ordinary density away from the origin is

$$\mathcal{L}_n(\boldsymbol{k},\mu)|_{\mu=\mu_p(k_T)|_+} = \frac{1}{\pi k_T^2}\ln^n\frac{\mu_p^2(k_T)}{k_T^2}\left(1 - \frac{\mathrm{d}\ln\mu_p}{\mathrm{d}\ln k_T}\right), \quad k_T > 0. \tag{66}$$

This is only the off-origin part of the weak distribution. A constant profile recovers the fixed-scale plus prescription. Canonical scale setting gives

$$\mu_p(K) = K, \qquad \zeta_p(K) = \mu_p^2(K) \tag{67}$$

and, for the logarithmic boundary factors,

$$\mathcal{L}_n(\boldsymbol{k},\mu)|_{\mu=k_T|_+}=0,\quad n\geq 0. \tag{68}$$

This does not set the generator-basis distributions to zero: their contact pieces survive. Their finite canonical boundary conversion is given in Sec. 13.6.

A possible profile pattern is

$$\mu_p(K)\to\mu_0\ (K\ll\mu_0),\quad \mu_p(K)\simeq K\ (\mu_0\ll K\ll Q),\quad \mu_p(K)\to Q\ (K\sim Q). \tag{69}$$

Freezing μ_p does not make the logarithmic zero-radius cumulant finite; it still behaves as powers of $\ln t$, which are covered by the weak definition. The running coupling, PDFs, matching coefficients, and scale-dependent contact coefficients must all be included before differentiating. For example, a contact coefficient $c(\mu)$ becomes $\mathrm{d}[\theta(t)c(\mu_p(\sqrt{t}))]/\mathrm{d}t$; it is not correctly represented by a formal $c(\mu_p(k_T))\delta^{(2)}$ alone.

For a TMD boundary define, at fixed symbolic scales,

$$\mathbb{B}^{\mathrm{dist}}_{i/h}(x,\boldsymbol{k})=\mathcal{S}_p\left\{\sum_j[\mathcal{C}_{i\leftarrow j}(\boldsymbol{k};\mu_T,\zeta_T)\otimes_x f_{j/h}(\mu_T)](x)\right\}. \tag{70}$$

This defines a profiled boundary block. It does *not* by itself define how to combine a momentum-dependent initial scale with an evolution kernel; that requires an explicitly ordered construction of the complete evolved operator.

Remark 11.4 (Profile operation versus convolution). The weak profile map is linear but generally

$$\mathcal{S}_p[A\otimes_T B]\neq(\mathcal{S}_pA)\otimes_T(\mathcal{S}_pB). \tag{71}$$

Leg-wise profiling and profiling the already-convolved spectrum are distinct prescriptions. In the iterative distribution-space RG construction, canonical rapidity scales must be set at the prescribed recurrence step, not moved through convolutions or substituted only at the end [14]. A proof for one placement is not a proof for the other.

For CSS2 families, the evolution connection must obey Eq. (14). The constraint $\zeta_{A,f}\zeta_{B,f}=Q^4$ applies to the two *final* TMDs entering a Drell–Yan factorization formula. It does not constrain two independent initial boundary scales to have product Q^4; for example both may start at $\zeta_i=\mu_0^2$. The mathematical chain rule itself does not require this physics constraint.

11.1 Reproducible tests and their limitations

The archived v1.4 audit script `validators/validate_v1p4.py` tests the weak operator on a pure contact term, fixed-scale plus distributions, a frozen profile with a divergent origin cumulative, a scale-dependent contact coefficient, and smooth profiles on bins including the origin. For the last class it checks *contact plus density*, not just the regular-density integral. It also checks the off-origin chain rule and the final rapidity-product identity. The complete structural audit, including the further tests described in Sec. 37, historically reports

$$N^{\mathrm{structural}}_{\mathrm{pass}}=102/102. \tag{72}$$

This historical count is for the complete v1.4 audit, not an eight-test profile certificate. Its scripts and outputs are preserved in the separately labeled archive in the v1.5 bundle. The generated

JSON/CSV records individual checks, precision, tolerances, source hashes, and residuals; the note does not retain a hard-coded legacy quadrature residual. Tests at high arithmetic precision are implementation checks, not physical uncertainty estimates.

The archived legacy profile script tested the ordinary derivative on $k_T > 0$. Its bin formula subtracted the origin cumulative even for a bin starting at zero. Consequently its eight passes, and the old note-to-repository numerical agreement, were not a verification of the contact-complete operator. The new pure-delta negative control rejects that formula. None of these tests supplies an all-order resummation solution, a physical three-loop coefficient import, or a data fit.

Working decision. Use the contact-complete weak profile definition in this construction. A formal scale replacement, a successful finite-order bin test, or a count of passing scripts cannot replace an all-order resummation argument. The argument for the spectral-canonical branch is supplied in Sec. 23.

12 One-loop KCSS2 kernels

This section performs the first explicit bridge calculation. We start from the CSS2 small-b_T coefficients and Collins-Soper kernel, then represent them as distributions in k_T space. Throughout this section

$$a_s(\mu) \equiv \frac{\alpha_s(\mu)}{4\pi}, \qquad b_0 \equiv 2e^{-\gamma_E}, \qquad L_b \equiv \ln\left(\frac{\mu^2 b_T^2}{b_0^2}\right), \qquad \ell_\zeta \equiv \ln\left(\frac{\zeta}{\mu^2}\right). \tag{73}$$

The leading-order collinear splitting functions use the standard $df/d\ln\mu^2 = (\alpha_s/2\pi)P^{(0)} \otimes f + \cdots$ convention:

$$P_{qq}^{(0)}(z) = C_F\left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2}\delta(1-z)\right], \tag{74}$$

$$P_{qg}^{(0)}(z) = T_F\left[z^2 + (1-z)^2\right]. \tag{75}$$

12.1 CSS2 input at the canonical small-bT point

Collins and Rogers give the one-loop CSS2 matching coefficients at the canonical point [39]

$$\mu_b = \frac{b_0}{b_T}, \qquad \zeta_b = \mu_b^2 = \frac{b_0^2}{b_T^2}, \qquad L_b = \ell_\zeta = 0. \tag{76}$$

In our notation their Eq. (74) gives

$$\widetilde{C}_{q\leftarrow q}^{(1),\mathrm{CSS2}}(z;\mu_b,\zeta_b) = C_F\left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z)\right], \tag{77}$$

$$\widetilde{C}_{q\leftarrow g}^{(1),\mathrm{CSS2}}(z;\mu_b,\zeta_b) = 2z(1-z), \tag{78}$$

$$\widetilde{C}_{q\leftarrow q'}^{(1),\mathrm{CSS2}} = \widetilde{C}_{q\leftarrow \bar{q}}^{(1),\mathrm{CSS2}} = \widetilde{C}_{q\leftarrow \bar{q}'}^{(1),\mathrm{CSS2}} = 0. \tag{79}$$

The coefficient is expanded as

$$\widetilde{C}_{i\leftarrow j}^{\mathrm{CSS2}} = \delta_{ij}\delta(1-z) + a_s\widetilde{C}_{i\leftarrow j}^{(1),\mathrm{CSS2}} + \mathcal{O}\left(a_s^2\right). \tag{80}$$

The $-\pi^2/6$ constant is part of the CSS2 separation between TMD matching and the Drell-Yan hard factor. A different finite TMD scheme would reshuffle this delta-function constant, but not the logarithmic distributional structure derived below.

12.2 Lifting the canonical coefficient to general mu and zeta

The logarithms at general (μ, ζ) are fixed by the CSS2 coefficient evolution equations [39]. To one loop, integrating those equations from the canonical point gives

$$\widetilde{C}^{(1),\mathrm{CSS2}}_{q\leftarrow q}(z, b_T; \mu, \zeta) = C_F \left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z)\right] - 2L_b P^{(0)}_{qq}(z) \tag{81}$$

$$+ C_F \delta(1-z)\left[(3 - 2\ell_\zeta)L_b - L_b^2\right], \tag{82}$$

$$\widetilde{C}^{(1),\mathrm{CSS2}}_{q\leftarrow g}(z, b_T; \mu, \zeta) = 2z(1-z) - 2L_b P^{(0)}_{qg}(z), \tag{83}$$

with the same zero entries as Eq. (79). These expressions obey two immediate checks:

$$\frac{\partial}{\partial \ln\sqrt{\zeta}} \widetilde{C}^{(1),\mathrm{CSS2}}_{q\leftarrow q} = -4C_F L_b\, \delta(1-z), \tag{84}$$

$$\frac{\partial}{\partial \ln\sqrt{\zeta}} \widetilde{C}^{(1),\mathrm{CSS2}}_{q\leftarrow g} = 0, \tag{85}$$

which is the one-loop CSS equation because

$$\widetilde{K}^{(1)}_q(b_T; \mu) = -4C_F L_b. \tag{86}$$

At fixed ζ, Eq. (82) also gives the required one-loop renormalization-scale derivative,

$$\frac{\partial}{\partial \ln\mu} \widetilde{C}^{(1),\mathrm{CSS2}}_{q\leftarrow j} = \left[6C_F - 4C_F\ell_\zeta\right]\delta_{qj}\delta(1-z) - 4P^{(0)}_{qj}(z), \tag{87}$$

which is Eq. (76) of Collins and Rogers written with the standard leading-order DGLAP kernels of Eq. (75).

12.3 Momentum-space matching coefficients

The $\mathcal{K}_{\mathrm{CSS2}}$ coefficient is the inverse transverse Fourier transform of Eq. (82). Using

$$L_b \mapsto \mathfrak{L}_0 = -\mathcal{L}_0, \qquad L_b^2 \mapsto \mathfrak{L}_1 = -2\mathcal{L}_1, \tag{88}$$

we obtain the one-loop momentum-space kernels

$$\mathcal{C}^{(1),\mathcal{K}_{\mathrm{CSS2}}}_{q\leftarrow q}(z, \boldsymbol{k}; \mu, \zeta) = C_F \left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z)\right]\delta^{(2)}(\boldsymbol{k}) \tag{89}$$

$$+ \left[2P^{(0)}_{qq}(z) - C_F(3 - 2\ell_\zeta)\delta(1-z)\right]\mathcal{L}_0(\boldsymbol{k}, \mu) \tag{90}$$

$$+ 2C_F\delta(1-z)\mathcal{L}_1(\boldsymbol{k}, \mu), \tag{91}$$

$$\mathcal{C}^{(1),\mathcal{K}_{\mathrm{CSS2}}}_{q\leftarrow g}(z, \boldsymbol{k}; \mu, \zeta) = 2z(1-z)\delta^{(2)}(\boldsymbol{k}) + 2P^{(0)}_{qg}(z)\mathcal{L}_0(\boldsymbol{k}, \mu), \tag{92}$$

$$\mathcal{C}^{(1),\mathcal{K}_{\mathrm{CSS2}}}_{q\leftarrow q'} = \mathcal{C}^{(1),\mathcal{K}_{\mathrm{CSS2}}}_{q\leftarrow \bar{q}} = \mathcal{C}^{(1),\mathcal{K}_{\mathrm{CSS2}}}_{q\leftarrow \bar{q}'} = 0. \tag{93}$$

The antiquark coefficients follow by charge conjugation. The gluon-initiated analogues for gluon TMDPDFs are not needed for the first fixed-target Drell-Yan validation, but the same map applies once the CSS2 gluon coefficients are inserted.

In generator form the same result is more compact:

$$\mathcal{C}_{q\leftarrow q}^{(1),\mathcal{K}_{\rm CSS2}} = c_{q\leftarrow q}^{\rm can}(z)\delta^{(2)} + \left[-2P_{qq}^{(0)}(z) + C_F(3-2\ell_\zeta)\delta(1-z)\right]\mathfrak{L}_0 - C_F\delta(1-z)\mathfrak{L}_1, \tag{94}$$

$$\mathcal{C}_{q\leftarrow g}^{(1),\mathcal{K}_{\rm CSS2}} = c_{q\leftarrow g}^{\rm can}(z)\delta^{(2)} - 2P_{qg}^{(0)}(z)\mathfrak{L}_0, \tag{95}$$

where

$$c_{q\leftarrow q}^{\rm can}(z) = C_F\left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z)\right], \qquad c_{q\leftarrow g}^{\rm can}(z) = 2z(1-z). \tag{96}$$

Equations (91) and (92) are the first concrete large-k_T TMD-tail kernels for the project.

12.4 Momentum-space Collins-Soper kernel

With our sign convention in Eq. (13),

$$D_q^{(1),\rm CSS2}(b_T;\mu) = -\widetilde{K}_q^{(1)}(b_T;\mu) = 4C_F L_b. \tag{97}$$

More generally,

$$D_i^{(1),\rm CSS2}(b_T;\mu) = \Gamma_0^i L_b, \qquad \Gamma_0^q = 4C_F, \qquad \Gamma_0^g = 4C_A. \tag{98}$$

Therefore

$$\mathcal{D}_i^{(1),\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = \Gamma_0^i\mathfrak{L}_0(\boldsymbol{k};\mu) = -\Gamma_0^i\mathcal{L}_0(\boldsymbol{k},\mu), \tag{99}$$

and

$$\mathcal{D}_q^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = -4C_F a_s(\mu)\mathcal{L}_0(\boldsymbol{k},\mu) + \mathcal{O}\left(a_s^2\right). \tag{100}$$

Although Eq. (100) is negative away from $k_T = 0$ with our plus-basis convention, this is not a positivity problem: $\mathcal{D}_i$ is a renormalized rapidity evolution distribution, not a probability density.

The RG check is immediate. Since

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\mathcal{L}_0(\boldsymbol{k},\mu) = -2\delta^{(2)}(\boldsymbol{k}), \tag{101}$$

Eq. (99) implies

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\mathcal{D}_i^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = 2\Gamma_0^i a_s(\mu)\delta^{(2)}(\boldsymbol{k}) + \mathcal{O}\left(a_s^2\right) = \gamma_{K,0}^i a_s(\mu)\delta^{(2)}(\boldsymbol{k}) + \mathcal{O}\left(a_s^2\right), \tag{102}$$

with $\gamma_{K,0}^i = 2\Gamma_0^i$, as required by Eq. (38).

12.5 First fixed-order singular-spectrum check

At order a_s, the $\mathcal{K}_{\rm CSS2}$ Drell-Yan W term follows by inserting Eqs. (91) and (92) into the two TMD factors in Eq. (20), adding the one-loop CSS2 hard factor, and keeping only terms through a_s. We use a reduced CSS2 hard factor with the Born charge e_q^2 pulled outside the flavor sum. At $\mu = Q$ it is [39]

$$\widehat{H}_{q\bar{q}}^{\rm DY,CSS2}(Q,Q) = 1 + a_s h_q^{(1)} + \mathcal{O}\left(a_s^2\right), \qquad h_q^{(1)} = C_F\left(-16 + \frac{7\pi^2}{3}\right). \tag{103}$$

For the first validation we take

$$\mu = Q, \qquad \zeta_A = \zeta_B = Q^2, \qquad \ell_{\zeta_A} = \ell_{\zeta_B} = 0, \tag{104}$$

and evaluate all collinear PDFs at $\mu = Q$ unless explicitly stated otherwise.

It is useful to introduce the $\mathcal{L}_0$ and delta-function coefficient kernels

$$b_{q\leftarrow q}(z) \equiv 2P_{qq}^{(0)}(z) - 3C_F\delta(1-z), \qquad b_{q\leftarrow g}(z) \equiv 2P_{qg}^{(0)}(z), \tag{105}$$

$$c_{q\leftarrow q}(z) \equiv C_F\left[-\frac{\pi^2}{6}\delta(1-z) + 2(1-z)\right], \qquad c_{q\leftarrow g}(z) \equiv 2z(1-z), \tag{106}$$

with identical antiquark kernels by charge conjugation. Then define the leading luminosity

$$\Phi_q^{(0)} \equiv f_{q/A}(x_A,Q)f_{\bar{q}/B}(x_B,Q) + f_{\bar{q}/A}(x_A,Q)f_{q/B}(x_B,Q), \tag{107}$$

and the one-leg insertions

$$\Phi_q^{b,A} \equiv \left[b_{q\leftarrow q}\otimes_x f_{q/A} + b_{q\leftarrow g}\otimes_x f_{g/A}\right](x_A)f_{\bar{q}/B}(x_B,Q) \tag{108}$$

$$+ \left[b_{q\leftarrow q}\otimes_x f_{\bar{q}/A} + b_{q\leftarrow g}\otimes_x f_{g/A}\right](x_A)f_{q/B}(x_B,Q), \tag{109}$$

$$\Phi_q^{b,B} \equiv f_{q/A}(x_A,Q)\left[b_{q\leftarrow q}\otimes_x f_{\bar{q}/B} + b_{q\leftarrow g}\otimes_x f_{g/B}\right](x_B) \tag{110}$$

$$+ f_{\bar{q}/A}(x_A,Q)\left[b_{q\leftarrow q}\otimes_x f_{q/B} + b_{q\leftarrow g}\otimes_x f_{g/B}\right](x_B), \tag{111}$$

and similarly $\Phi_q^{c,A}$ and $\Phi_q^{c,B}$ by replacing b with c in Eqs. (109) and (111). In this notation the complete one-loop singular W term in the $d^2\boldsymbol{q}$ distribution is

$$W_{AB}^{(0+1),\mathrm{sing}}(Q,y,\boldsymbol{q}) = \sum_q e_q^2\Bigg\{\Phi_q^{(0)}\delta^{(2)}(\boldsymbol{q}) \tag{112}$$

$$+ a_s(Q)\Big[4C_F\Phi_q^{(0)}\mathcal{L}_1(\boldsymbol{q},Q) + (\Phi_q^{b,A}+\Phi_q^{b,B})\mathcal{L}_0(\boldsymbol{q},Q) \tag{113}$$

$$+ \left(h_q^{(1)}\Phi_q^{(0)} + \Phi_q^{c,A} + \Phi_q^{c,B}\right)\delta^{(2)}(\boldsymbol{q})\Big]\Bigg\} + \mathcal{O}\left(a_s^2\right). \tag{114}$$

Equation (114) is the first fully explicit object that the code should reproduce from pure k_T convolutions. The coefficient of $\mathcal{L}_1$ is fixed only by the two incoming quark TMDs,

$$2C_F + 2C_F = 4C_F, \tag{115}$$

while all flavor mixing at this order enters through the $\mathcal{L}_0$ and $\delta^{(2)}(\boldsymbol{q})$ one-leg insertions.

12.6 Relation to the usual CSS $A^{(1)}$ and $B^{(1)}$ check

The standard $A^{(1)}$–$B^{(1)}$ comparison is a coefficient-level check. One must not apply the polynomial dictionary to a factor such as $f(x,\mu_b)$ while treating it as independent of b_T, because $\mu_b = b_0/b_T$ itself carries b_T dependence. A literal transform of the full canonical expression requires either keeping the PDF evolution inside the transform or first re-expanding the PDFs at a fixed reference scale.

To isolate the universal diagonal logarithms, freeze the Born luminosity at a fixed scale and define the one-loop coefficient polynomial

$$\widetilde{\mathcal{S}}_{q\bar{q}}^{(1)}(L_b) = -2C_FL_b^2 + 6C_FL_b + h_q^{(1)} - \frac{\pi^2}{3}C_F. \tag{116}$$

The last term is the sum of the two canonical one-loop TMD matching constants. The exact distribution map of the polynomial is

$$\mathcal{S}^{(1)}_{q\bar{q}}(\boldsymbol{q}) = 4C_F\mathcal{L}_1(\boldsymbol{q},Q) - 6C_F\mathcal{L}_0(\boldsymbol{q},Q) + \left(h^{(1)}_q - \frac{\pi^2}{3}C_F\right)\delta^{(2)}(\boldsymbol{q}). \tag{117}$$

For $q_T > 0$,

$$\pi a_s\left[4C_F\mathcal{L}_1(\boldsymbol{q},Q) - 6C_F\mathcal{L}_0(\boldsymbol{q},Q)\right] = \frac{\alpha_s}{\pi}\frac{1}{q_T^2}\left[C_F\ln\frac{Q^2}{q_T^2} - \frac{3}{2}C_F\right], \tag{118}$$

which gives $A^{(1)}_q = C_F$ and $B^{(1)}_q = -3C_F/2$ in the $\mathrm{d}q_T^2$ spectrum [5, 8]. The complete fixed-Q result remains Eq. (114); its DGLAP terms restore the scale dependence that was deliberately frozen in this coefficient-level check.

12.7 Binned plus-distribution form

Because $\mathcal{L}_n$ and $\delta^{(2)}$ are distributions, the validation should be performed on bin integrals rather than pointwise values at the origin. Define an azimuthally integrated bin functional

$$\mathcal{B}_n(q_a,q_b;Q) \equiv \int_{q_a^2}^{q_b^2}\mathrm{d}q_T^2\,\pi\mathcal{L}_n(\boldsymbol{q},Q), \qquad \mathcal{B}_\delta(q_a,q_b) \equiv \int_{q_a^2}^{q_b^2}\mathrm{d}q_T^2\,\pi\delta^{(2)}(\boldsymbol{q}). \tag{119}$$

For bins that do not touch the origin, $0 < q_a < q_b$, the plus prescription is inactive and

$$\mathcal{B}_n(q_a,q_b;Q) = \frac{1}{n+1}\left[\ln^{n+1}\frac{Q^2}{q_a^2} - \ln^{n+1}\frac{Q^2}{q_b^2}\right], \qquad \mathcal{B}_\delta(q_a,q_b) = 0. \tag{120}$$

For a cumulant bin that starts at the origin and has $q_b < Q$, the plus prescription gives

$$\mathcal{B}_n(0,q_b;Q) = -\frac{1}{n+1}\ln^{n+1}\frac{Q^2}{q_b^2}, \qquad \mathcal{B}_\delta(0,q_b) = 1. \tag{121}$$

Therefore the binned one-loop singular prediction corresponding to Eq. (114) is

$$W^{(0+1),\mathrm{sing}}_{AB,\mathrm{bin}}(q_a,q_b) = \sum_q e_q^2\Bigg\{\Phi^{(0)}_q\mathcal{B}_\delta(q_a,q_b) \tag{122}$$

$$+ a_s(Q)\Big[4C_F\Phi^{(0)}_q\mathcal{B}_1(q_a,q_b;Q) + (\Phi^{b,A}_q + \Phi^{b,B}_q)\mathcal{B}_0(q_a,q_b;Q) \tag{123}$$

$$+ \Big(h^{(1)}_q\Phi^{(0)}_q + \Phi^{c,A}_q + \Phi^{c,B}_q\Big)\mathcal{B}_\delta(q_a,q_b)\Big]\Bigg\} + \mathcal{O}\Big(a_s^2\Big). \tag{124}$$

This is the preferred unit-test expression. A pointwise comparison at $q_T > 0$ should agree with the ordinary-function part of Eq. (114), but it cannot validate the delta-function constants or the plus-prescription signs.

Working decision. For the first implementation, use Eqs. (91), (92), (100), and (124) as the reference one-loop $\mathcal{K}_{\mathrm{CSS2}}$ test set. Any later child scheme must reproduce these results after applying its finite conversion factors.

12.8 Executable one-loop bin validator

This subsection promotes Eq. (124) from a formal expression to an executable regression target. The validator is intentionally narrower than a physical cross-section code. It contains no PDFs, no Y term, no fiducial cuts, no resummation, and no nonperturbative model. Its purpose is to test the distribution algebra that all later implementations must inherit.

Let a bin partition be

$$0 = q_0 < q_1 < \cdots < q_N, \qquad B_{n,r} \equiv \mathcal{B}_n(q_r, q_{r+1}; Q), \qquad B_{\delta,r} \equiv \mathcal{B}_\delta(q_r, q_{r+1}). \tag{125}$$

The first validation layer consists of four algebraic tests.

Plus-null test. For each $n \geq 0$, the plus prescription requires

$$\mathcal{B}_n(0, Q; Q) = 0. \tag{126}$$

This is the most basic sign check for the zero bin. In particular, for $q_b < Q$,

$$\mathcal{B}_0(0, q_b; Q) = -\ln \frac{Q^2}{q_b^2}, \qquad \mathcal{B}_1(0, q_b; Q) = -\frac{1}{2} \ln^2 \frac{Q^2}{q_b^2}, \tag{127}$$

so the singular plus-distribution contribution to the first bin is not positive by itself. This is expected: the zero-bin plus terms, the Born delta term, virtual constants, resummation, the nonsingular remainder, and experimental binning together define the physical prediction.

Additivity test. For any $0 \leq q_a < q_b < q_c$,

$$\mathcal{B}_n(q_a, q_c; Q) = \mathcal{B}_n(q_a, q_b; Q) + \mathcal{B}_n(q_b, q_c; Q), \qquad \mathcal{B}_\delta(q_a, q_c) = \mathcal{B}_\delta(q_a, q_b) + \mathcal{B}_\delta(q_b, q_c). \tag{128}$$

This checks that the zero-bin prescription is compatible with ordinary binning and that a rebinning operation does not change the cumulant.

Off-origin ordinary-function test. For bins with $q_a > 0$, the plus prescription is inactive. The diagonal one-loop kernel must reduce to the ordinary integral

$$a_s \left[4C_F \mathcal{B}_1(q_a, q_b; Q) - 6C_F \mathcal{B}_0(q_a, q_b; Q)\right] = \frac{\alpha_s}{\pi} C_F \left[\mathcal{B}_1(q_a, q_b; Q) - \frac{3}{2} \mathcal{B}_0(q_a, q_b; Q)\right]. \tag{129}$$

This is the bin-integrated version of Eq. (118). It fixes the $a_s = \alpha_s/(4\pi)$ convention, the factor of π from $d^2\boldsymbol{q} = \pi\, dq_T^2$, and the normalization of $\mathcal{L}_n$.

Delta-support test. For a partition that contains the origin in exactly one bin,

$$\sum_{r=0}^{N-1} B_{\delta,r} = 1, \qquad B_{\delta,r} = \begin{cases} 1, & q_r = 0, \\ 0, & q_r > 0. \end{cases} \tag{130}$$

This isolates the hard factor and the delta-supported pieces of the TMD matching coefficients from the ordinary $q_T > 0$ spectrum.

Combining these tests, the canonical diagonal one-loop test weight for a single $q\bar{q}$ luminosity is

$$S_r^{(1),\mathrm{diag}} = a_s(Q) \left[4C_F B_{1,r} - 6C_F B_{0,r} + \left(h_q^{(1)} - \frac{\pi^2}{3} C_F\right) B_{\delta,r}\right]. \tag{131}$$

The term proportional to $h_q^{(1)} - \pi^2 C_F/3$ is present in the canonical $\mu = Q$ diagonal CSS2 check in Eq. (117). The full general flavor-vector test in Eq. (124) is implemented in Sec. 12.9 with analytic toy PDFs.

The accompanying script `kt_kcss_one_loop_validator.py` implements Eqs. (119)–(131). With the sample choice $Q = 8\,\mathrm{GeV}$, $\alpha_s(Q) = 0.25$, and bin edges $(0, 0.25, 0.5, 1, 2, 4, 8)\,\mathrm{GeV}$, the tests give

$$\max_n |\mathcal{B}_n(0, Q; Q)| = 0, \qquad \max |\Delta_{\text{additivity}}| = 0, \qquad \max_{q_a>0} |\Delta_{\text{offzero}}| \simeq 1.1 \times 10^{-16}. \tag{132}$$

These numbers are not phenomenological predictions; they are regression-test tolerances for the distribution implementation.

Working decision. The first executable validation target is now fixed: any pure-k_T implementation of one-loop $\mathcal{K}_{\text{CSS2}}$ must reproduce Eqs. (126), (128), (129), and (130) before adding PDFs, flavor mixing, resummation, or a Y term.

12.9 Flavor-vector toy-PDF x-convolution validator

The next regression layer extends the diagonal bin test to the full flavor-vector expression in Eq. (124). The point is still not phenomenology. The new validator uses analytic toy PDFs only to test the longitudinal convolution algebra and the x-space plus prescription independently of the transverse plus prescription.

For a test PDF $f(x)$ define

$$\phi_f(z; x) \equiv \frac{1}{z} f\left(\frac{x}{z}\right), \qquad \phi_f(1; x) = f(x), \tag{133}$$

and the ordinary regular-kernel convolution functional

$$\mathcal{R}_K[f](x) \equiv \int_x^1 \frac{\mathrm{d}z}{z}\, K(z)\, f\left(\frac{x}{z}\right). \tag{134}$$

The basic endpoint-subtracted plus convolution is

$$\mathcal{P}[f](x) \equiv \int_x^1 \mathrm{d}z\, \frac{\phi_f(z; x) - f(x)}{1 - z} + f(x) \ln(1 - x), \tag{135}$$

which implements

$$\mathcal{P}[f](x) = \left[\frac{1}{1 - z}\right]_+ \otimes_x f. \tag{136}$$

We use the compact convention in which the plus prescription acts on the full endpoint singular function,

$$P_{qq}^{(0)}(z) = C_F \left[\frac{1 + z^2}{1 - z}\right]_+, \tag{137}$$

which is equivalent to Eq. (75). In this convention the validator checks the decomposed identity

$$\left[P_{qq}^{(0)} \otimes_x f\right](x) = C_F \left\{2\mathcal{P}[f](x) - \mathcal{R}_{1+z}[f](x) + \frac{3}{2} f(x)\right\}. \tag{138}$$

Equivalently, a direct implementation of the full plus distribution gives

$$\left[P_{qq}^{(0)} \otimes_x f\right]_{\text{direct}}(x) = C_F \left[\int_x^1 \mathrm{d}z\, \frac{1 + z^2}{1 - z} \left(\phi_f(z; x) - f(x)\right) + f(x) \left(2 \ln(1 - x) + x + \frac{x^2}{2}\right)\right]. \tag{139}$$

Equations (138) and (139) must agree numerically. The same test also verifies the endpoint cancellation in the $\mathcal{L}_0$ coefficient kernel,

$$[b_{q\leftarrow q}\otimes_x f](x) = \left[(2P_{qq}^{(0)} - 3C_F\delta(1-z))\otimes_x f\right](x) = C_F\left\{4\mathcal{P}[f](x) - 2\mathcal{R}_{1+z}[f](x)\right\}. \tag{140}$$

The remaining one-loop kernels in the validation formula are regular or delta-supported:

$$[b_{q\leftarrow g}\otimes_x g](x) = 2T_F\,\mathcal{R}_{z^2+(1-z)^2}[g](x), \tag{141}$$

$$[c_{q\leftarrow q}\otimes_x f](x) = C_F\left[-\frac{\pi^2}{6}f(x) + 2\mathcal{R}_{1-z}[f](x)\right], \tag{142}$$

$$[c_{q\leftarrow g}\otimes_x g](x) = \mathcal{R}_{2z(1-z)}[g](x). \tag{143}$$

The toy PDFs are beta-like analytic functions,

$$f_{i/h}^{\rm toy}(x) = N_{i/h}\,x^{a_{i/h}}(1-x)^{b_{i/h}}(1+c_{i/h}x), \qquad 0<x<1, \tag{144}$$

with independent parameter choices for the two incoming hadrons and for $u, \bar{u}, d, \bar{d}, s, \bar{s}, g$. The exact parameter values are stored in the validator JSON output; they are not intended to approximate a physical PDF set.

The second validation layer has four regression tests:

$$\Delta_P \equiv \max_{h,i,x}\left|\left[P_{qq}^{(0)}\otimes_x f_{i/h}^{\rm toy}\right]_{\rm direct}(x) - \left[P_{qq}^{(0)}\otimes_x f_{i/h}^{\rm toy}\right]_{\rm decomp}(x)\right|, \tag{145}$$

$$\Delta_b \equiv \max_{h,i,x}\left|\left[(2P_{qq}^{(0)} - 3C_F\delta)\otimes_x f_{i/h}^{\rm toy}\right](x) - C_F\{4\mathcal{P}[f_{i/h}^{\rm toy}] - 2\mathcal{R}_{1+z}[f_{i/h}^{\rm toy}]\}(x)\right|, \tag{146}$$

$$\Delta_W \equiv \max_r\left|W_r^{\rm master} - W_r^{\text{explicit leg}}\right|, \tag{147}$$

$$\Delta_{\rm add} \equiv \max\left|W(q_a,q_c) - W(q_a,q_b) - W(q_b,q_c)\right|. \tag{148}$$

Here $W_r^{\rm master}$ uses the five aggregated luminosities entering Eq. (124). By contrast, $W_r^{\text{explicit leg}}$ constructs the two partonic channels $q_A\bar{q}_B$ and $\bar{q}_Aq_B$ from one-leg insertions before summing flavors and charges. This duplicate path deliberately checks the flavor bookkeeping and the charge-conjugation assumptions in Eqs. (109)–(111).

The accompanying script `kt_kcss_toypdf_xconv_validator.py` implements Eqs. (135)–(148). With the sample choice $Q = 8\,\text{GeV}$, $\alpha_s(Q) = 0.25$, $x_A = 0.32$, $x_B = 0.21$, and bin edges $(0, 0.25, 0.5, 1, 2, 4, 8)\,\text{GeV}$, it gives

$$\Delta_P \simeq 8.0\times10^{-14}, \qquad \Delta_b \simeq 1.8\times10^{-15}, \tag{149}$$

$$\Delta_W \simeq 0, \qquad \Delta_{\rm add} \simeq 1.4\times10^{-17}. \tag{150}$$

These values are numerical integration and floating-point regression checks. They do not validate a physical cross section; they validate the algebraic separation of longitudinal plus distributions, transverse plus distributions, flavor-vector luminosities, and one-loop $\mathcal{K}_{\rm CSS2}$ hard/matching constants.

Working decision. The second executable validation target is now fixed: before attaching physical PDFs, resummation, or a Y term, the code must pass the longitudinal plus-distribution tests in Eqs. (145)–(148) with analytic toy PDFs.

13 Generator convolution algebra

The generator identity of Theorem 10.1 provides the algebraic bridge from fixed-order singular distributions to convolution-exponential evolution. The construction is exact on the admissible generator subspace and is used as a formal perturbative algebra outside the region where the all-order multiplier is known to be tempered.

13.1 Algebra in the generator subspace

The generator and derivative basis obey

$$\mathcal{F}_{k\to b}[\mathcal{G}_\eta] = e^{-\eta L_b}, \tag{151}$$

$$\mathfrak{L}_n(\boldsymbol{k};\mu) = (-1)^{n+1}\,\frac{\partial^{n+1}}{\partial\eta^{n+1}}\mathcal{G}_\eta(\boldsymbol{k};\mu)\bigg|_{\eta=0}, \qquad \mathcal{F}_{k\to b}[\mathfrak{L}_n] = L_b^{n+1}. \tag{152}$$

Equation (47) then gives

$$\boxed{\mathfrak{L}_m \otimes_T \mathfrak{L}_n = \mathfrak{L}_{m+n+1}.} \tag{153}$$

The identity element is

$$\delta^{(2)} \otimes_T A = A \otimes_T \delta^{(2)} = A. \tag{154}$$

Both statements are equalities in $\mathcal{S}'(\mathbb{R}^2)$ under the multiplier-domain assumptions of Sec. 5.1. They can also be read as identities of formal perturbative distributions.

Working decision. Use the derivative basis or $\mathcal{G}_\eta$ internally. Use the conventional $\mathcal{L}_n$ basis for published fixed-order coefficients and bin-integrated output.

13.2 Projection onto the conventional Ln basis

For fixed-order comparison, we still need the conventional basis

$$\mathcal{L}_n(\boldsymbol{k},\mu) = \frac{1}{\pi\mu^2}\left[\frac{\mu^2}{k_T^2}\ln^n\left(\frac{\mu^2}{k_T^2}\right)\right]_+. \tag{155}$$

Let

$$P_n(X) \equiv \mathcal{F}_{k\to b}[\mathcal{L}_n], \qquad X \equiv L_b. \tag{156}$$

The generator expansion gives a compact formula for the Fourier dictionary:

$$P_n(X) = (-1)^n n!\,[\eta^{n+1}]\frac{e^{-\eta X}}{h(\eta)}, \qquad h(\eta) = e^{-2\gamma_E\eta}\frac{\Gamma(1-\eta)}{\Gamma(1+\eta)} = \exp\left[2\sum_{r=1}^{\infty}\frac{\zeta_{2r+1}}{2r+1}\eta^{2r+1}\right]. \tag{157}$$

The first entries are

$$P_0(X) = -X, \tag{158}$$

$$P_1(X) = -\frac{1}{2}X^2, \tag{159}$$

$$P_2(X) = -\frac{1}{3}X^3 - \frac{4}{3}\zeta_3, \tag{160}$$

$$P_3(X) = -\frac{1}{4}X^4 - 4\zeta_3 X, \tag{161}$$

$$P_4(X) = -\frac{1}{5}X^5 - 8\zeta_3 X^2 - \frac{48}{5}\zeta_5. \tag{162}$$

Equivalently,

$$\mathfrak{L}_0 = -\mathcal{L}_0, \tag{163}$$
$$\mathfrak{L}_1 = -2\mathcal{L}_1, \tag{164}$$
$$\mathfrak{L}_2 = -3\mathcal{L}_2 - 4\zeta_3\delta^{(2)}, \tag{165}$$
$$\mathfrak{L}_3 = -4\mathcal{L}_3 + 16\zeta_3\mathcal{L}_0, \tag{166}$$
$$\mathfrak{L}_4 = -5\mathcal{L}_4 + 80\zeta_3\mathcal{L}_1 - 48\zeta_5\delta^{(2)}. \tag{167}$$

The appearance of $\zeta_3, \zeta_5, \ldots$ is not new dynamics. These constants are a basis-conversion effect caused by packaging the Fourier-transform constants into $h(\eta)$.

13.3 Conventional-basis convolution table

The conventional table is obtained by multiplying $P_m(X)P_n(X)$ and projecting the result back onto $\{\delta^{(2)}, \mathcal{L}_0, \mathcal{L}_1, \ldots\}$. The first entries are

$$\mathcal{L}_0 \otimes_T \mathcal{L}_0 = -2\mathcal{L}_1, \tag{168}$$
$$\mathcal{L}_0 \otimes_T \mathcal{L}_1 = -\frac{3}{2}\mathcal{L}_2 - 2\zeta_3\delta^{(2)}, \tag{169}$$
$$\mathcal{L}_0 \otimes_T \mathcal{L}_2 = -\frac{4}{3}\mathcal{L}_3 + 4\zeta_3\mathcal{L}_0, \tag{170}$$
$$\mathcal{L}_1 \otimes_T \mathcal{L}_1 = -\mathcal{L}_3 + 4\zeta_3\mathcal{L}_0, \tag{171}$$
$$\mathcal{L}_0 \otimes_T \mathcal{L}_3 = -\frac{5}{4}\mathcal{L}_4 + 12\zeta_3\mathcal{L}_1 - 12\zeta_5\delta^{(2)}, \tag{172}$$
$$\mathcal{L}_1 \otimes_T \mathcal{L}_2 = -\frac{5}{6}\mathcal{L}_4 + 12\zeta_3\mathcal{L}_1 - 8\zeta_5\delta^{(2)}. \tag{173}$$

The table is symmetric under $m \leftrightarrow n$. Higher entries should be generated by Eq. (157), not typed by hand.

13.4 First convolution-exponential check

The one-loop quark Collins-Soper kernel in Eq. (100) is proportional to $\mathcal{L}_0$. Therefore the first nontrivial rapidity-evolution Green function is already controlled by the generator algebra. For any scalar λ,

$$\mathrm{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} = \delta^{(2)} + \lambda\mathcal{L}_0 + \frac{\lambda^2}{2}\mathcal{L}_0 \otimes_T \mathcal{L}_0 + \frac{\lambda^3}{6}\mathcal{L}_0 \otimes_T \mathcal{L}_0 \otimes_T \mathcal{L}_0 + \cdots . \tag{174}$$

Since $\mathcal{F}[\mathcal{L}_0] = -L_b$, the equality

$$\boxed{\mathrm{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} = \mathcal{G}_\lambda} \tag{175}$$

holds as an analytic-distribution identity where $e^{-\lambda L_b}$ is admissible and, more generally, as a formal Taylor identity about $\lambda = 0$. Expanding in the conventional basis gives

$$\begin{aligned}\mathrm{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} = \delta^{(2)} + \lambda\mathcal{L}_0 - \lambda^2\mathcal{L}_1 + \lambda^3\left(\frac{1}{2}\mathcal{L}_2 + \frac{2}{3}\zeta_3\delta^{(2)}\right) \\ + \lambda^4\left(-\frac{1}{6}\mathcal{L}_3 + \frac{2}{3}\zeta_3\mathcal{L}_0\right) + \mathcal{O}\left(\lambda^5\right).\end{aligned} \tag{176}$$

This check is useful because it verifies the signs in Eqs. (168)–(169). For example,

$$\mathcal{L}_0 \otimes_T \mathcal{L}_0 \otimes_T \mathcal{L}_0 = 3\mathcal{L}_2 + 4\zeta_3\delta^{(2)}, \tag{177}$$

which is the λ^3 coefficient of Eq. (176) multiplied by 3!.

13.5 Executable generator-algebra validator

The accompanying script `kt_kcss_generator_convolution_validator.py` implements Eq. (157), generates the table through $\mathcal{L}_3 \otimes_T \mathcal{L}_3$, and checks three identities:

1. the exact generator-basis rule $\mathfrak{L}_m \otimes_T \mathfrak{L}_n = \mathfrak{L}_{m+n+1}$;
2. the conventional-basis projection by verifying $P_m(X)P_n(X) - \mathcal{F}[\mathcal{L}_m \otimes_T \mathcal{L}_n] = 0$;
3. the one-loop exponential identity $\mathrm{Exp}_{\otimes_T}\{\lambda\mathcal{L}_0\} = \mathcal{G}_\lambda$ through the tested order.

The regression report gives

$$\text{passed} = \text{True}, \qquad N_{\text{frak failures}} = 0, \qquad \Delta_{\text{projection}} = 0, \qquad \Delta_{\exp} = 0. \tag{178}$$

This is the first validation step that directly targets the convolution-exponential structure needed for k_T-space Sudakov evolution.

Working decision. The generator algebra has passed the stated fixed-order symbolic checks. It supplies the engine used in Sec. 14 to build the first one-loop evolution Green function; the next extension is the running-coupling and two-loop version of $\mathcal{D}_i$ and $\mathcal{C}_{i\leftarrow j}$.

13.6 Canonical momentum boundary constants through three loops

A boundary constant at $L_b = 0$ is not, beyond one loop, the same as the canonical momentum cumulant at $K = \mu$. This distinction is already present in the distribution-space construction of Ref. [14], Sec. 4.3 and Appendix C. It is indispensable for the primed accuracy convention.

Proposition 13.1 (Finite canonical boundary map)**.** *Define* $J_0 = 1$ *and*

$$J_m = \int_{|\boldsymbol{k}|\le\mu} \mathrm{d}^2\boldsymbol{k}\, \mathfrak{L}_{m-1}(\boldsymbol{k};\mu), \quad m \ge 1. \tag{179}$$

For the generator normalization used in this note,

$$J_m = (-1)^m \left.\frac{\mathrm{d}^m h(\eta)}{\mathrm{d}\eta^m}\right|_{\eta=0}, \qquad (J_0,\dots,J_6) = (1,0,0,-4\zeta_3,0,-48\zeta_5,160\zeta_3^2). \tag{180}$$

If a fixed-scale matching coefficient is $C^{(n)}(z,L_b,0) = \sum_m c^{(n,m)}(z;0)L_b^m$*, its canonical momentum boundary is*

$$c_k^{(n)}(z) = \sum_{m=0}^{2n} J_m c^{(n,m)}(z;0). \tag{181}$$

In particular,

$$\begin{aligned} c_k^{(2)} &= c^{(2,0)} - 4\zeta_3 c^{(2,3)}, \\ c_k^{(3)} &= c^{(3,0)} - 4\zeta_3 c^{(3,3)} - 48\zeta_5 c^{(3,5)} + 160\zeta_3^2 c^{(3,6)}. \end{aligned} \tag{182}$$

All terms remain distributions in z*; products are not arbitrary numerical coefficients. The rapidity log is zero only for this particular canonical choice.*

Proof. The disk integral of every $\mathcal{L}_n(\boldsymbol{k},\mu)$ vanishes at radius μ. Consequently Eq. (50), extended to the needed order, gives $\int_{|\boldsymbol{k}|\le\mu}\mathcal{G}_\eta = h(\eta)$. Differentiate and use Eq. (49). Linearity then proves the coefficient map. Equivalently the polynomial dictionary gives

$$\mathfrak{L}_5 = -6\mathcal{L}_5 + 240\zeta_3\mathcal{L}_2 + 288\zeta_5\mathcal{L}_0 + 160\zeta_3^2\delta^{(2)}, \tag{183}$$

which independently checks J_6 against the tabulated inverse transform in Ref. [14] after reversing its logarithm convention. □

Applied to the three-loop CS polynomial this yields

$$d^{(3)}_{k,i} = d^{(3)}_i - \frac{4}{3}\zeta_3\beta_0^2\Gamma^i_0. \tag{184}$$

The quantity denoted D^δ_i in a *derivative-basis* decomposition is therefore not the total coefficient of $\delta^{(2)}$ after conversion to the conventional plus basis. These finite shifts do not change the CSS2 operator scheme. They translate between two boundary conventions inside its representations. They must be included before using a momentum-canonical recurrence; merely setting $L_b = 0$ in an imported coefficient does not perform that translation. If the coupling or PDFs subsequently run with the cumulative scale, their derivatives remain part of the weak profile action.

14 One-loop evolution Green function

This section turns the generator algebra of Sec. 13 into the first explicit k_T-space evolution Green function. The purpose here is not yet to claim a phenomenological NLL implementation. The purpose is to validate the exact one-loop, fixed-coupling convolution dynamics that any higher-logarithmic implementation must reduce to. Running-coupling evolution, profile scales, and two-loop kernels are the next layer.

14.1 One-loop anomalous dimensions and variables

At one loop we write

$$\mathcal{D}_i^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = a_s\Gamma^i_0\mathfrak{L}_0(\boldsymbol{k};\mu) + \mathcal{O}\Big(a_s^2\Big) = -a_s\Gamma^i_0\mathcal{L}_0(\boldsymbol{k},\mu) + \mathcal{O}\Big(a_s^2\Big), \tag{185}$$

and

$$\gamma^i_F(\mu,\zeta) = a_s\left[\gamma^i_{F,0} - \Gamma^i_0\ln\Big(\frac{\zeta}{\mu^2}\Big)\right] + \mathcal{O}\Big(a_s^2\Big). \tag{186}$$

For the quark TMD in the CSS2 convention used above,

$$\Gamma^q_0 = 4C_F, \qquad \gamma^q_{F,0} = 6C_F, \tag{187}$$

where the second equality follows from the coefficient-evolution check in Eq. (87). The gluon channel is obtained by replacing these constants by the corresponding CSS2 gluon anomalous-dimension coefficients.

For a path from (μ_i,ζ_i) to (μ_f,ζ_f) define

$$L_\mu \equiv \ln\frac{\mu_f}{\mu_i}, \qquad \rho \equiv \ln\sqrt{\frac{\zeta_f}{\zeta_i}}, \qquad \ell_i \equiv \ln\frac{\zeta_i}{\mu_i^2}, \qquad \ell_f \equiv \ln\frac{\zeta_f}{\mu_f^2}. \tag{188}$$

These variables obey

$$\ell_f = \ell_i + 2\rho - 2L_\mu. \tag{189}$$

We hold a_s fixed in this subsection. This is an algebraic one-loop Green function; NLL evolution will replace the fixed a_s factors by RG integrals.

14.2 Rapidity evolution at fixed mu

At fixed μ, Eq. (36) gives

$$F_i(x,\boldsymbol{k};\mu,\zeta_f) = \Big[\mathcal{U}_i^{\zeta}(\mu;\zeta_f,\zeta_i)\otimes_T F_i(x;\mu,\zeta_i)\Big](\boldsymbol{k}), \tag{190}$$

with

$$\mathcal{U}_i^{\zeta}(\boldsymbol{k};\mu;\zeta_f,\zeta_i) = \mathrm{Exp}_{\otimes_T}\{-\rho\,\mathcal{D}_i(\mu)\} = \mathrm{Exp}_{\otimes_T}\Big\{a_s\Gamma_0^i\rho\,\mathcal{L}_0(\mu)\Big\} + \mathcal{O}\Big(a_s^2 \text{ in the kernel}\Big). \tag{191}$$

Using Eq. (175), the fixed-μ rapidity Green function is therefore

$$\boxed{\mathcal{U}_i^{\zeta}(\boldsymbol{k};\mu;\zeta_f,\zeta_i) = \mathcal{G}_{\eta_i}(\boldsymbol{k};\mu), \qquad \eta_i = a_s\Gamma_0^i\rho.} \tag{192}$$

In b_T space this is simply

$$\widetilde{\mathcal{U}}_i^{\zeta}(b;\mu;\zeta_f,\zeta_i) = \exp\Big[-a_s\Gamma_0^i\rho\,L_b(\mu)\Big], \tag{193}$$

which solves

$$\frac{\partial}{\partial\rho}\widetilde{\mathcal{U}}_i^{\zeta} = -a_s\Gamma_0^i L_b(\mu)\,\widetilde{\mathcal{U}}_i^{\zeta}. \tag{194}$$

14.3 Adding the mu leg

Choose the path that first evolves in μ at fixed ζ_i, then evolves in ζ at fixed μ_f. The μ-leg is local in transverse momentum, so it contributes a scalar factor. Integrating Eq. (186) gives

$$\begin{aligned} R_i^{(1)}(\mu_f,\mu_i;\zeta_i) &\equiv \int_{\ln\mu_i}^{\ln\mu_f} \mathrm{d}\ln\mu\, a_s\left[\gamma_{F,0}^i - \Gamma_0^i\ln\Big(\frac{\zeta_i}{\mu^2}\Big)\right] \\ &= a_s L_\mu\left[\gamma_{F,0}^i - \Gamma_0^i(\ell_i - L_\mu)\right]. \end{aligned} \tag{195}$$

The resulting one-loop fixed-coupling Green function, represented with the final scale μ_f in the generator, is

$$\boxed{\mathcal{U}_i^{(1\ell)}(\boldsymbol{k};\mu_f,\zeta_f;\mu_i,\zeta_i) = \exp\Big\{R_i^{(1)}(\mu_f,\mu_i;\zeta_i)\Big\}\,\mathcal{G}_{\eta_i}(\boldsymbol{k};\mu_f), \qquad \eta_i = a_s\Gamma_0^i\rho.} \tag{196}$$

Equivalently, the b_T-space image is

$$\widetilde{\mathcal{U}}_i^{(1\ell)} = \exp\Big\{a_s L_\mu\left[\gamma_{F,0}^i - \Gamma_0^i(\ell_i - L_\mu)\right] - a_s\Gamma_0^i\rho\,L_b(\mu_f)\Big\}. \tag{197}$$

The corresponding fixed-order expansion in the conventional $\mathcal{L}_n$ basis is

$$\mathcal{U}_i^{(1\ell)} = \delta^{(2)} + a_s L_\mu\left[\gamma_{F,0}^i - \Gamma_0^i(\ell_i - L_\mu)\right]\delta^{(2)} + a_s\Gamma_0^i\rho\,\mathcal{L}_0(\boldsymbol{k},\mu_f) + \mathcal{O}\Big(a_s^2\Big). \tag{198}$$

Equation (198) is the object that should appear when the resummed kernel is expanded to first order.

14.4 Path-independence check

The same endpoint evolution can be obtained by first evolving in ζ at μ_i and then evolving in μ at fixed ζ_f. That path gives

$$\mathcal{U}^{(1\ell)}_{i,\zeta\text{-first}} = \exp\Big\{a_s L_\mu \Big[\gamma^i_{F,0} - \Gamma^i_0(\ell_f + L_\mu)\Big]\Big\}\, \mathcal{G}_{\eta_i}(\boldsymbol{k};\mu_i). \tag{199}$$

The two forms are identical. In b_T space the generator satisfies

$$L_b(\mu_f) = L_b(\mu_i) + 2L_\mu, \qquad \mathcal{G}_\eta(\mu_i) = e^{2\eta L_\mu}\mathcal{G}_\eta(\mu_f), \tag{200}$$

while Eq. (189) relates ℓ_i and ℓ_f. These two relations give

$$\mathcal{U}^{(1\ell)}_{i,\mu\text{-first}} = \mathcal{U}^{(1\ell)}_{i,\zeta\text{-first}}. \tag{201}$$

This is the one-loop fixed-coupling version of path independence in the (μ,ζ) plane. It is the momentum-space analogue of the usual Collins-Soper/RG consistency condition.

14.5 Differential-equation and semigroup checks

The Green function in Eq. (196) satisfies the endpoint differential equations. Holding the initial point fixed,

$$\frac{\partial}{\partial\rho}\mathcal{U}^{(1\ell)}_i = -\mathcal{D}_i(\mu_f)\otimes_T \mathcal{U}^{(1\ell)}_i + \mathcal{O}\Big(a_s^2 \text{ in the kernel}\Big), \tag{202}$$

and

$$\frac{\partial}{\partial \ln\mu_f}\mathcal{U}^{(1\ell)}_i = \gamma^i_F(\mu_f,\zeta_f)\,\mathcal{U}^{(1\ell)}_i + \mathcal{O}\Big(a_s^2 \text{ in the kernel}\Big). \tag{203}$$

The second check uses both the explicit scalar derivative and the scale derivative of $\mathcal{G}_\eta(\boldsymbol{k};\mu_f)$.

The Green function also obeys the semigroup property. For three points (μ_0,ζ_0), (μ_1,ζ_1), and (μ_2,ζ_2),

$$\mathcal{U}_i(2;0) = \mathcal{U}_i(2;1)\otimes_T \mathcal{U}_i(1;0), \tag{204}$$

where the product is a transverse convolution in k_T space and ordinary multiplication in b_T space. This check will become nontrivial numerically once the kernel is used on a finite k_T grid.

14.6 Executable evolution-Green validator

The accompanying script `kt_kcss_one_loop_evolution_validator.py` validates Eqs. (196)–(204) symbolically in b_T space. It checks

1. equality of the μ-first and ζ-first paths;
2. the rapidity differential equation;
3. the μ differential equation at the final endpoint;
4. the semigroup law for three arbitrary points;
5. the $\mathcal{O}(a_s)$ conventional-basis expansion in Eq. (198).

The regression report gives

$$\text{passed} = \text{True}, \qquad \Delta_{\text{path}} = 0, \qquad \Delta_{\rho} = 0, \qquad \Delta_{\mu} = 0, \qquad \Delta_{\text{semi}} = 0. \tag{205}$$

For a quark-like smoke point with $\Gamma_0^q = 16/3$, $\gamma_{F,0}^q = 8$, and $a_s = 0.02$, the script finds

$$\eta_q \simeq 2.13 \times 10^{-2}, \qquad R_q^{(1)} \simeq 3.52 \times 10^{-2}, \tag{206}$$

which is only a numerical smoke check, not a phenomenological prediction.

Working decision. The first evolution Green function is now specified and regression-tested at one loop, fixed-coupling level. The next formal layer is to replace the fixed a_s factors by running-coupling RG integrals and then include the two-loop Collins-Soper kernel and matching coefficients needed for the first NLL/NNLL W-term prototype.

15 Two-loop Collins-Soper kernel and NNLO matching interface

This section adds the first genuinely higher-order perturbative ingredient. The Collins-Soper kernel is compact enough to write explicitly in the $\mathcal{K}_{\text{CSS2}}$ basis. The full two-loop TMD matching coefficients are much longer; for them we define the import interface, scheme checks, and k_T projection map rather than duplicating pages of known coefficient functions. This is the right split for the code: the CS kernel is an internal analytic kernel, while NNLO matching tables should be imported from a vetted source and then transformed into our distribution basis.

15.1 Conventions and anomalous dimensions

We keep the expansion parameter

$$a_s(\mu) \equiv \frac{\alpha_s(\mu)}{4\pi}, \qquad \frac{\mathrm{d}a_s}{\mathrm{d}\ln\mu} = -2\beta_0 a_s^2 + \mathcal{O}\left(a_s^3\right), \qquad \beta_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f. \tag{207}$$

For a parton in representation i, with $C_i = C_F$ for quarks and $C_i = C_A$ for gluons,

$$\Gamma_0^i = 4C_i, \tag{208}$$

$$\Gamma_1^i = 4C_i \left[\left(\frac{67}{9} - \frac{\pi^2}{3}\right) C_A - \frac{20}{9} T_F n_f\right]. \tag{209}$$

The two-loop boundary coefficient of the CSS2 Collins-Soper kernel, in the present normalization $D_i = -\widetilde{K}_i$, is

$$d_i^{(2)} = C_i \left[C_A \left(\frac{808}{27} - 28\zeta_3\right) - \frac{224}{27} T_F n_f\right]. \tag{210}$$

For quarks this follows directly from the two-loop $\widetilde{K}$ result compiled by Collins and Rogers. Their Eq. (69) gives $\widetilde{K}_q$ through a_s^2 with $\ln(b_T\mu/(2e^{-\gamma_E})) = L_b/2$; after setting $D_q = -\widetilde{K}_q$ one obtains Eqs. (209)–(210). The same constant is equivalently the negative of the two-loop rapidity anomalous-dimension boundary term in the Li-Zhu normalization, after translating between $\widetilde{K}$ and D conventions [39, 18].

15.2 Two-loop kernel in bT space

Solving the RG equation

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}D_i(b_T;\mu) = 2\Gamma^i_{\rm cusp}(\alpha_s) = 2\left[a_s\Gamma^i_0 + a_s^2\Gamma^i_1 + \mathcal{O}\Big(a_s^3\Big)\right] \tag{211}$$

with $L_b = \ln(\mu^2 b_T^2 e^{2\gamma_E}/4)$ gives

$$\boxed{D_i^{\rm CSS2}(b_T;\mu) = a_s\Gamma^i_0 L_b + a_s^2\left[\frac{1}{2}\beta_0\Gamma^i_0 L_b^2 + \Gamma^i_1 L_b + d_i^{(2)}\right] + \mathcal{O}\Big(a_s^3\Big)\,.} \tag{212}$$

The coefficient of L_b^2 is not independent. It is fixed by the running of a_s in the one-loop term. The coefficient of L_b is fixed by the two-loop cusp anomalous dimension. The only new constant at this order is $d_i^{(2)}$.

The RG check is useful enough to record explicitly. Using $\mathrm{d}L_b/\mathrm{d}\ln\mu = 2$ and Eq. (207),

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}D_i^{\rm CSS2}(b_T;\mu) = 2a_s\Gamma^i_0 + a_s^2\left[-2\beta_0\Gamma^i_0 L_b + 2\beta_0\Gamma^i_0 L_b + 2\Gamma^i_1\right] + \mathcal{O}\Big(a_s^3\Big) \tag{213}$$

$$= 2a_s\Gamma^i_0 + 2a_s^2\Gamma^i_1 + \mathcal{O}\Big(a_s^3\Big)\,. \tag{214}$$

15.3 Two-loop kernel in the generator and conventional kT bases

The $\mathcal{K}_{\rm CSS2}$ momentum-space kernel is obtained by the same Fourier map used at one loop:

$$L_b^m \quad\longmapsto\quad \begin{cases}\delta^{(2)}(\boldsymbol{k}), & m=0,\\ \mathfrak{L}_{m-1}(\boldsymbol{k};\mu), & m\geq 1.\end{cases} \tag{215}$$

Therefore

$$\boxed{\mathcal{D}_i^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = a_s\Gamma^i_0\mathfrak{L}_0 + a_s^2\left[\frac{1}{2}\beta_0\Gamma^i_0\mathfrak{L}_1 + \Gamma^i_1\mathfrak{L}_0 + d_i^{(2)}\delta^{(2)}\right] + \mathcal{O}\Big(a_s^3\Big)\,.} \tag{216}$$

Projecting onto the conventional plus basis with $\mathfrak{L}_0 = -\mathcal{L}_0$ and $\mathfrak{L}_1 = -2\mathcal{L}_1$ gives

$$\boxed{\mathcal{D}_i^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = -a_s\Gamma^i_0\mathcal{L}_0 + a_s^2\left[-\beta_0\Gamma^i_0\mathcal{L}_1 - \Gamma^i_1\mathcal{L}_0 + d_i^{(2)}\delta^{(2)}\right] + \mathcal{O}\Big(a_s^3\Big)\,.} \tag{217}$$

This is the first nontrivial test of the generator basis. The $\mathfrak{L}$ form keeps the RG structure transparent, while the $\mathcal{L}_n$ form is the one to use for binned singular spectra.

In the $\mathfrak{L}$ basis, the scale derivative is

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\mathfrak{L}_n(\boldsymbol{k};\mu) = 2(n+1)\mathfrak{L}_{n-1}(\boldsymbol{k};\mu), \qquad \mathfrak{L}_{-1}\equiv\delta^{(2)}, \tag{218}$$

so Eq. (216) obeys the momentum-space RG equation

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\mathcal{D}_i^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = 2\left[a_s\Gamma^i_0 + a_s^2\Gamma^i_1\right]\delta^{(2)}(\boldsymbol{k}) + \mathcal{O}\Big(a_s^3\Big)\,. \tag{219}$$

15.4 NNLO matching coefficients: import architecture

The NNLO matching coefficients are needed for NNLL′-type accuracy and for a stable large-k_T perturbative tail. They are known in the literature; Gehrmann, Lübbert, and Yang give the two-loop transverse-PDF calculation and its scale-logarithm recursion, Echevarria, Scimemi, and Vladimirov provide NNLO unpolarized TMDPDF/TMDFF matching coefficients for all flavor combinations, and Luo, Yang, Zhu, and Zhu provide unpolarized quark and gluon TMDPDF and TMDFF matching coefficients through N^3LO, including ancillary machine-readable files [16, 17, 22]. We should import those coefficients rather than hand-transcribing them.

For the $\mathcal{K}_{\rm CSS2}$ engine, the required interface is the polynomial representation

$$\widetilde{C}_{i\leftarrow j}^{(2),\mathrm{CSS2}}(z,b_T;\mu,\zeta)=\sum_{m=0}^{4}\widetilde{c}_{i\leftarrow j}^{(2,m)}(z;\ell_\zeta)L_b^m,\qquad \ell_\zeta=\ln\frac{\zeta}{\mu^2}. \tag{220}$$

The upper power $m=4$ is allowed because the one-loop coefficient already contains L_b^2 terms. The momentum-space projection is then purely mechanical:

$$\boxed{\mathcal{C}_{i\leftarrow j}^{(2),\mathcal{K}_{\rm CSS2}}(z,\boldsymbol{k};\mu,\zeta)=\widetilde{c}_{i\leftarrow j}^{(2,0)}(z;\ell_\zeta)\delta^{(2)}+\sum_{m=1}^{4}\widetilde{c}_{i\leftarrow j}^{(2,m)}(z;\ell_\zeta)\mathfrak{L}_{m-1}(\boldsymbol{k};\mu).} \tag{221}$$

For output in the conventional basis, use Eq. (167); explicitly,

$$\begin{aligned}\mathcal{C}_{i\leftarrow j}^{(2),\mathcal{K}_{\rm CSS2}}=&\left[\widetilde{c}_{ij}^{(2,0)}-4\zeta_3\widetilde{c}_{ij}^{(2,3)}\right]\delta^{(2)}-\left[\widetilde{c}_{ij}^{(2,1)}-16\zeta_3\widetilde{c}_{ij}^{(2,4)}\right]\mathcal{L}_0-2\widetilde{c}_{ij}^{(2,2)}\mathcal{L}_1\\&-3\widetilde{c}_{ij}^{(2,3)}\mathcal{L}_2-4\widetilde{c}_{ij}^{(2,4)}\mathcal{L}_3,\end{aligned} \tag{222}$$

where the common arguments $(z;\ell_\zeta)$ and $(\boldsymbol{k};\mu)$ have been suppressed. This formula is an important implementation checkpoint: all zeta constants in Eq. (222) are basis-conversion constants, not new hard physics.

15.5 Logarithmic reconstruction of the NNLO matching coefficient

The finite functions $\widetilde{c}_{i\leftarrow j}^{(2,0)}(z)$ must be imported from a vetted CSS2-compatible source. However, the L_b and ℓ_ζ dependence is not an independent input. It is fixed by the Collins-Soper equation, the μ-RG equation, the DGLAP kernels, and the two-loop kernel above. This gives a strong convention check on any imported NNLO table.

For a compact scalar-channel projection, suppress the flavor matrices and x-convolutions and write

$$\widetilde{C}=1+a_sC_1+a_s^2C_2+\mathcal{O}\left(a_s^3\right),\qquad P_\mu=4a_sP_0+8a_s^2P_1+\mathcal{O}\left(a_s^3\right), \tag{223}$$

with

$$D_1(L_b)=\Gamma_0L_b, \tag{224}$$

$$D_2(L_b)=\frac{1}{2}\beta_0\Gamma_0L_b^2+\Gamma_1L_b+d^{(2)}, \tag{225}$$

$$\gamma_F(a_s,\ell_\zeta)=a_s(\gamma_0-\Gamma_0\ell_\zeta)+a_s^2(\gamma_1-\Gamma_1\ell_\zeta)+\mathcal{O}\left(a_s^3\right). \tag{226}$$

The one-loop coefficient may be written as

$$C_1(L_b,\ell_\zeta)=C_1^{[0]}(L_b)-\frac{1}{2}\Gamma_0\ell_\zeta L_b,\qquad C_1^{[0]}(L_b)=c_1+\left(\frac{\gamma_0}{2}-2P_0\right)L_b-\frac{1}{4}\Gamma_0L_b^2. \tag{227}$$

Then the NNLO logarithmic terms are reconstructed from the finite boundary $c_2 \equiv C_2(0,0)$ by

$$C_2^{[0]}(L_b) = c_2 + \frac{1}{2}\int_0^{L_b} \mathrm{d}L \left[\gamma_1 - D_2(L) - 8P_1 + (\gamma_0 + 2\beta_0)C_1^{[0]}(L) - D_1(L)C_1^{[0]}(L) - 4C_1^{[0]}(L)P_0\right], \tag{228}$$

$$C_2(L_b, \ell_\zeta) = C_2^{[0]}(L_b) - \frac{\ell_\zeta}{2}\left[D_2(L_b) + D_1(L_b)C_1^{[0]}(L_b)\right] + \frac{\Gamma_0^2}{8}\ell_\zeta^2 L_b^2. \tag{229}$$

Equations (228) and (229) are not intended to replace the full NNLO coefficient tables. Their role is to reconstruct and validate all logarithmic terms once the finite boundary functions are imported.

The full flavor-vector form is obtained by restoring matrix multiplication under x-convolution. Thus P_0 and P_1 are DGLAP matrices, products such as $C_1^{[0]}P_0$ become $C_1^{[0]} \otimes_x P_0$, and the quantities $D_n^i, \Gamma_n^i, \gamma_n^i$ multiply the outgoing TMD row i. The ordering in x-space should not be changed when translating the scalar check into code.

The two evolution equations checked by this reconstruction are

$$2\frac{\partial}{\partial \ell_\zeta}\widetilde{C}_{i\leftarrow j} = -D_i\,\widetilde{C}_{i\leftarrow j}, \tag{230}$$

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\widetilde{C}_{i\leftarrow j} = \gamma_F^i\,\widetilde{C}_{i\leftarrow j} - \sum_k \widetilde{C}_{i\leftarrow k} \otimes_x (P_\mu)_{k\leftarrow j}. \tag{231}$$

Here $P_\mu = 4a_sP^{(0)} + 8a_s^2P^{(1)} + \cdots$ is the DGLAP kernel for evolution with respect to $\ln\mu$. The factors of four and eight follow from the standard $\mathrm{d}f/\mathrm{d}\ln\mu^2 = (\alpha_s/2\pi)P^{(0)} \otimes f + (\alpha_s/2\pi)^2P^{(1)} \otimes f + \cdots$ convention together with $a_s = \alpha_s/(4\pi)$.

15.6 Ordered flavor-matrix reconstruction through arbitrary fixed order

The scalar check above extends constructively to the full flavor matrices. This extension fixes logarithms but cannot determine the finite physical matching functions. Use

$$\begin{aligned} C &= \sum_{n\geq 0} a_s^n C_n(L,\ell), & C_0 &= \mathbf{1}, \\ D &= \sum_{r\geq 1} a_s^r D_r(L), & P_\mu &= \sum_{r\geq 1} a_s^r p_r, \qquad p_r = 2^{r+1}P^{(r-1)}, \end{aligned} \tag{232}$$

where $L = L_b$, $\ell = \ell_\zeta$, and $D_r, \gamma_{r-1}, \Gamma_{r-1}$ are diagonal operators on the outgoing TMD index. The unit includes $\delta_{ij}\delta(1-z)$. Matrix products below include $\otimes_x$; this convolution is suppressed only for diagonal anomalous dimensions.

Proposition 15.1 (Ordered logarithmic recurrence)**.** *Let $\bar{C}_n(L) = C_n(L,0)$ and supply finite boundaries $\bar{C}_n(0) = c_n(z)$ and $D_n(0) = d_n$. The two scale equations determine all logarithms by*

$$D_n'(L) = \Gamma_{n-1} + \sum_{r=1}^{n-1}(n-r)\beta_{r-1}D_{n-r}(L), \tag{233}$$

$$\begin{aligned} 2\bar{C}_n'(L) = &\sum_{r=1}^{n}(\gamma_{r-1} - D_r)\bar{C}_{n-r} - \sum_{r=1}^{n}\bar{C}_{n-r} \otimes_x p_r \\ &+ 2\sum_{r=1}^{n-1}(n-r)\beta_{r-1}\bar{C}_{n-r}, \end{aligned} \tag{234}$$

$$C(a_s, L, \ell) = \exp[-\ell D(a_s, L)/2]\,\bar{C}(a_s, L). \tag{235}$$

These equations are formal power-series identities. The exponential in the last equation is an ordinary diagonal matrix exponential at fixed L, and is expanded to the requested order; it is not an unqualified resummed momentum kernel.

Proof. At fixed b_T and ζ, the scale derivative is $2\partial_L - 2\partial_\ell + \beta(a_s)\partial_{a_s}$. Use $2\partial_\ell C = -DC$ in $\mathrm{d}C/\mathrm{d}\ln\mu = \gamma_F C - C \otimes_x P_\mu$ and set $\ell = 0$. Comparing coefficients gives Eq. (234); the running-coupling term has the displayed positive sign after it is moved to the right-hand side. The same comparison in $\mathrm{d}D/\mathrm{d}\ln\mu = 2\Gamma$ gives Eq. (233). Integration in L from zero determines each coefficient uniquely from lower orders and its finite boundary. Finally integrating the CS equation in ℓ gives Eq. (235). This also proves uniqueness of the *logarithmic reconstruction*, not uniqueness of a TMD extracted from data. □

At three loops the explicit zero-rapidity recurrence reads

$$\begin{aligned} 2\bar{C}_3' = {} & (\gamma_2 - D_3)\mathbf{1} - p_3 + (\gamma_1 - D_2)\bar{C}_1 - \bar{C}_1 \otimes_x p_2 \\ & + (\gamma_0 - D_1)\bar{C}_2 - \bar{C}_2 \otimes_x p_1 + 2\beta_1\bar{C}_1 + 4\beta_0\bar{C}_2. \end{aligned} \tag{236}$$

The finite $c_3(z)$ is independent input. Replacing $\bar{C}_j \otimes_x p_r$ by $p_r \otimes_x \bar{C}_j$ is not allowed for a flavor matrix. The new audit uses two noncommuting 2×2 rational Mellin-moment matrices to verify both RG equations, finite boundaries, and the degree bound $\deg_L C_n \le 2n$ through $n = 3$. These matrices are explicitly labeled nonphysical fixtures; they do not stand in for the published NNLO or N^{3}LO matching functions.

For completeness, if the evolved perturbative legs are $A = \sum a_s^n A_n$, $B = \sum a_s^n B_n$ and $H = \sum a_s^n H_n$, the complete fixed-order process assembly is

$$W_n = \sum_{r+s+t=n} H_r\,[A_s \otimes_T B_t]. \tag{237}$$

Thus

$$\begin{aligned} W_2 = {} & H_2[A_0 \otimes_T B_0] + H_1[A_1 \otimes_T B_0 + A_0 \otimes_T B_1] \\ & + H_0[A_2 \otimes_T B_0 + A_1 \otimes_T B_1 + A_0 \otimes_T B_2], \\ W_3 = {} & H_3[A_0 \otimes_T B_0] + H_2[A_1 \otimes_T B_0 + A_0 \otimes_T B_1] \\ & + H_1[A_2 \otimes_T B_0 + A_1 \otimes_T B_1 + A_0 \otimes_T B_2] \\ & + H_0[A_3 \otimes_T B_0 + A_2 \otimes_T B_1 + A_1 \otimes_T B_2 + A_0 \otimes_T B_3]. \end{aligned} \tag{238}$$

Longitudinal convolutions, flavor sums, and the measurement map remain implicit here. The formulas include all product partitions; the available finite coefficient payload, not these identities, decides whether a physical NNLO/N^{3}LO prediction has been made.

15.7 Flavor channels required at two loops

The import layer must be flavor-vector valued. For a quark TMD, the NNLO channels are schematically

$$q \leftarrow q, \qquad q \leftarrow \bar{q}, \qquad q \leftarrow q', \qquad q \leftarrow \bar{q}', \qquad q \leftarrow g, \tag{239}$$

with $q' \neq q$. For a gluon TMD, the corresponding channels are

$$g \leftarrow g, \qquad g \leftarrow q, \qquad g \leftarrow \bar{q}. \tag{240}$$

Drell-Yan fixed-target phenomenology will mostly stress the quark channels, but the formalism and code must carry the gluon entries from the start so that the same $\mathcal{K}_{\mathrm{CSS2}}$ objects can be used for collider Drell-Yan, Higgs-like color-singlet production, SIDIS, and eventually e^+e^- fragmentation observables.

15.8 Validation gates before using NNLO matching in a fit

The two-loop matching interface should not be considered active until the following tests pass:

1. **Scheme test.** The imported coefficients must be in CSS2 or converted to CSS2 by explicit finite factors. If an SCET/RRG table is used, the conversion must be applied before Eq. (221).
2. **One-loop regression.** Applying the same import/projection machinery to the one-loop table must reproduce Eqs. (91) and (92).
3. **RG test.** The imported $\widetilde{C}^{(2)}$ must satisfy the two-loop μ and ζ evolution equations with the kernel in Eq. (212) and the NLO DGLAP kernels.
4. **Distribution projection test.** The b_T-space polynomial and the k_T-space distribution representation must agree after transforming back to b_T space.
5. **Observable test.** The expanded NNLO singular W term must agree with the known small-q_T singular coefficients bin by bin before any Y term or nonperturbative model is attached.

Working decision. For the next implementation stage, the CS kernel is internal and analytic. The NNLO matching coefficients are external perturbative data with a strict import, conversion, and projection interface. This prevents transcription errors and keeps the $\mathcal{K}_{\mathrm{CSS2}}$ formalism from becoming tied to a single paper's coefficient notation.

15.9 Executable two-loop ingredient validator

The accompanying script `kt_kcss_two_loop_ingredients_validator.py` checks the compact CS-kernel part of this section and the scalar projection of the NNLO matching-interface equations. It verifies

1. equivalence between Collins-Rogers Eq. (69), rewritten with L_b, and Eq. (212);
2. the two-loop RG identity in Eq. (214);
3. the projection from Eq. (216) to Eq. (217);
4. Casimir scaling through two loops, $D_g/D_q = C_A/C_F$, for the perturbative coefficients;
5. the scalar-channel CS and μ evolution equations for the NNLO matching reconstruction, with the finite $c^{(2,0)}(z)$ table left as external input.

The regression report gives

$$\text{passed} = \text{True}, \qquad \Delta_{K\to D} = 0, \qquad \Delta_{\mathrm{RG}} = 0, \qquad \Delta_{\mathrm{proj}} = 0, \qquad \Delta_{\mathrm{Casimir}} = 0, \qquad \Delta_{C^{(2)}\,\mathrm{RG}} = 0. \tag{241}$$

16 Running-coupling evolution kernel

This section promotes the fixed-coupling Green function of Sec. 14 to a running-coupling endpoint kernel. The goal is still formal and algorithmic: define the RG-integral object that the pure-k_T code will evaluate, expose the path-independence checks, and obtain the first two-loop rapidity-leg projection in the generator and conventional distribution bases.

16.1 General RG-integral form

We write

$$\Gamma_i(a_s) = \sum_{n\geq 0} a_s^{n+1}\Gamma_n^i, \qquad \gamma_i(a_s) = \sum_{n\geq 0} a_s^{n+1}\gamma_n^i, \tag{242}$$

$$\frac{\mathrm{d}a_s}{\mathrm{d}\ln\mu} = -2\sum_{n\geq 0}\beta_n a_s^{n+2} = -2\beta_0 a_s^2 - 2\beta_1 a_s^3 + \cdots . \tag{243}$$

Here Γ_i is the cusp anomalous dimension for the TMD parton representation and γ_i is the noncusp piece in

$$\gamma_F^i(\mu,\zeta) = \gamma_i(a_s(\mu)) - \Gamma_i(a_s(\mu))\ln\!\left(\frac{\zeta}{\mu^2}\right). \tag{244}$$

For reference,

$$\beta_1 = \frac{34}{3}C_A^2 - \frac{20}{3}C_A T_F n_f - 4C_F T_F n_f. \tag{245}$$

The coefficient γ_1^i is a CSS2 scheme input. We keep it as an imported perturbative coefficient rather than hard-coding a convention before the complete CSS2 coefficient interface is active.

For endpoint evolution from (μ_i,ζ_i) to (μ_f,ζ_f) define

$$L \equiv \ln\frac{\mu_f}{\mu_i}, \qquad \rho \equiv \ln\sqrt{\frac{\zeta_f}{\zeta_i}}, \qquad \ell_i \equiv \ln\frac{\zeta_i}{\mu_i^2}. \tag{246}$$

It is useful to introduce the RG integrals

$$A_{\Gamma_i}(\mu_f,\mu_i) \equiv \int_{\ln\mu_i}^{\ln\mu_f} \mathrm{d}\ln\mu\,\Gamma_i(a_s(\mu)), \tag{247}$$

$$A_{\gamma_i}(\mu_f,\mu_i) \equiv \int_{\ln\mu_i}^{\ln\mu_f} \mathrm{d}\ln\mu\,\gamma_i(a_s(\mu)), \tag{248}$$

$$S_{\Gamma_i}(\mu_f,\mu_i) \equiv \int_{\ln\mu_i}^{\ln\mu_f} \mathrm{d}\ln\mu\,\ln\!\left(\frac{\mu}{\mu_i}\right)\Gamma_i(a_s(\mu)). \tag{249}$$

Then the μ leg at fixed ζ_i gives the scalar exponent

$$\boxed{R_i(\mu_f,\mu_i;\zeta_i) = A_{\gamma_i} - \ell_i A_{\Gamma_i} + 2S_{\Gamma_i}.} \tag{250}$$

This reduces to Eq. (195) when a_s is fixed and only the one-loop anomalous dimensions are retained.

Using a μ-first path, the parent CSS2 image of the evolution kernel is

$$\boxed{\widetilde{\mathcal{U}}_i(b_T;\mu_f,\zeta_f;\mu_i,\zeta_i) = \exp[R_i(\mu_f,\mu_i;\zeta_i) - \rho D_i(b_T;\mu_f)]\,.} \tag{251}$$

This parent-space expression establishes the endpoint identity and supplies a regression reference. The implementation remains in k_T space by replacing the last factor with a transverse convolution exponential.

16.2 Path independence

The ζ-first path gives

$$\widetilde{\mathcal{U}}_{i,\zeta\text{-first}} = \exp[-\rho D_i(b_T;\mu_i) + A_{\gamma_i} - (\ell_i + 2\rho)A_{\Gamma_i} + 2S_{\Gamma_i}]\,. \tag{252}$$

Equations (251) and (252) are equal exactly for an integrable RG solution, or through the common fixed order for consistently truncated polynomials, because

$$D_i(b_T;\mu_f) - D_i(b_T;\mu_i) = 2A_{\Gamma_i}(\mu_f,\mu_i), \tag{253}$$

which is the integrated Collins-Soper-kernel RG equation $\mathrm{d}D_i/\mathrm{d}\ln\mu = 2\Gamma_i$. In k_T space this is a statement about convolution kernels. The kernels commute because their Fourier images are ordinary functions of L_b.

16.3 Momentum-space representation through two loops

Separate the $\mathcal{K}_{\rm CSS2}$ kernel into its identity and plus-distribution pieces,

$$\mathcal{D}_i^{\mathcal{K}_{\rm CSS2}}(\boldsymbol{k};\mu) = D_i^{\delta}(a_s(\mu))\,\delta^{(2)}(\boldsymbol{k}) + \sum_{n\geq 0} D_{i,n}(a_s(\mu))\,\mathfrak{L}_n(\boldsymbol{k};\mu). \tag{254}$$

Through two loops, Eq. (216) gives

$$D_i^{\delta}(a_f) = a_f^2 d_i^{(2)} + \mathcal{O}\Big(a_f^3\Big)\,, \tag{255}$$

$$D_{i,0}(a_f) = a_f\Gamma_0^i + a_f^2\Gamma_1^i + \mathcal{O}\Big(a_f^3\Big)\,, \tag{256}$$

$$D_{i,1}(a_f) = \frac{1}{2}a_f^2\beta_0\Gamma_0^i + \mathcal{O}\Big(a_f^3\Big)\,, \tag{257}$$

where $a_f \equiv a_s(\mu_f)$. It is useful to isolate the new two-loop generator associated with the L_b^2 term:

$$\mathcal{H}_\kappa(\boldsymbol{k};\mu) \equiv \mathrm{Exp}_{\otimes_T}[-\kappa\mathfrak{L}_1(\boldsymbol{k};\mu)]\,, \qquad \mathcal{F}_{k\to b}[\mathcal{H}_\kappa] = \exp[-\kappa L_b^2]. \tag{258}$$

Then the rapidity leg through the two-loop CS kernel may be written as

$$\mathrm{Exp}_{\otimes_T}[-\rho\mathcal{D}_i(\mu_f)] = e^{-\rho D_i^{\delta}(a_f)}\,\mathcal{H}_{\kappa_i}(\mu_f)\otimes_T \mathcal{G}_{\eta_i}(\mu_f) + \mathcal{O}\Big(a_f^3 \text{ in } \mathcal{D}_i\Big)\,, \tag{259}$$

with

$$\eta_i = \rho D_{i,0}(a_f), \qquad \kappa_i = \rho D_{i,1}(a_f). \tag{260}$$

For $\mathrm{Re}\,\kappa > 0$, the multiplier $e^{-\kappa L_b^2}$ is bounded and its inverse Fourier transform is a tempered distribution. For other signs or complex profile choices, $\mathcal{H}_\kappa$ is used only through its formal perturbative expansion unless temperedness is established separately. No new scheme information is hidden in $\mathcal{H}_\kappa$; it represents the L_b^2 term in the two-loop Collins–Soper kernel. The pure-k_T running-coupling kernel is therefore

$$\boxed{\mathcal{U}_i^{\rm run}(\boldsymbol{k};f,i) = \exp\Big[R_i(\mu_f,\mu_i;\zeta_i) - \rho D_i^{\delta}(a_f)\Big]\,\mathrm{Exp}_{\otimes_T}\left\{-\rho\sum_{n\geq 0} D_{i,n}(a_f)\mathfrak{L}_n(\boldsymbol{k};\mu_f)\right\}.} \tag{261}$$

The expression is diagonal in the TMD parton representation. Flavor mixing enters in the matching coefficients and collinear PDFs, not in the Collins-Soper rapidity kernel.

Expanding the rapidity part of Eq. (261) through a_f^2 gives

$$\mathcal{U}_i^\zeta(\boldsymbol{k};\mu_f) = \delta^{(2)} - a_f\rho\Gamma_0^i\mathfrak{L}_0 + a_f^2\Big[-\rho\Gamma_1^i\mathfrak{L}_0 + \Big(\frac{1}{2}\rho^2(\Gamma_0^i)^2 - \frac{1}{2}\rho\beta_0\Gamma_0^i\Big)\mathfrak{L}_1 - \rho d_i^{(2)}\delta^{(2)}\Big] + \mathcal{O}\Big(a_f^3\Big)\,. \tag{262}$$

Projecting to the conventional basis gives

$$\mathcal{U}_i^\zeta(\boldsymbol{k};\mu_f) = \delta^{(2)} + a_f\rho\Gamma_0^i\mathcal{L}_0 + a_f^2\Big[\rho\Gamma_1^i\mathcal{L}_0 + \Big(\rho\beta_0\Gamma_0^i - \rho^2(\Gamma_0^i)^2\Big)\mathcal{L}_1 - \rho d_i^{(2)}\delta^{(2)}\Big] + \mathcal{O}\Big(a_f^3\Big)\,. \tag{263}$$

This equation is a useful regression target for the binned distribution layer. The $\mathcal{L}_1$ term contains both the genuine two-loop L_b^2 contribution in D_i and the square of the one-loop rapidity kernel.

16.4 Expansion around the initial coupling

For algebraic checks it is convenient to expand around $a_i \equiv a_s(\mu_i)$. To $\mathcal{O}(a_i^2)$,

$$a_f = a_i - 2\beta_0 a_i^2 L + \mathcal{O}\Big(a_i^3\Big)\,. \tag{264}$$

The RG integrals are

$$A_{\Gamma_i} = a_i\Gamma_0^i L + a_i^2\Big(\Gamma_1^i L - \beta_0\Gamma_0^i L^2\Big) + \mathcal{O}\Big(a_i^3\Big)\,, \tag{265}$$

$$A_{\gamma_i} = a_i\gamma_0^i L + a_i^2\Big(\gamma_1^i L - \beta_0\gamma_0^i L^2\Big) + \mathcal{O}\Big(a_i^3\Big)\,, \tag{266}$$

$$S_{\Gamma_i} = \frac{1}{2}a_i\Gamma_0^i L^2 + a_i^2\Big(\frac{1}{2}\Gamma_1^i L^2 - \frac{2}{3}\beta_0\Gamma_0^i L^3\Big) + \mathcal{O}\Big(a_i^3\Big)\,. \tag{267}$$

Thus

$$\begin{aligned} R_i = {} & a_i L\Big(\gamma_0^i - \Gamma_0^i\ell_i\Big) + a_i\Gamma_0^i L^2 \\ & + a_i^2\Big[\gamma_1^i L - \beta_0\gamma_0^i L^2 - \ell_i\Big(\Gamma_1^i L - \beta_0\Gamma_0^i L^2\Big) + \Gamma_1^i L^2 - \frac{4}{3}\beta_0\Gamma_0^i L^3\Big] + \mathcal{O}\Big(a_i^3\Big)\,. \end{aligned} \tag{268}$$

Setting $\beta_0 = \Gamma_1^i = \gamma_1^i = d_i^{(2)} = 0$ reduces the full endpoint exponent to

$$a_i L\Big[\gamma_0^i - \Gamma_0^i(\ell_i - L)\Big] - a_i\rho\Gamma_0^i L_b(\mu_f), \tag{269}$$

which is Eq. (197).

16.5 Executable running-coupling validator

The accompanying script `kt_kcss_running_coupling_evolution_validator.py` checks the $\mathcal{O}(a_i^2)$ algebra symbolically. It verifies

1. the fixed-coupling limit of the scalar exponent;
2. the integrated identity $D_i(\mu_f) - D_i(\mu_i) = 2A_{\Gamma_i}$;
3. equality of the μ-first and ζ-first paths;
4. the rapidity endpoint equation;

5. the μ endpoint equation;

6. the semigroup law for two consecutive endpoint evolutions;

7. the projection of Eq. (259) back to the two-loop b_T-space rapidity exponent.

The regression report gives

$$\begin{aligned} &\text{passed} = \text{True}, && \Delta_{\text{fixed}} = 0, && \Delta_D = 0, && \Delta_{\text{path}} = 0, \\ &\Delta_\rho = 0, && \Delta_\mu = 0, && \Delta_{\text{semi}} = 0, && \Delta_{\text{HG}} = 0. \end{aligned} \tag{270}$$

For the quark-like smoke point used by the script,

$$R_q \simeq 9.49 \times 10^{-2}, \qquad \eta_q \simeq 2.14 \times 10^{-2}, \qquad \kappa_q \simeq 1.12 \times 10^{-3}, \tag{271}$$

which is only a numerical regression check, not a phenomenological prediction. The coefficient β_1 is already part of the running-coupling interface, but it first contributes to this explicit initial-coupling expansion at $\mathcal{O}(a_i^3)$.

Working decision. The evolution layer is formulated as a running-coupling k_T-space convolution kernel. The next section attaches this kernel to a fixed-auxiliary-scale binned W-term prototype and validates its one-loop fixed-order expansion. That prototype is an algebraic endpoint test and is not the distributional-profile implementation required for phenomenology.

17 Binned resummed W-term prototype

This is the first point where the perturbative ingredients are assembled into a forward-operator prototype. The purpose is deliberately narrow. We do not yet attach a nonperturbative core, a Y term, fiducial recoil corrections, or external NNLO matching tables. Instead, we define a fixed auxiliary-scale map and the perturbative boundary blocks, then check that its fixed-order expansion reproduces the canonical CSS2 singular spectrum bin by bin. This is an algebraic endpoint test; it is not yet the logarithmically minimized distributional-profile implementation of Sec. 11.

17.1 Endpoint scale map

For the first Drell–Yan prototype use the symmetric endpoint choice

$$\mu_H = Q, \qquad \mu_f = Q, \qquad \zeta_{A,f} = \zeta_{B,f} = Q^2, \qquad \zeta_{A,f}\zeta_{B,f} = Q^4. \tag{272}$$

The perturbative TMD boundary is evaluated at an auxiliary transverse scale

$$\mu_i = \mu_T, \qquad \zeta_i = \mu_T^2, \qquad \rho_T \equiv \ln \frac{Q}{\mu_T}, \qquad \ell_i = \ln \frac{\zeta_i}{\mu_i^2} = 0. \tag{273}$$

For this fixed-order regression test, μ_T is held fixed over all bins. This makes bin additivity manifest and isolates the RG and DGLAP cancellations. Within the cumulative prescription proposed here, a phenomenological resummed calculation would instead implement the weak profile of Eq. (60), with a canonical region satisfying Eq. (67). One must not replace the fixed scale by a pointwise function $\mu_T(q_T)$ in the differential plus distributions. The constant-scale test and the distributional-profile test are separate validation gates.

17.2 Perturbative boundary block and evolved TMD leg

Define the perturbative boundary block

$$\mathbb{B}_{i/h}^{[m_C]}(x,\boldsymbol{k};\mu_T) \equiv \sum_j \left[\mathcal{C}_{i\leftarrow j}^{[m_C],\mathcal{K}_{\mathrm{CSS2}}}\left(\boldsymbol{k};\mu_T,\mu_T^2\right) \otimes_x f_{j/h}(\mu_T)\right](x), \tag{274}$$

where m_C is the matching order retained. The evolved perturbative TMD leg is

$$\mathbb{F}_{i/h}^{[N,m_C]}(x,\boldsymbol{k};Q,\mu_T) = \mathcal{U}_i^{[N]}(\boldsymbol{k};Q,Q^2;\mu_T,\mu_T^2) \otimes_T \mathbb{B}_{i/h}^{[m_C]}(x,\boldsymbol{k};\mu_T). \tag{275}$$

Here N denotes the perturbative order of the evolution kernel and anomalous-dimension integrals. The first prototype uses the running-coupling kernel of Eq. (261) and the one-loop matching block of Sec. 12.

The binned W-term prototype is

$$W_{AB,r}^{\mathrm{proto}}(Q,y) = \int_{q_a<|\boldsymbol{q}|<q_b} \mathrm{d}^2\boldsymbol{q}\, \widehat{H}_{q\bar{q}}^{\mathrm{CSS2}}(Q,Q) \sum_q e_q^2 \Big[\mathbb{F}_{q/A} \otimes_T \mathbb{F}_{\bar{q}/B} + \mathbb{F}_{\bar{q}/A} \otimes_T \mathbb{F}_{q/B}\Big](\boldsymbol{q}), \tag{276}$$

with $x_A = (Q/\sqrt{s})e^y$ and $x_B = (Q/\sqrt{s})e^{-y}$ understood. In the symmetric quark channel the two evolution kernels can be combined before the boundary convolution:

$$\begin{aligned}\mathbb{U}_q^{\mathrm{pair}}(\boldsymbol{k};Q,\mu_T) &\equiv \mathcal{U}_q(\boldsymbol{k};Q,Q^2;\mu_T,\mu_T^2) \otimes_T \mathcal{U}_q(\boldsymbol{k};Q,Q^2;\mu_T,\mu_T^2) \\ &= \exp\Big[2R_q(Q,\mu_T;\mu_T^2) - 2\rho_T D_q^\delta(a_Q)\Big]\, \mathrm{Exp}_{\otimes_T}\left\{-2\rho_T \sum_{n\geq 0} D_{q,n}(a_Q)\mathfrak{L}_n(\boldsymbol{k};Q)\right\}. \end{aligned} \tag{277}$$

This is still a pure-k_T object: the expression is evaluated by generator algebra and then projected onto bin-integrated distributions.

17.3 One-loop fixed-order expansion gate

The first nontrivial check is independence of the auxiliary scale μ_T at $\mathcal{O}(a_s)$. Work in the parent Fourier image only as a proof device and define

$$X \equiv L_b(Q), \qquad L \equiv \ln\frac{Q}{\mu_T}, \qquad L_i \equiv L_b(\mu_T) = X - 2L. \tag{278}$$

Let P_A denote the action of the one-loop DGLAP kernel on the leg-A luminosity, and let c_A denote the finite $\delta^{(2)}$ matching contribution on that leg. The one-loop boundary coefficient in the diagonal quark channel is

$$C_A^{(1)}(\mu_T) = c_A + 3C_F L_i - C_F L_i^2 - 2P_A L_i. \tag{279}$$

Since our DGLAP convention is $\mathrm{d}f/\mathrm{d}\ln\mu = 4a_s P^{(0)} \otimes_x f + \cdots$, expressing the boundary PDF at μ_T through a PDF at Q gives

$$f(\mu_T) = f(Q) - 4a_s L\, P^{(0)} \otimes_x f(Q) + \mathcal{O}\left(a_s^2\right). \tag{280}$$

The one-loop evolution factor for a quark leg from (μ_T,μ_T^2) to (Q,Q^2) is

$$E_q^{(1)} = 6C_F L + 4C_F L^2 - 4C_F LX. \tag{281}$$

Therefore

$$C_A^{(1)}(\mu_T) - 4LP_A + E_q^{(1)} = c_A + 3C_F X - C_F X^2 - 2P_A X, \tag{282}$$

so the auxiliary μ_T dependence cancels at the first fixed-order gate. For two Drell–Yan legs and one-loop hard coefficient $h_q^{(1)}$,

$$W_{q\bar{q},b}^{(1)}(X) = h_q^{(1)} + c_A + c_B + 6C_F X - 2C_F X^2 - 2(P_A + P_B)X. \tag{283}$$

Using $\mathcal{F}[\mathcal{L}_0] = -X$ and $\mathcal{F}[\mathcal{L}_1] = -X^2/2$, this maps back to

$$\boxed{W_{q\bar{q},k}^{(1)} = 4C_F\mathcal{L}_1 + [2(P_A + P_B) - 6C_F]\,\mathcal{L}_0 + \left[h_q^{(1)} + c_A + c_B\right]\delta^{(2)}.} \tag{284}$$

This is the same diagonal/splitting structure as the one-loop binned singular-spectrum master formula in Sec. 12, now derived from an auxiliary-scale evolved boundary. Its bin action is

$$W_{q\bar{q},r}^{(1)} = 4C_F\mathcal{B}_{1,r} + [2(P_A + P_B) - 6C_F]\,\mathcal{B}_{0,r} + \left[h_q^{(1)} + c_A + c_B\right]\mathcal{B}_{\delta,r}. \tag{285}$$

17.4 Two-loop rapidity projection for the prototype

The NNLL-style bridge begins with the pair rapidity kernel. Let

$$\rho_{\text{pair}} = 2\rho_T. \tag{286}$$

Then the two-loop CS-kernel projection in the conventional basis is

$$\begin{aligned}\mathbb{U}_{q,\zeta}^{\text{pair}}(\boldsymbol{k}; Q, \mu_T) = \delta^{(2)} &+ a_Q \rho_{\text{pair}} \Gamma_0^q \mathcal{L}_0 \\ &+ a_Q^2\left[\rho_{\text{pair}}\Gamma_1^q\mathcal{L}_0 + \left(\rho_{\text{pair}}\beta_0\Gamma_0^q - \rho_{\text{pair}}^2(\Gamma_0^q)^2\right)\mathcal{L}_1 - \rho_{\text{pair}} d_q^{(2)}\delta^{(2)}\right] + \mathcal{O}\left(a_Q^3\right).\end{aligned} \tag{287}$$

This is not yet the complete $\mathcal{O}(a_s^2)$ Drell–Yan singular spectrum: completing that requires the two-loop hard factor, the imported NNLO matching coefficients, the $P^{(1)}$ PDF evolution term, and products of one-loop matching and evolution pieces. Equation (287) is the first kernel checked sub-block needed for that calculation.

17.5 Executable binned W-term prototype validator

The accompanying script `kt_kcss_binned_wterm_prototype_validator.py` checks the algebra above. It verifies

1. the one-leg auxiliary-scale identity in Eq. (282);
2. the two-leg identity in Eq. (283);
3. the projection to the distribution coefficients in Eq. (284);
4. the two-loop pair rapidity projection in Eq. (287);
5. bin-level equality for several auxiliary μ_T values.

The regression report gives

$$\begin{aligned} &\text{passed} = \text{True}, && \Delta_{\text{leg}} = 0, && \Delta_{\text{pair}} = 0, \\ &\Delta_{\text{coeffs}} = 0, && \Delta_{\text{rapidity}} = 0, && \max_r \left| W_r^{\text{aux}} - W_r^{\text{canonical}} \right| = 0. \end{aligned} \tag{288}$$

The final equality refers to the stated smoke-test bins. The last check is not a physical prediction. It is a regression test that the binned distribution layer, the DGLAP normalization, and the constant auxiliary-scale cancellation are consistent before the NNLO matching import, the distributional-profile gate, and the Y-term are activated. The legacy script variable name "profile" refers only to the sampled constant values of μ_T and must not be confused with the distributional profile of Sec. 11.

Working decision. The prototype establishes the algebraic and binned fixed-scale skeleton for a pure-k_T resummed W operator; it is not yet a logarithmically minimized resummed prediction. The contact-complete cumulative profile operator has been tested independently in Sec. 11.1, but it has not yet been integrated into this process-level W prototype. The next required gates are that production integration and the complete $\mathcal{O}(a_s^2)$ singular expansion: import the CSS2-compatible NNLO matching tables, include $H^{(2)}$ and $P^{(1)}$, and check the result against the CSS2 expansion before benchmarking any fixed-order Y term.

18 Perturbative matching onto collinear PDFs

The rigorous starting point is the small-b_T operator product expansion of the parent CSS2 TMD,

$$\widetilde{F}_{i/h}(x, \boldsymbol{b}; \mu, \zeta) = \sum_j \left[\widetilde{C}_{i\leftarrow j}(\boldsymbol{b}; \mu, \zeta) \otimes_x f_{j/h}(\mu) \right](x) + \mathcal{O}((b_T \Lambda_{\text{QCD}})^p)\,, \tag{289}$$

for $b_T \Lambda_{\text{QCD}} \ll 1$, with p determined by the leading omitted operators. The k_T-space matching coefficient is defined distributionally by inverse Fourier transformation,

$$\mathcal{C}_{i\leftarrow j}^{\mathcal{K}_{\text{CSS2}}} \equiv \mathcal{F}_{b\to k}^{-1}[\widetilde{C}_{i\leftarrow j}^{\text{CSS2}}], \qquad F_{i/h}^{\text{OPE}} = \sum_j \mathcal{C}_{i\leftarrow j}^{\mathcal{K}_{\text{CSS2}}} \otimes_x f_{j/h}. \tag{290}$$

For $k_T \gg \Lambda_{\text{QCD}}$, the large-transverse-momentum collinear matching calculation determines the ordinary tail away from $\boldsymbol{k} = 0$. Inferring a pointwise power bound from the small-b_T OPE alone additionally requires regularity and large-b_T control of its remainder; it is not a general Fourier-transform theorem. Such bounds are imposed explicitly for the admissible smearing class below. The power correction in Eq. (289) should not be read as a globally defined additive function after Fourier transformation.

At fixed perturbative order N, the logarithmic part has the finite distributional form

$$\mathcal{C}_{i\leftarrow j}^{[N]}(z, \boldsymbol{k}; \mu, \zeta) = c_{ij}^{\delta,[N]}(z; \mu, \zeta)\delta^{(2)}(\boldsymbol{k}) + \sum_{n=0}^{n_{\max}(N)} c_{ij}^{n,[N]}(z; \mu, \zeta)\mathfrak{L}_n(\boldsymbol{k}; \mu), \tag{291}$$

with z-space distributions retained as such. Any regular finite-k_T completion from exact kinematics or a process-level fixed-order calculation is not part of the universal leading-power TMD matching coefficient and must not be confused with the Y term.

A flexible long-distance model must therefore not be trained to absorb the known tail in Eq. (290). The large-k_T behavior is a perturbative matching condition, and the transition to a nonperturbative module requires an explicit overlap prescription.

18.1 TMD tail versus cross-section tail

The universal TMD tail is $\mathcal{C}_{i\leftarrow j} \otimes_x f_j$. The process-dependent large-q_T remainder is

$$Y_{AB}^{[\mathcal{K}_{\rm CSS2}]} = \left[\frac{1}{\sigma_0}\frac{\mathrm{d}\sigma}{\mathrm{d}Q^2\,\mathrm{d}y\,\mathrm{d}^2\boldsymbol{q}}\right]_{\rm FO} - \left[W_{AB,\rm pert}^{[\mathcal{K}_{\rm CSS2}],\rm LP}\right]_{\rm FO\,exp.}. \tag{292}$$

DYTurbo, SCETlib, CuTe–MCFM/MCFM, or equivalent tools can provide fixed-order inputs or independent benchmarks for this subtraction [40, 41, 42, 37]; they do not define the universal TMD coefficient.

19 Model-agnostic nonperturbative module

The perturbative accuracy label belongs to the $\mathcal{K}_{\rm CSS2}$ forward operator, not to a chosen long-distance parameterization. Define a nonperturbative module

$$\mathcal{M}_{\rm NP} : (i, h, x, \boldsymbol{k}; \mu_0, \zeta_0, \theta) \longmapsto \text{an admissible long-distance distribution or kernel.} \tag{293}$$

A DNN, spline, Gaussian mixture, analytic ansatz, or lattice-informed input may realize this map. The following placements are alternative model classes; they are not generally equivalent.

1. **Overlap-subtracted boundary decomposition:**

$$F_{i/h} = F_{i/h}^{\rm pert} + F_{i/h}^{\rm core}(\theta) - F_{i/h}^{\rm overlap}(\theta). \tag{294}$$

   The overlap term must be defined in the same scheme and must make the perturbative large-k_T asymptotics independent of the arbitrary transition scale to the retained order.

2. **Convolutional broadening:**

$$F_{i/h} = S_{i/h}^{\rm NP}(\theta) \otimes_T F_{i/h}^{\rm pert} + F_{i/h}^{\rm power}. \tag{295}$$

   This form is admissible only if the short-distance moments of $S^{\rm NP}$ are controlled so that the known perturbative tail and regulated collinear cumulants are not altered at the claimed leading power.

3. **Nonperturbative Collins–Soper kernel:**

$$\mathcal{D}_i = \mathcal{D}_i^{\rm pert} + \mathcal{D}_i^{\rm NP}(\theta). \tag{296}$$

   In the convention of Eq. (38), RG consistency requires $\mathrm{d}\mathcal{D}_i^{\rm NP}/\mathrm{d}\ln\mu = 0$ for an RG-consistent perturbative subtraction, up to its declared truncation error. It must also be independent of the hadron and of x for the same CS operator and parton representation. A scheme-dependent split with different running requires an explicitly compensating redefinition.

An admissible module must satisfy all of the following.

1. It is an element of a stated distribution/test-function domain and has well-defined bin or smooth-measurement actions.

2. It uses the same flavor basis, Wilson-line process class, operator scheme, and (μ, ζ) convention as the perturbative TMD it modifies.

3. It preserves the perturbative OPE and does not duplicate its large-k_T tail.
4. It includes an explicit overlap or transition prescription whenever perturbative and fitted components coexist.
5. It preserves the RG integrability conditions of Proposition 9.1.
6. Any positivity or normalization condition is imposed only on a quantity for which that property is theoretically meaningful; a renormalized perturbative distribution and its separate plus components need not be positive pointwise.

These requirements are aligned with the large-k_T consistency problem emphasized in Refs. [26, 27]. They are constraints on a later extraction, not consequences of Fourier representation alone.

Working decision. A DNN is one possible implementation of $\mathcal{M}_{\rm NP}$. It is neither a replacement for the perturbative kernel nor a source of the logarithmic accuracy label.

19.1 A constructive admissible boundary class

The interface can be made nonempty without choosing a neural network. Let $S \in C_c^\infty(\mathbb{R}^2)$ be radial, supported inside $|\boldsymbol{k}| \le \Lambda_S$, and normalized by $\int \mathrm{d}^2\boldsymbol{k}\, S = 1$. Positivity may be imposed on this *smearing kernel*, not on each perturbative plus distribution. Convolution with a tempered fixed-order TMD is well defined.

Proposition 19.1 (Preservation of the perturbative tail under localized smearing)**.** *Suppose $F^{\rm pert}(\boldsymbol{k})$ is an ordinary C^4 function for $k_T > \Lambda_S$ and its derivatives have the inverse-power and logarithmic bounds of the perturbative tail. For $k_T > 2\Lambda_S$,*

$$[S \otimes_T F^{\rm pert}](\boldsymbol{k}) = F^{\rm pert}(\boldsymbol{k}) + \frac{m_2}{4}\nabla_k^2 F^{\rm pert}(\boldsymbol{k}) + R_4(\boldsymbol{k}), \qquad m_2 = \int \mathrm{d}^2\boldsymbol{p}\, p_T^2 S(\boldsymbol{p}), \tag{297}$$

where R_4 is bounded by a constant times the fourth absolute moment of S times the supremum of fourth derivatives of $F^{\rm pert}$ over the smearing ball. For a $1/k_T^2$ logarithmic tail the first correction is relatively of order Λ_S^2/k_T^2, with the associated logarithms.

Proof. Taylor expand $F^{\rm pert}(\boldsymbol{k}-\boldsymbol{p})$ through third order with integral remainder. Radial symmetry removes the odd moments and gives $\int p_a p_b S = \delta_{ab} m_2/2$. The Taylor factor $1/2$ then gives $m_2/4$. The support condition keeps every argument away from the contact singularity. The fourth-derivative remainder proves the estimate. No assertion about an unregulated full-k_T integral of $F^{\rm pert}$ is needed. □

An independent illustrative nonperturbative evolution model is

$$\mathcal{D}_i^{\rm NP} = -g_i \nabla_k^2 \delta^{(2)}, \qquad g_i \ge 0, \tag{298}$$

with g_i of dimension mass squared and independent of μ, x, and the external hadron for a common CS operator/parton representation. For forward rapidity evolution $\rho > 0$,

$$U_i^{\rm NP}(\boldsymbol{k};\rho) = \frac{1}{4\pi g_i \rho}\exp[-k_T^2/(4g_i\rho)], \tag{299}$$

with the delta limit at $g_i\rho = 0$. Direct differentiation gives $\partial_\rho U = g_i\nabla_k^2 U = -\mathcal{D}_i^{\rm NP} \otimes_T U$; the Gaussian integral proves unit normalization, the semigroup law, and $\langle k_T^2\rangle = 4g_i\rho$. Backward heat evolution is not defined on an arbitrary tempered boundary and requires a smaller domain. This example supplies an explicit k_T-only long-distance module, not a prediction for the physical CS kernel or a solution of the perturbative resummation problem. More general modules must meet the same operator universality and domain requirements.

20 Regulated normalization and transverse moments

The perturbative tail makes an unrestricted $\int \mathrm{d}^2\boldsymbol{k}\, F$ and its transverse moments ultraviolet sensitive. Define a regulated action

$$\mathcal{I}_w[F_i] = \int \mathrm{d}^2\boldsymbol{k}\, w_K(\boldsymbol{k}) F_{i/h}(x, \boldsymbol{k}; \mu, \zeta). \tag{300}$$

Being an experimentally meaningful weight does not by itself make this action calculable from collinear PDFs. In particular, a window selecting only $k_T \sim \Lambda_{\mathrm{QCD}}$ retains leading nonperturbative dependence even if one introduces an unrelated parameter called K.

Proposition 20.1 (A sufficient class of short-distance integral functionals)**.** *Let $w_K(\boldsymbol{k}) = w(\boldsymbol{k}/K)$ with $w \in \mathcal{S}(\mathbb{R}^2)$, and use $w(0) = 1$ for a broad normalization-type cumulant. Assume the small-b_T OPE remainder obeys a uniform bound proportional to $(b_T\Lambda)^p$ in a fixed small-distance region and has controlled polynomial growth outside it. Then, at fixed x away from unresummed endpoints and with the corresponding perturbative scale hierarchy, its contribution to $\mathcal{I}_w$ is $\mathcal{O}((\Lambda/K)^p)$, allowing the logarithms in the stated remainder bound. The leading term is*

$$\mathcal{I}_w[F_i] = \sum_j [Z^w_{i\leftarrow j}(K; \mu, \zeta) \otimes_x f_{j/h}(\mu)](x) + \mathcal{O}((\Lambda/K)^p)\,, \qquad Z^w_{ij} = \langle \mathcal{C}_{ij}, w_K \rangle\,. \tag{301}$$

Proof. For proof only, the dual weight is $K^2\widetilde{w}(K\boldsymbol{b})$ in the Fourier pairing. Set $\boldsymbol{u} = K\boldsymbol{b}$. In the small-$b_T$ region the assumed OPE bound gives a factor $(\Lambda/K)^p$ times an integrable moment of $\widetilde{w}(\boldsymbol{u})$. Schwartz decay suppresses the complement to arbitrarily high powers of $1/K$ under the stated polynomial-growth bound. The leading coefficient is precisely the k_T-space test action displayed above. This is a sufficient-condition argument, not a claim for arbitrary acceptance functions. □

Sharp disk cumulants require the endpoint/truncation analysis of the relevant integral functional rather than this smooth-weight proof; controlled versions are constructed in Ref. [25]. Narrow soft bins, non-smooth acceptances, and fiducial measurements must be treated with their actual TMD and measurement operator or with a separately justified expansion. The original blanket matching assertion for every bin or acceptance weight is not used.

A regulated width can still be defined by

$$\langle k_T^2 \rangle_w = \frac{\int \mathrm{d}^2\boldsymbol{k}\, w_2(k_T; K) k_T^2 F_{i/h}}{\int \mathrm{d}^2\boldsymbol{k}\, w_0(k_T; K) F_{i/h}}, \tag{302}$$

provided the actions exist and the denominator is nonzero. This is a scheme- and weight-dependent quantity, not automatically a probability variance. The weights, cutoff, scales, and any power corrections must accompany a quoted value. Normalizing a finite phenomenological profile is a separate model constraint, not the exact unregulated TMD-to-collinear identity. Numerically one may combine analytic singular test actions with a radial basis for ordinary long-distance components.

21 Accuracy theorem and $N^3\mathrm{LL}'$ ingredient map

The logarithmic label belongs to a specified perturbative singular operator. It does not label a DNN, a nonperturbative ansatz, or the quality of a fit. We use the conventional $N^k\mathrm{LL}'$ counting stated explicitly in Table 5; other codes may use different names for the same ingredient set.

21.1 Definition of $N^k\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$

Let W_N^{CSS2} be a parent CSS2 singular operator truncated according to a stated ingredient manifest and scale prescription. At fixed symbolic scales let $W_N^{\mathcal{K}_{\mathrm{CSS2}}}$ be obtained by

$$1 \mapsto \delta^{(2)}(\boldsymbol{k}), \qquad L_b^m \mapsto \mathfrak{L}_{m-1}(\boldsymbol{k};\mu) \quad (m \geq 1), \tag{303}$$

and by replacing every admissible b_T product with the convolution defined in Eq. (11). If a scale varies with transverse momentum, it must either remain inside an exact inverse transform of the full parent multiplier or be applied with the distributional scale operator of Sec. 11. The finite polynomial dictionary alone does not map an unexpanded $\alpha_s(\mu_b)$, $f(x,\mu_b)$, or b_T-dependent profile. Section 23 provides a convergent momentum-only spectral realization for a specified complete canonical solution.

Definition 21.1. A singular calculation is labeled $N^k\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$ when it contains the same renormalized anomalous dimensions, running coupling, hard functions, TMD matching coefficients, and DGLAP ingredients as the stated parent $N^k\mathrm{LL}'$ CSS2 calculation, together with a well-defined resummed solution and either exact representation or a proved logarithmically equivalent direct-space prescription. The fixed-scale coefficient map is Eq. (303); varying scales are governed by Sec. 11 for the cumulative branch, or by the spectral functional calculus of Sec. 23. A matched full-spectrum label additionally requires a fixed-order Y subtraction at the stated order.

The prime denotes the higher-order boundary functions specified in Table 5; it does not imply a perturbative expansion of the nonperturbative module.

21.2 Exact representation theorem and accuracy corollary

Theorem 21.2 (Fixed-truncation representation identity)**.** *Assume that all parent kernels, boundary TMDs, and measurement factors entering a fixed truncation W_N^{CSS2} lie in the common multiplier domain of Sec. 5.1, and that the $\mathcal{K}_{\mathrm{CSS2}}$ objects are their inverse Fourier images. Scales are held symbolic/fixed. Alternatively, all scale dependence may be kept inside the complete parent multiplier before that multiplier is inverse transformed as one object. Then*

$$W_N^{\mathcal{K}_{\mathrm{CSS2}}} = W_N^{\mathrm{CSS2}} \quad \text{in } \mathcal{S}'(\mathbb{R}^2). \tag{304}$$

Consequently their actions agree for every $\varphi \in \mathcal{S}(\mathbb{R}^2)$. The corresponding sharp bin actions agree when defined by the common cumulative plus prescription.

Proof. The dictionary

$$\mathcal{F}_{k\to b}[\delta^{(2)}] = 1, \qquad \mathcal{F}_{k\to b}[\mathfrak{L}_n] = L_b^{n+1} \tag{305}$$

turns every transverse convolution into the corresponding parent product. In particular, whenever the exponential multiplier is admissible, or coefficient by coefficient as a formal perturbative series,

$$\mathcal{F}_{k\to b}\left[\mathrm{Exp}_{\otimes_T}\left\{\sum_{n\geq 0} r_n \mathfrak{L}_n\right\}\right] = \exp\left\{\sum_{n\geq 0} r_n L_b^{n+1}\right\}. \tag{306}$$

The hard and longitudinal factors are unchanged. Fourier injectivity therefore proves Eq. (304). □

Corollary 21.3 (Conditional inheritance of resummed accuracy)**.** *Let the parent* W_N^{CSS2} *be a* well-defined resummed *prediction of the stated logarithmic accuracy. If its complete admissible multiplier is represented exactly in* k_T*, its measurement actions inherit that accuracy. Alternatively, a direct momentum-space construction inherits it only after its all-order evolution, boundary prescription, and resummed remainder have been shown to reproduce the parent's retained logarithmic towers. Matching through a finite power of* a_s *is necessary but not sufficient.*

Proof. For the exact representation, Fourier injectivity gives the same distribution and therefore the same resummed remainder on each admitted measurement. For a separately derived direct-space construction the conclusion follows from the stated resummed equivalence and error estimate, not from a finite Taylor test. Proposition 11.3 supplies only the latter. □

The fixed-scale theorem is exact *order by order*. It remains valid for an unexpanded operator only if its existence and product domain are established. A formal series must not be identified with a summed tempered distribution without that additional step. The earlier formal-polynomial candidate does not meet that all-order hypothesis. The spectral-canonical construction in Sec. 23 now meets it on the stated domain, with logarithmic counting proved in Theorem 23.4. In particular, one cannot promote an $\mathcal{O}(a_s^3)$ expansion check to an N^{3}LL$'$ proof; see Proposition 22.2. Any separate numerical Fourier calculation used for benchmarking also does not satisfy the production requirement of no numerical b_T transform.

21.3 Generic counting table

The counting convention used in this note is summarized in Table 5. It follows the common SCET/CSS practice that the cusp anomalous dimension is needed one loop higher than the noncusp anomalous dimensions, while the prime means that fixed-order boundary functions are kept through the same order as the logarithmic label. Different codes sometimes vary in naming conventions, so a paper using this framework must state this table explicitly.

Table 5: Working logarithmic accuracy convention for the singular W operator. Here $k = 0, 1, 2, 3, \ldots$ corresponds to LL, NLL, NNLL, N^{3}LL, etc. The boundary order refers to hard functions and TMD matching coefficients.

Label	Γ_{cusp}	β function	noncusp γ_F	CS kernel / rapidity anomalous dimension	hard and TMD matching boundary
LL	1 loop	1 loop	none/tree	tree or RG-generated one-loop log	tree
NLL	2 loops	2 loops	1 loop	1 loop	tree
NLL$'$	2 loops	2 loops	1 loop	1 loop	1 loop
NNLL	3 loops	3 loops	2 loops	2 loops	1 loop
NNLL$'$	3 loops	3 loops	2 loops	2 loops	2 loops
N^{3}LL	4 loops	4 loops	3 loops	3 loops	2 loops
N^{3}LL$'$	4 loops	4 loops	3 loops	3 loops	3 loops

The two-loop import interface in Eq. (220) generalizes directly: an n-loop matching coefficient contains powers up to L_b^{2n}. The $\mathcal{K}_{\mathrm{CSS2}}$ import rule is therefore purely mechanical:

$$\widetilde{C}_{i\leftarrow j}^{(n)}(z, L_b, \ell_\zeta) = \sum_{m=0}^{2n} \widetilde{c}_{i\leftarrow j}^{(n,m)}(z; \ell_\zeta) L_b^m \quad \Longrightarrow \quad \mathcal{C}_{i\leftarrow j}^{(n),\mathcal{K}_{\mathrm{CSS2}}} = \widetilde{c}^{(n,0)}\delta^{(2)} + \sum_{m=1}^{2n} \widetilde{c}^{(n,m)} \mathfrak{L}_{m-1}. \tag{307}$$

No Bessel integral is required once the coefficient tables are expressed as polynomials in L_b.

21.4 Required ingredients for N3LL-prime-KCSS

Table 6 is the project manifest for the perturbative operator. The entries are phrased as requirements for the singular TMD operator; the final row adds the extra requirement for a matched full-spectrum Drell–Yan prediction.

Table 6: Ingredient manifest for a $N^3\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$ singular TMD operator.

Ingredient	Required order	Role in the pure-k_T prescription
Cusp anomalous dimension Γ^i_{cusp}	4 loops, $\Gamma^i_0, \ldots, \Gamma^i_3$	Controls the double-logarithmic part of the μ evolution and the RG evolution of the CS kernel. The four-loop cusp anomalous dimension is available in QCD [33].
QCD beta function	4 loops, $\beta_0, \ldots, \beta_3$	Required for the running-coupling integrals A_Γ, A_γ, and S_Γ at N^{3}LL accuracy [2].
TMD noncusp anomalous dimension γ^i_F	3 loops, $\gamma^i_{F,0}, \gamma^i_{F,1}, \gamma^i_{F,2}$	Enters the scalar endpoint factor multiplying the transverse convolution exponential.
Collins–Soper kernel / rapidity anomalous dimension D_i	3-loop boundary, with RG-generated logarithms; the boundary convention is declared	Use the parent canonical boundary in the spectral branch, or the converted momentum boundary in the cumulative branch. Do not double count their finite shifts or exponentiate an uncontrolled fixed-scale cubic polynomial. The three-loop rapidity anomalous dimension was extracted by Li and Zhu [18]; four-loop results provide the path beyond N^{3}LL [23, 24].
Hard function $H_{ij\to F}$	3 loops for the relevant channel	Provides the short-distance virtual coefficient. For Drell–Yan and Higgs color-singlet channels, the needed three-loop hard functions are obtained from finite parts of the quark and gluon form factors [34].
TMD matching coefficients $\mathcal{C}_{i\leftarrow j}$ and, where needed, TMDFF coefficients	3 loops, i.e. N^{3}LO boundary functions	Fix the perturbative large-k_T tail and the primed boundary. Unpolarized quark and gluon TMDPDF/TMDFF matching coefficients through N^{3}LO are available in machine-readable or ancillary form in the modern TMD literature [20, 22].

Ingredient	Required order	Role in the pure-k_T prescription
Collinear DGLAP kernels for x evolution inside the matching logs	$P^{(0)}, P^{(1)}, P^{(2)}$ for the $\mathrm{N^3LL'}$ singular operator	Needed to reconstruct the μ dependence of the three-loop matching coefficients and to convert boundary PDFs between profile scales. The three-loop unpolarized splitting functions are the NNLO DGLAP kernels [35, 36].
Flavor matrix structure	Full quark, antiquark, and gluon matrix	The formalism is not a fixed-target DY ansatz. The matching and evolution act on parton flavor vectors, with process-specific hard functions and charge/color projectors.
Canonical/profile scale operator	Admissible resummed solution with canonical boundary conversion and controlled logarithmic error	Requires more than a finite-order chain rule. Include the contact-complete weak profile, its placement in the ordered evolution, and an all-order accuracy argument; these final resummed steps remain open in this note.
Distribution basis	All $\mathfrak{L}_n$ required by the imported coefficients and the expanded convolution exponential	At three loops, matching polynomials can reach L_b^6, hence the import map can require $\mathfrak{L}_5$ before convolutions with evolution are expanded.
Fixed-order subtraction for full-spectrum matching	$\mathcal{O}(\alpha_s^3)$ for an $\mathrm{N^3LL'}$+$\mathrm{N^3LO}$ Drell–Yan spectrum	Not part of the universal TMD definition. It is required only when the prediction is extended beyond the singular region through $Y = d\sigma_{\mathrm{FO}} - [W]_{\mathrm{FO}}$. Modern Drell–Yan benchmarks already combine $\mathrm{N^3LL'}$ resummation with available $\mathcal{O}(\alpha_s^3)$ fixed-order information [37].

The DGLAP entry should not be confused with the PDF-set order. For a self-contained $\mathrm{N^3LL'}$ singular operator, $P^{(0)}$ through $P^{(2)}$ are sufficient to reconstruct the three-loop matching-scale logarithms. A phenomenological implementation may still use NNLO, approximate $\mathrm{N^3LO}$, or future full $\mathrm{N^3LO}$ PDF evolution depending on the chosen PDF input and on whether one aims at an approximate $\mathrm{N^4LL}$ or full-spectrum $\mathrm{N^3LO}$ benchmark.

21.5 What the prime does and does not mean

The prime in $N^3\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$ means

$$\mathrm{N^3LL\ evolution} + \text{three-loop hard and TMD matching boundary functions.} \tag{308}$$

It does *not* mean that the nonperturbative model is determined to third order, and it does not certify the quality of a DNN fit. The most precise statement is

$$W^{\mathcal{K}_{\mathrm{CSS2}}}_{\mathrm{theory}} = \mathfrak{W}^{\mathcal{K}_{\mathrm{CSS2}},N^3\mathrm{LL}'}_{\mathrm{pert}}[\mathcal{M}_{\mathrm{NP}}] + \text{power corrections}, \tag{309}$$

where the brackets denote one explicitly declared interface—boundary correction, convolutional smearing, or nonperturbative CS-kernel contribution—and $\mathcal{M}_{\mathrm{NP}}$ is a separately specified nonperturbative module. These interfaces are alternatives and are not algebraically interchangeable without an overlap/evolution proof. A DNN is one possible representation of $\mathcal{M}_{\mathrm{NP}}$, but so are splines, Gaussian mixtures, renormalon-inspired models, lattice-informed inputs, or process-specific effective models.

21.6 Executable ingredient-manifest validator

The companion script `kt_kcss_accuracy_ingredient_validator.py` encodes the counting convention in Table 5. It checks that the generic rule gives the manifest orders in Table 6 for $N^3\mathrm{LL}'$, namely

$$\Gamma_{\mathrm{cusp}} : 4\text{ loops}, \quad \beta : 4\text{ loops}, \quad \gamma_F : 3\text{ loops}, \quad D_i : 3\text{ loops}, \quad H, C : 3\text{ loops}, \quad P_{ij} : P^{(0)} \dots P^{(2)}. \tag{310}$$

This validator is intentionally simple. Its purpose is to prevent later sections and code from silently changing the project accuracy convention.

Working decision. The accuracy convention is now fixed. The next layer adds the N^{3}LO matching-import interface and manifest-driven kernel object needed to turn the convention into an executable perturbative operator.

22 N3LO matching-import interface and manifest-driven kernel object

This section turns the accuracy manifest of Sec. 21 into an operator contract. The purpose is to make the high-order implementation reproducible without hand-transcribing multi-page coefficient functions into the text. The note should state the interface precisely, while the code should import vetted coefficient tables from external ancillary files or equivalent libraries. Relevant published inputs include the N^{3}LO unpolarized TMDPDF and TMDFF matching calculations in Refs. [20, 21, 22]. The high-order rapidity anomalous dimensions are discussed in Refs. [18, 23, 24].

Working decision. The paper should not print the full N^{3}LO matching tables. It should print the scheme, normalization, basis, and import map, and require that an implementation import machine-readable CSS2-compatible coefficient data satisfying the checks below.

22.1 Coefficient-table contract

An external matching table accepted by the $\mathcal{K}_{\rm CSS2}$ importer must be convertible to the CSS2 parent convention before the k_T map is applied. The minimal metadata contract is

$$\mathbb{C}_{\rm in} = \left\{ \begin{array}{l} \text{scheme, operator type, } a_s \text{ normalization,} \\ L_b \text{ convention, } \ell_\zeta \text{ convention,} \\ \text{flavor basis, coefficient terms} \end{array} \right\}. \tag{311}$$

The first implementation uses

$$a_s = \frac{\alpha_s}{4\pi}, \qquad L_b = \ln\!\left(\frac{\mu^2 b_T^2}{b_0^2}\right), \qquad \ell_\zeta = \ln\!\left(\frac{\zeta}{\mu^2}\right). \tag{312}$$

The table may be supplied in a full flavor-vector basis or in an equivalent singlet, non-singlet, and gluon basis, but the loader must expose a full parton-vector action on

$$\mathbf{f}_h = (g, u, \bar{u}, d, \bar{d}, s, \bar{s}, \ldots)_h. \tag{313}$$

A source table in another TMD scheme is allowed only if a finite conversion kernel is provided:

$$\widetilde{C}^{\rm CSS2}_{i\leftarrow j} = \sum_a \widetilde{Z}^{{\rm CSS2}\leftarrow S}_{i\leftarrow a} \otimes_x \widetilde{C}^{S}_{a\leftarrow j}. \tag{314}$$

Only after this conversion is the $\mathcal{K}_{\rm CSS2}$ distribution map applied. A finite conversion can also redistribute finite terms between the two TMD legs and the hard factor; therefore the importer must verify invariance of the complete process-level W operator, not only a one-leg coefficient equation. This keeps the accuracy claim tied to a single CSS2 parent convention rather than to an implicit mixture of rapidity-regulator schemes.

For each loop order $n \le 3$, the CSS2-compatible input must be expressible as

$$\widetilde{C}^{(n),{\rm CSS2}}_{i\leftarrow j}(z, L_b, \ell_\zeta) = \sum_{m=0}^{2n} \widetilde{c}^{(n,m)}_{i\leftarrow j}(z;\ell_\zeta)\, L_b^m, \tag{315}$$

where the coefficients $\widetilde{c}^{(n,m)}$ are still distributions or functions in z. They must not be flattened into ordinary numbers unless the action on an x-space test function has already been specified. The x-space plus prescriptions and the transverse plus prescriptions are independent layers.

22.2 Mechanical map into the KCSS2 basis

Define the import operator $\mathsf{I}_{\mathcal{K}_{\rm CSS2}}$ by

$$\mathsf{I}_{\mathcal{K}_{\rm CSS2}}[\widetilde{c}(z;\ell_\zeta)] = \widetilde{c}(z;\ell_\zeta)\,\delta^{(2)}(\boldsymbol{k}), \tag{316}$$

$$\mathsf{I}_{\mathcal{K}_{\rm CSS2}}[\widetilde{c}(z;\ell_\zeta) L_b^m] = \widetilde{c}(z;\ell_\zeta)\,\mathfrak{L}_{m-1}(\boldsymbol{k};\mu), \qquad m \ge 1. \tag{317}$$

Then the imported three-loop matching operator is

$$\mathcal{C}^{\mathcal{K}_{\rm CSS2},[3]}_{i\leftarrow j} = \delta_{ij}\delta(1-z)\delta^{(2)} + \sum_{n=1}^{3} a_s^n \left[\widetilde{c}^{(n,0)}_{i\leftarrow j}\delta^{(2)} + \sum_{m=1}^{2n} \widetilde{c}^{(n,m)}_{i\leftarrow j}\mathfrak{L}_{m-1}\right]. \tag{318}$$

At N^{3}LO the largest logarithmic power is L_b^6, so the matching import can require $\mathfrak{L}_5$. Subsequent convolutions with the evolution Green function can generate higher $\mathfrak{L}_n$ in fixed-order expansions,

which is why the implementation should use the generator algebra rather than a hand-coded finite convolution table. This mechanical rule applies to coefficients expressed as fixed-scale polynomials in L_b and ℓ_ζ. If a source leaves $\alpha_s(\mu_b)$, $f(x,\mu_b)$, or another b_T-dependent scale unexpanded, the full multiplier must first be converted to a fixed- scale expansion or treated by an exact-transform/distributional-scale construction; it cannot be imported by a finite L_b^m replacement alone.

22.3 Manifest-driven N3LL-prime kernel object

The perturbative object assembled by the implementation is not a single function. It is a manifest-driven operator

$$\mathbb{K}_F^{N^3\mathrm{LL}'} = \Big[H_{ij\to F}^{[3]},\ \mathcal{U}_i^{N^3\mathrm{LL}},\ \mathcal{C}_{i\leftarrow j}^{[3]},\ P_{ij}^{(0,1,2)},\ \mathcal{B}_{\mathrm{dist}},\ \mathcal{S}_{\mathrm{dist}}\Big]_{\mathcal{K}_{\mathrm{CSS2}}}, \tag{319}$$

where $\mathcal{B}_{\mathrm{dist}}$ denotes the distribution-basis engine for bins, cumulants, or test functions, and $\mathcal{S}_{\mathrm{dist}}$ is the canonical/profile scale-setting operator of Sec. 11. The rapidity part of the N^{3}LL kernel uses the three-loop CS kernel in the derivative basis,

$$\mathcal{D}_i^{[3]}(\boldsymbol{k};\mu) = D_i^{\delta,[3]}\,\delta^{(2)} + D_{i,0}^{[3]}\,\mathfrak{L}_0 + D_{i,1}^{[3]}\,\mathfrak{L}_1 + D_{i,2}^{[3]}\,\mathfrak{L}_2, \tag{320}$$

with

$$D_i^{\delta,[3]} = a_s^2 d_i^{(2)} + a_s^3 d_i^{(3)}, \tag{321}$$

$$D_{i,0}^{[3]} = a_s\Gamma_0^i + a_s^2\Gamma_1^i + a_s^3\left(\Gamma_2^i + 2\beta_0 d_i^{(2)}\right), \tag{322}$$

$$D_{i,1}^{[3]} = a_s^2\frac{\beta_0\Gamma_0^i}{2} + a_s^3\left(\beta_0\Gamma_1^i + \frac{\beta_1\Gamma_0^i}{2}\right), \tag{323}$$

$$D_{i,2}^{[3]} = a_s^3\frac{\beta_0^2\Gamma_0^i}{3}. \tag{324}$$

These coefficients follow from the RG equation

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu}\,D_i(b_T;\mu) = 2\Gamma_i(\alpha_s) \tag{325}$$

with $\mathrm{d}a_s/\mathrm{d}\ln\mu = -2\beta_0 a_s^2 - 2\beta_1 a_s^3 + \mathcal{O}(a_s^4)$. In particular, the $2\beta_0 d_i^{(2)}$ term in Eq. (322) is required by running-coupling consistency; omitting it would break path independence at $\mathcal{O}(a_s^3)$.

The endpoint evolution kernel is then represented as

$$\mathcal{U}_{i,\mathrm{formal}}(\boldsymbol{k};f,i) = \exp\Big[R_i^{[3]}(f,i) - \rho D_i^{\delta,[3]}(a_f)\Big]\,\mathrm{Exp}_{\otimes_T}\left\{-\rho\sum_{r=0}^{2} D_{i,r}^{[3]}(a_f)\mathfrak{L}_r(\boldsymbol{k};\mu_f)\right\}, \tag{326}$$

where $R_i^{[3]}$ is built from $\Gamma_{0\ldots3}$, $\gamma_{F,0\ldots2}$, and $\beta_{0\ldots3}$ through the RG integrals in Sec. 16. The subscript "formal" is essential: this is a generator for fixed-order checks, not an already defined resummed N^{3}LL kernel. Its literal cubic exponent has the obstruction proved next.

22.4 Fixed-order existence is not resummed existence

The finite perturbative series and its unexpanded exponential are different mathematical objects. The two-loop example $\exp[-\kappa L_b^2]$ is an admissible regular tempered multiplier for real $\kappa > 0$. The analogous assertion is false for the literal three-loop polynomial in Eq. (320).

Proposition 22.1 (Obstruction to literal cubic exponentiation)**.** *For real $a_s > 0$, $\Gamma_0^i > 0$, $\beta_0 \neq 0$, and $\rho > 0$, the bare multiplier $\exp[-\rho D_i^{[3]}(L_b;\mu)]$ obtained by exponentiating the fixed-scale cubic polynomial is not locally integrable at $b_T = 0$, nor does its restriction away from zero admit an ordinary finite-order distribution extension there. For $\rho < 0$ it instead grows faster than any power as $b_T \to \infty$ and is not a regular tempered multiplier. These statements concern this literal extrapolation of the polynomial, not the properly resummed CSS2 kernel.*

Proof. The leading coefficient in $D^{[3]}$ is $a_s^3\beta_0^2\Gamma_0^i L_b^3/3$. Write $u = -L_b \to +\infty$ near the origin. The radial measure gives, up to constants, the positive integral $\int^\infty \mathrm{d}u\, \exp[cu^3 - u + \mathcal{O}(u^2)]$ with $c > 0$, which diverges. Equivalently, pairing with nonnegative test functions supported on successively smaller annuli grows faster than any fixed inverse power of the annulus radius. This contradicts the finite-order bound of a distribution on a compact neighborhood of zero. Reversing ρ moves the super-polynomial growth to infinity, violating the Schwartz-dual bound. Adding lower-degree logarithms does not change either conclusion. □

Equation (326) must consequently be read as a *formal series* until a resummed weak operator or an admissible complete multiplier is constructed. Checking that its coefficients are tempered does not prove convergence of the series in $\mathcal{S}'(\mathbb{R}^2)$. Multiplication by some specially chosen boundary may improve its domain, but that must be demonstrated for the full product, not assumed universally.

22.5 RG-completed kernels and what they do establish

There is also a truncation distinction: the derivative of the three-loop D polynomial agrees with 2Γ through a_s^3, but not exactly with a four-loop cusp used in resummed integrals. At a fixed reference scale μ_r, an explicitly RG-completed candidate is

$$\mathcal{D}_i^{\mathrm{RG}}(\boldsymbol{k};\mu) = \mathcal{D}_i^{[3]}(\boldsymbol{k};\mu_r) + 2A_{\Gamma_i^{[4]}}(\mu,\mu_r)\delta^{(2)}(\boldsymbol{k}). \tag{327}$$

Its derivative is exactly the chosen truncated $2\Gamma_i^{[4]}\delta^{(2)}$, and its common expansion reproduces the quoted three-loop kernel. Together with the same $\Gamma_i^{[4]}$ in γ_F, it makes the endpoint connection curl-free within this chosen truncated system. This resolves the mismatch of loop orders in the RG identity; it does not cure the cubic-exponent domain problem or minimize logarithms in the reference boundary. A momentum-dependent μ_r additionally requires the ordered distributional prescription; it cannot be substituted into this fixed-reference identity after convolution.

Proposition 22.2 (Finite-order checks do not certify logarithmic accuracy)**.** *Equality of two spectra through $\mathcal{O}(a_s^3)$, including all their bin actions, is not sufficient to imply equality at N^3LL'.*

Proof. The modification of an exponent by $c\,a_s^4L^5$ leaves all coefficients through a_s^3 unchanged. It changes a leading exponent logarithm at the next order. In the resummation region $a_sL = \mathcal{O}(1)$ the extra term is of order $1/a_s$, not a small fixed-order remainder. The same logical counterexample applies to a cumulant generating a distribution. A finite-order Taylor test cannot bound the omitted tower. □

To finish a transform-free resummed prescription one must specify and solve an ordered momentum-space evolution problem, including its momentum-canonical finite boundaries, and prove the retained all-order logarithmic structure or a uniform resummed remainder estimate. The recurrence-level scale-setting treatment in Ref. [14] is an appropriate starting point: it cannot be replaced by moving a variable rapidity scale through the final convolution exponential. The resulting

TMD-level construction must also be well defined on the supplied nonperturbative boundary domain. These are theoretical completion tasks in addition to acquiring coefficient files. They are not established by the formal polynomial exponential or its tests. The next subsection supplies the general evaluator. Section 23 then constructs an RG-improved canonical QCD symbol, proves its admitted domain, and establishes its conventional perturbative logarithmic accuracy.

22.6 A constructive transform-free evaluator for an admitted resummed class

There is a sufficient construction that is stronger than a purely formal exponential. It applies to a *completed* radial multiplier, not directly to the divergent cubic example above. It supplies a concrete route to evaluating such a multiplier using only momentum-space convolutions and provides a numerical error bound.

Let $T = -\nabla_k^2$ on $L^2(\mathbb{R}^2)$, with its nonnegative self-adjoint realization. The parameter $\tau > 0$ has the same dimensions as T (inverse momentum squared). Define the resolvent

$$R_\tau = \tau(\tau + T)^{-1}, \qquad [R_\tau f](\boldsymbol{k}) = \int \mathrm{d}^2\boldsymbol{p}\, r_\tau(\boldsymbol{k} - \boldsymbol{p}) f(\boldsymbol{p}), \tag{328}$$

where

$$r_\tau(\boldsymbol{k}) = \frac{\tau}{2\pi} K_0(\sqrt{\tau}\, k_T), \qquad \int \mathrm{d}^2\boldsymbol{k}\, r_\tau(\boldsymbol{k}) = 1. \tag{329}$$

Here K_0 is the modified Bessel function, evaluated as an ordinary function, not as a Bessel-transform operation. The kernel is locally integrable at zero. It follows directly by integrating the two-dimensional heat kernel against $\tau e^{-\tau s}\mathrm{d}s$. The spectrum of R_τ lies in $[0, 1]$.

Theorem 22.3 (Convergent momentum-resolvent representation)**.** *Let $M(t)$ be a bounded continuous radial multiplier on $t \geq 0$ with a finite limit at infinity. Define the continuous function on $[0, 1]$*

$$m(r) = M\left(\tau \frac{1 - r}{r}\right), \qquad m(0) = M(\infty). \tag{330}$$

For $f \in L^2$, set

$$F_N = \sum_{j=0}^{N} m(j/N) \binom{N}{j} R_\tau^j (I - R_\tau)^{N-j} f. \tag{331}$$

Then F_N converges in L^2 to the operator with Fourier multiplier $M(b_T^2)$ acting on f. Every finite N expression is evaluated by k_T-space resolvent convolutions (or equivalent elliptic solves), with no numerical Fourier or Hankel transformation. With $\epsilon_N = \|B_N[m] - m\|_{L^\infty[0,1]}$,

$$\|F_N - M(T)f\|_2 \leq \epsilon_N \|f\|_2, \qquad \epsilon_N \longrightarrow 0. \tag{332}$$

If $m \in C^2[0, 1]$, $\epsilon_N \leq \|m''\|_\infty/(8N)$.

Proof. For proof only, T has Fourier multiplier b_T^2, so R_τ has multiplier $r = \tau/(\tau + b_T^2)$. Equation (331) is the Bernstein polynomial $B_N[m](r)$ applied to f. Its uniform convergence follows from uniform continuity and the binomial variance $r(1 - r)/N$: split the expectation into $|j/N - r| \leq \delta$ and its complement, then apply the variance bound. Uniform multiplier convergence and Plancherel give Eq. (332). For C^2 functions Taylor's theorem and $\mathbb{E}(j/N - r) = 0$ give the stated $1/(8N)$ bound. The heat-kernel resolvent identity proves that all operations on the right of Eq. (331) have a direct momentum-space realization. The Fourier transform is used only in this proof. □

For fixed longitudinal variables after the collinear convolutions, an admitted perturbative leg multiplier $M_i(x,t;Q)$ can act in this way on an L^2 nonperturbative smearing $S_{i/h}(x,\boldsymbol{k})$. The hard function remains outside both legs. If $\|F_{i,N}-F_i\|_2 \le \varepsilon_i$ and a bin $\mathcal{B}$ has finite transverse area, Cauchy–Schwarz in the convolution gives

$$\left|\int_{\mathcal{B}} \mathrm{d}^2\boldsymbol{q}\,(F_{A,N}\otimes_T F_{B,N} - F_A\otimes_T F_B)\right| \le |\mathcal{B}|(\varepsilon_A\|F_B\|_2 + \varepsilon_B\|F_A\|_2 + \varepsilon_A\varepsilon_B). \tag{333}$$

A finite flavor sum and its hard factors can then be bounded termwise. Numerical approximation error is thus separated from perturbative truncation error.

This theorem alone is a *sufficient class*, not a proof that the bare three-loop candidate is bounded or resummed correctly. The separate construction in Sec. 23 supplies a completed RG symbol satisfying the bounded canonical-leg hypotheses and treats unbounded endpoint evolution separately. A chosen CSS2 resummed multiplier, its ultraviolet/infrared prescription, and its nonperturbative boundary must first satisfy the hypotheses and retain the required logarithmic accuracy. Imposing artificial cutoffs solely to make M bounded can change the physical prescription and cannot be declared harmless without a power/counting argument. Unbounded but admissible multipliers need a separate domain and convergence analysis. The logarithmic fixed-order dictionary remains useful independently.

The companion `validators/resolvent_operator_demo.py` demonstrates this construction on a bounded analytic test multiplier with a direct finite-difference elliptic solver, comparing it to an independent matrix exponential on the same grid. It checks the approximation bound, not a QCD coefficient payload or the continuum limit of that grid. This adds a constructive possible evaluation method without misidentifying it as the completed $\mathrm{N^3LL'}$ phenomenology.

22.7 Assembly of the perturbative W operator

At fixed perturbative boundary scales (μ_T,ζ_T), the imported matching coefficients construct the boundary block

$$\mathbb{B}^{[3]}_{i/h}(x,\boldsymbol{k};\mu_T,\zeta_T) = \sum_j \left[\mathcal{C}^{\mathcal{K}_{\mathrm{CSS2}},[3]}_{i\leftarrow j}(\boldsymbol{k};\mu_T,\zeta_T)\otimes_x f_{j/h}(\mu_T)\right](x). \tag{334}$$

Equation (334) is a fixed-scale formula. A varying boundary can be defined by Eq. (70), but inserting it into the following fixed-endpoint convolution is not yet a valid variable-scale evolution prescription. That operation must be derived at the level of the ordered evolution recurrence or of a complete process cumulant. Conditionally on a defined resummed $\mathcal{U}$, the fixed-endpoint perturbative TMD leg is

$$\mathbb{F}^{\mathrm{pert},[3]}_{i/h}(x,\boldsymbol{k};Q) = \mathcal{U}_i^{N^3\mathrm{LL}}(\boldsymbol{k};Q,Q^2;\mu_T,\zeta_T)\otimes_T \mathbb{B}^{[3]}_{i/h}(x,\boldsymbol{k};\mu_T,\zeta_T). \tag{335}$$

For a generic color-singlet process $AB\to F$,

$$W^{\mathcal{K}_{\mathrm{CSS2}},N^3\mathrm{LL}'}_{AB\to F}(\boldsymbol{q}) = \sum_{ij} H^{[3]}_{ij\to F}(Q,\mu_H)\left[\mathbb{F}^{\mathrm{pert},[3]}_{i/A}\otimes_T \mathbb{F}^{\mathrm{pert},[3]}_{j/B}\right](\boldsymbol{q}), \tag{336}$$

with process-specific measurement maps and PDF/FF normalizations for SIDIS or back-to-back hadron production in e^+e^-. A scalar radial dictionary alone does not supply tensor gluon or polarized harmonic kernels; those require their own projector and matching analysis. The formula is not a factorization claim for arbitrary colored final states. A nonperturbative module may be

convolved with the perturbative leg or supplied as a boundary correction, but it is not part of the N^{3}LL$'$ accuracy label:

$$\mathbb{F}_{i/h} = \mathbb{F}_{i/h}^{\mathrm{pert},[3]} \otimes_T \mathcal{M}_{i/h}^{\mathrm{NP}} + \text{power-suppressed or overlap-subtracted terms.} \tag{337}$$

This is the desired separation for the eventual paper: the perturbative prescription is generic, while DNNs, splines, analytic ansatzes, and lattice-informed inputs are choices for $\mathcal{M}_{\mathrm{NP}}$.

22.8 Import and kernel validation gates

Before any extracted TMD is assigned an N^{3}LL$'$ label, the following gates must pass:

1. **Metadata gate:** the coefficient source must specify its scheme, coupling normalization, L_b, ℓ_ζ, operator type, and flavor basis.

2. **Scheme gate:** if the source is not CSS2, the finite conversion in Eq. (314) must be supplied and tested.

3. **Power gate:** an n-loop coefficient may contain no power higher than L_b^{2n}, and L_b^6 must map to $\mathfrak{L}_5$ at N^{3}LO.

4. **Flavor gate:** the loaded basis must act on the full quark, antiquark, and gluon vector, even if a first Drell–Yan application uses only a subset of channels.

5. **RG gate:** the logarithmic coefficients must satisfy the CS, μ-RG, and DGLAP reconstruction equations through $\mathcal{O}(a_s^3)$.

6. **Distribution gate:** transverse convolutions must be validated by generator algebra and by bin/test-function actions, not by pointwise evaluation at $q_T = 0$.

7. **Scale-setting gate:** every point-dependent cumulative profile acting on a delta or plus distribution must be implemented through the cumulative operator in Eq. (60), and its fixed-order expansion must reproduce the declared logarithmic counting. The spectral-canonical alternative is specified and proved separately in Sec. 23; it must not be mixed with a second momentum-disk boundary shift.

8. **Resummation gate:** the unexpanded weak kernel exists on the declared input domain, and the ordered scale prescription reproduces the retained all-order logarithmic towers with the converted canonical boundaries. This gate is not implemented by the mock importer.

9. **Expansion gate:** the fixed-order expansion of Eq. (336) through $\mathcal{O}(a_s^3)$ must agree with the equivalent CSS2 singular expansion bin by bin.

The companion script `kt_kcss_n3lo_matching_import_validator.py` implements a symbolic version of these checks. It builds a mock CSS2-compatible three-loop coefficient table, maps L_b^m to $\delta^{(2)}$ or $\mathfrak{L}_{m-1}$, constructs a manifest-driven N^{3}LL$'$ kernel object, and verifies the three-loop CS-kernel RG identity. The mock coefficients are placeholders; the point of the validator is the interface and algebra, not phenomenology.

Working decision. The next nontrivial physics task is to acquire the raw ancillary files into the local coefficient directory, convert the Mathematica tables into the canonical coefficient schema, and run the $\mathcal{O}(a_s^2)$ and $\mathcal{O}(a_s^3)$ singular-spectrum gates with physical coefficients.

23 An admitted unexpanded evolution and its logarithmic accuracy

This section completes one of the two construction routes left open by the v1.4 audit. It uses the momentum-space functional calculus of $T = -\nabla_k^2$, not pointwise k_T-dependent scale substitution and not a numerical Fourier transform. It is a spectral realization of a specified CSS2 canonical-scale solution. The scalar RG solution is inherited from CSS2; the contribution here is its admitted operator domain, constructive momentum-only evaluation, and accuracy argument. The cumulative-profile route of Sec. 11 remains a different possible realization. Its production implementation is not assumed in this construction.

23.1 Scope, conventions, and an explicit infrared completion

Work first in massless QCD at fixed n_f, with $\beta_0 > 0$ and $\Gamma_0^i > 0$. Keep the note's $D = -\widetilde{K}$ and $a = \alpha_s/(4\pi)$ conventions. The retained scalar functions are

$$\beta_{[4]}(a) = -2\sum_{r=0}^{3} \beta_r a^{r+2}, \qquad \Gamma_{i,[4]}(a) = \sum_{r=0}^{3} \Gamma_r^i a^{r+1},$$
$$\gamma_{i,[3]}(a) = \sum_{r=0}^{2} \gamma_r^i a^{r+1}, \qquad d_{i,[3]}(a) = d_i^{(2)} a^2 + d_i^{(3)} a^3. \tag{338}$$

Here $d_{i,[3]}$ is the *parent canonical* boundary at $L_b = 0$, not the momentum-disk boundary shifted by the J_m coefficients. These are published perturbative inputs, with the exact convention map specified earlier [39, 18, 33, 2]. The scalar running coupling is the *unexpanded* solution of $\mathrm{d}a/\mathrm{d}\ln\nu = \beta_{[4]}(a)$. Select $\mu_0 > 0$ and $a(\mu_0) = a_0$ on an asymptotically free branch: $\beta_{[4]}(a) < 0$ for $0 < a \le a_0$. All external virtuality scales in this construction are at least μ_0. There is then no Landau pole on any integration path. This is a checkable condition on a finite polynomial, not an assumption about the unknown infrared coupling. Finite heavy-flavor matching and mass-dependent hard functions require additional inputs and are not silently included in the fixed-n_f theorem.

Write s for the spectral value of T; s has units of inverse momentum squared. It is *not* the measured k_T^2. Set $s_0 = b_0^2/\mu_0^2$, and define

$$\chi(u) = \begin{cases} 0, & u \le 0, \\ \dfrac{e^{-1/u}}{e^{-1/u} + e^{-1/(1-u)}}, & 0 < u < 1, \\ 1, & u \ge 1. \end{cases} \qquad c(y) = \chi(2y-1). \tag{339}$$

An explicit smooth scale on $s > 0$ is

$$\mu_c(s) = \mu_0 \exp\left[-\frac{1 - c(s/s_0)}{2} \ln(s/s_0)\right]. \tag{340}$$

Thus $\mu_c = b_0/\sqrt{s}$ for $0 < s \le s_0/2$, and $\mu_c = \mu_0$ for $s \ge s_0$. In between it remains at least μ_0. There is *no ultraviolet cap*: $\mu_c(s) \to \infty$ as $s \to 0$. This matters for preserving the ultraviolet canonical solution. The scale is independent of Q, x, the hadron, and the endpoint path, so it does not turn a universal TMD evolution kernel into a process-specific object.

Equation (340) specifies a perturbative infrared *completion*, not the nonperturbative QCD kernel. In the infrared it amounts to a smooth globally frozen separation used to define perturbative and nonperturbative contributions. The need to declare that separation is standard in precision TMD analyses[37]. Two such completions have CS kernels differing by a μ-independent function; the fitted nonperturbative CS contribution and boundary must be changed consistently. No claim is made that their infrared predictions agree without that change.

23.2 RG-improved CS kernel and endpoint evolution

Define, for $s > 0$,

$$D_i^c(s;\mu) = d_{i,[3]}(a(\mu_c(s))) + 2\int_{\mu_c(s)}^{\mu} \frac{\mathrm{d}\nu}{\nu}\Gamma_{i,[4]}(a(\nu)). \tag{341}$$

This resums the $\ln(\mu/\mu_c)$ terms *before* exponentiation. In the canonical region its expansion through $a(\mu)^3$ reproduces Eqs. (212) and (320), including $2\beta_0 d_i^{(2)}$. Its exact derivative within the retained RG system is

$$\partial_{\ln\mu} D_i^c(s;\mu) = 2\Gamma_{i,[4]}(a(\mu)). \tag{342}$$

It is not the exponentiation of the fixed-scale cubic polynomial of Proposition 22.1.

For endpoints $e_a = (\mu_a, \zeta_a)$ and $e_b = (\mu_b, \zeta_b)$, put

$$\begin{aligned} \rho_{ba} &= \ln\sqrt{\zeta_b/\zeta_a}, \\ R_{i;ba}(\zeta_a) &= \int_{\mu_a}^{\mu_b} \frac{\mathrm{d}\nu}{\nu}\left[\gamma_{i,[3]}(a(\nu)) - \Gamma_{i,[4]}(a(\nu))\ln\frac{\zeta_a}{\nu^2}\right], \\ u_{i;ba}(s) &= \exp[R_{i;ba}(\zeta_a) - \rho_{ba} D_i^c(s;\mu_b)]\,. \end{aligned} \tag{343}$$

The notation μ_b in this equation means endpoint b, not b_0/b_T. The operator is

$$\mathbb{U}_i(e_b;e_a) = u_{i;ba}(T), \qquad \mathbb{D}_i^c(\mu) = D_i^c(T;\mu), \tag{344}$$

defined by the spectral calculus of the nonnegative self-adjoint momentum-space Laplacian [50]. The exact meaning of this possibly unbounded operator is fixed next.

Theorem 23.1 (Admitted endpoint evolution)**.** *Fix $\tau > 0$ with the units of s, let*

$$w(s) = 1 + \ln(1+\tau/s), \qquad \mathscr{D}_{\log} = \bigcap_{m=0}^{\infty} \operatorname{Dom} w(T)^m \subset L^2(\mathbb{R}^2). \tag{345}$$

For the running branch and completion above, $\mathscr{D}_{\log}$ is dense, contains $\mathcal{S}(\mathbb{R}^2)$, and is invariant under all finite-endpoint operators $\mathbb{U}_i(e_b;e_a)$. These are continuous maps on this common Fréchet domain. Their scalar symbols define regular tempered distributions, and their inverse Fourier images therefore define admitted transverse evolution kernels. They obey, exactly within the retained RG system,

$$\begin{aligned} \mathbb{U}_i(e_c;e_b)\mathbb{U}_i(e_b;e_a) &= \mathbb{U}_i(e_c;e_a), & \mathbb{U}_i(e_a;e_a) &= I, \\ \partial_{\ln\sqrt{\zeta_b}}\mathbb{U}_i(e_b;e_a) &= -\mathbb{D}_i^c(\mu_b)\mathbb{U}_i(e_b;e_a), \\ \partial_{\ln\mu_b}\mathbb{U}_i(e_b;e_a) &= \left[\gamma_{i,[3]} - \Gamma_{i,[4]}\ln(\zeta_b/\mu_b^2)\right]\mathbb{U}_i(e_b;e_a). \end{aligned} \tag{346}$$

The derivatives and products are taken on $\mathscr{D}_{\log}$. A claim of boundedness on all of L^2 is neither needed nor generally true.

Proof. Asymptotic freedom gives, for any fixed positive reference ν_*, $a(\nu) = [2\beta_0\ln(\nu/\nu_*)]^{-1} + O(\ln\ln\nu/\ln^2\nu)$. It follows that

$$D_i^c(s;\mu) = -\frac{\Gamma_0^i}{\beta_0}\ln\ln\frac{\mu_c(s)}{\nu_*} + O(1) \quad (s\to 0). \tag{347}$$

The boundary $d_{i,[3]}(a(\mu_c))$ tends to zero. Hence, for each compact set of external endpoints, $|u_{i;ba}(s)|$ and $|u_{i;ba}(s)^{-1}|$ are bounded by $Cw(s)^p$ for some finite p. For $s \ge s_0$ they are constant in s. Powers

of $\ln s$ are locally integrable in the two-dimensional spectral measure $\pi \mathrm{d}s$, establishing temperedness. These bounds also show $\|w(T)^m \mathbb{U}_i f\|_2 \le C\|w(T)^{m+p} f\|_2$; replacing p by its ceiling proves invariance and continuity. Schwartz Fourier images are bounded near zero and rapidly decreasing at infinity, proving the stated inclusion and density. Spectral calculus gives the closed maximal realization on $\{f : u_{i;ba}(T) f \in L^2\}$, whose common invariant restriction is used here. Finally differentiate Eq. (343) using Eq. (342). The endpoint μ_b derivative combines the initial ζ_a in R with $2\rho_{ba}$ to produce the final ζ_b. The same identity makes the two rectangular paths equal and gives the composition law. All steps are dominated by a higher seminorm of $\mathscr{D}_{\log}$. □

At one-loop running the ultraviolet repair can already be seen without a formal series. If $d = 0$, then $D_i^c(s;\mu) = (\Gamma_0^i/\beta_0)\ln[a(\mu_c(s))/a(\mu)]$. Its exponential grows only as a power of a logarithm near $s = 0$, rather than as $\exp(c|\ln s|^3)$. The higher retained loops do not change this leading domain classification.

23.3 A bounded canonical leg without an ultraviolet cutoff

Define the complete canonical evolution symbol by

$$v_i(s;\mu,\zeta) = \exp\left\{ - d_{i,[3]}(a(\mu_c)) \ln \frac{\sqrt{\zeta}}{\mu_c} + \int_{\mu_c}^{\mu} \frac{\mathrm{d}\nu}{\nu} \left[\gamma_{i,[3]}(a(\nu)) - \Gamma_{i,[4]}(a(\nu)) \ln \frac{\zeta}{\nu^2} \right] \right\}. \tag{348}$$

This is a solution from the spectral canonical point (μ_c, μ_c^2), rather than the insertion of a variable scale into a pre-existing plus distribution. For $s > 0$ it satisfies

$$u_{i;ba}(s) = v_i(s;e_b)/v_i(s;e_a). \tag{349}$$

Theorem 23.2 (Bounded canonical evolution)**.** *For each finite endpoint, v_i extends continuously to $s = 0$ by $v_i(0) = 0$; it is bounded and strictly positive for $s > 0$, and is constant for $s \ge s_0$. Its ultraviolet behavior is*

$$\ln v_i(s;\mu,\zeta) = -\frac{\Gamma_0^i}{\beta_0} \ln \frac{\mu_c(s)}{\nu_*} + O\Big((\ln\ln \mu_c(s))^2\Big). \tag{350}$$

Thus for each $0 < \epsilon < p_i = \Gamma_0^i/(2\beta_0)$, $v_i(s) = O(s^{p_i - \epsilon})$ near zero. The canonical operators $v_i(T;e)$ are bounded on L^2, have dense range, and map L^2 into $\mathscr{D}_{\log}$.

Proof. In the large-ν integrand of Eq. (348), the cusp term $2\Gamma_{i,[4]}(a(\nu))\ln\nu$ approaches Γ_0^i/β_0. Integrating from μ_c down to a fixed endpoint gives the negative leading term in Eq. (350); the subleading running terms integrate to at most $O((\ln\ln\mu_c)^2)$. The noncusp integral grows only logarithmically in $\ln\mu_c$, and $d(a(\mu_c))\ln\mu_c = O(1/\ln\mu_c)$ since $d_1 = 0$. These estimates prove the limit and power bound. Continuity on the remaining finite interval and the constant infrared limit give boundedness. Multiplying by any power of w preserves boundedness, proving the range inclusion. Positivity almost everywhere implies a trivial kernel and dense range. □

This result is the key distinction between an unbounded endpoint ratio and a bounded, physically assembled canonical leg. A fit can parameterize a canonical input g and evaluate $v_i(T;e)g$ directly; it need not invert an ill-conditioned reference operator. On the range spaces $\mathscr{H}_e = v_i(T;e)L^2$, with norm $\|F\|_{\mathscr{H}_e} = \|v_i(T;e)^{-1}F\|_2$, Eq. (349) also defines an isometry from $\mathscr{H}_{e_a}$ to $\mathscr{H}_{e_b}$.

For a perturbative TMD leg include the parent matching boundary and longitudinal input in the order

$$B^{[3]}_{i/h}(x;s)=\sum_j\left[C^{\mathrm{can},[3]}_{i\leftarrow j}(a(\mu_c(s)))\otimes_x f_{j/h}(\mu_c(s))\right](x),\qquad M_{i/h}(x,s;e)=v_i(s;e)B^{[3]}_{i/h}(x;s).\tag{351}$$

Require the longitudinal convolutions to exist, be continuous for finite μ_c, and have at most polynomial growth in $\ln\mu_c$, with corresponding derivative bounds. This is a sufficient input class, not a uniform assertion at $x=0$ or 1. For example it holds for fixed Mellin moments away from splitting-kernel singularities: Gronwall's inequality applied to the finite-order DGLAP matrix gives polynomial growth in $\ln\mu_c$. Uniform x-space use requires the analogous bounds on the chosen PDFs and coefficient convolutions. Under these conditions $M_{i/h}$ has all the boundedness and endpoint properties needed below, even if the matching boundary is signed. The factor $s^{p_i-\epsilon}$ dominates the stated longitudinal growth.

The value $M(0)=0$ has a specific consequence: a nonzero integrable canonical output cannot be nonnegative everywhere. Proposition 25.1 states the finite-measure obstruction and its qualifications. This endpoint is not a probability normalization or the inclusive physical cross section. The unmodified small-distance Sudakov solution is a singular-region construction; its integration over all q_T is not an inclusive prediction. The $W+Y$ subtraction and any high-q_T turnoff remain separately declared, process-level operations. An ultraviolet cap depending on Q has deliberately not been hidden inside the universal evolution kernel.

23.4 Constructive evaluation on ordinary inputs and contact distributions

Theorem 22.3 now applies to the actual canonical RG symbol, not merely an arbitrary bounded test function. A stronger form also evaluates its action on $\delta^{(2)}$, which is not in L^2. For any of the admitted $M(s)$ above set

$$M_\infty=\lim_{s\to\infty}M(s),\quad m(r)=M\!\left(\tau\frac{1-r}{r}\right),\quad h(r)=\frac{m(r)-M_\infty}{r}\quad(r>0),\quad h(0)=0.\tag{352}$$

The infrared completion makes h identically zero near $r=0$. The ultraviolet result gives $h(1)=-M_\infty$, so $h\in C[0,1]$. Let $B_N[h]$ denote its Bernstein polynomial and define

$$\mathbb{M}_N=M_\infty I+R_\tau B_N[h](R_\tau),\qquad R_\tau=\tau(\tau+T)^{-1}.\tag{353}$$

Every operation on an input in Eq. (353) is a momentum-space convolution or an elliptic solve. The numbers $h(j/N)$ require ordinary one-dimensional RG integrals evaluated at spectral nodes; they do not require an input function sampled in b_T, an FFT, or a Hankel/Bessel transformation. Operator de Casteljau recurrence is one stable way to evaluate the Bernstein polynomial without large monomial coefficients.

Theorem 23.3 (Contact-preserving continuum evaluator)**.** *Let $\epsilon_N=\|B_N[h]-h\|_\infty$. Then*

$$\begin{aligned}\|\mathbb{M}_N-M(T)\|_{2\to2}&\le\epsilon_N\longrightarrow0,\\ M(T)\delta^{(2)}&=M_\infty\delta^{(2)}+g,\qquad g\in L^2(\mathbb{R}^2),\\ \|(\mathbb{M}_N-M(T))\delta^{(2)}\|_2&\le\sqrt{\frac{\tau}{4\pi}}\,\epsilon_N.\end{aligned}\tag{354}$$

The contact coefficient is exact at every N. For any finite-area bin $\mathcal{B}$, with a stated convention for including the origin,

$$\left|\int_{\mathcal{B}}\mathrm{d}^2\boldsymbol{k}\,(\mathbb{M}_N-M(T))\delta^{(2)}\right|\le\sqrt{|\mathcal{B}|}\sqrt{\frac{\tau}{4\pi}}\,\epsilon_N.\tag{355}$$

If h is Hölder continuous with exponent $0 < \nu \le 1$ and constant H_ν, then $\epsilon_N \le H_\nu (4N)^{-\nu/2}$. The C^2 bound of the earlier theorem must not be assumed at the ultraviolet endpoint without checking it.

Proof. The symbol error equals $r\{B_N[h](r) - h(r)\}$, establishing the operator bound and exact $r = 0$ limit. Uniform Bernstein convergence follows from continuity. For proof only, Plancherel with the note's Fourier convention gives

$$\|g_N - g\|_2^2 \le \frac{\epsilon_N^2}{(2\pi)^2} \int \mathrm{d}^2\boldsymbol{b} \left(\frac{\tau}{\tau + b_T^2} \right)^2 = \frac{\tau}{4\pi} \epsilon_N^2. \tag{356}$$

Here $M - M_\infty$ is bounded and supported at finite s, so $g \in L^2$; the approximate remainder is also L^2 because of the leading resolvent. Cauchy–Schwarz proves the bin bound. For the last statement use $B_N[h](r) = \mathbb{E}[h(J/N)]$, $J \sim \mathrm{Binomial}(N, r)$, and Jensen's inequality to bound $\mathbb{E}|J/N - r|^\nu$ by $[r(1-r)/N]^{\nu/2}$. The spectral/elliptic construction, not a transform, is the evaluation algorithm. □

For the RG-only canonical symbol, its differentiated ultraviolet asymptotics imply Hölder continuity for every $0 < \nu < \min(1, p_i)$ after compactification. The same holds for longitudinal inputs satisfying the derivative bounds above. This supplies an existence rate, although the quark exponent is small and the operator de Casteljau implementation can require quadratically many right-hand-side applications. It is no longer the preferred production proposal. The constrained rational replacement and its own contact/bin error theorem are developed in Sec. 26, with measured stability limitations in Sec. 29.

For an arbitrary endpoint input $f \in \mathscr{D}_{\log}$ one can also evaluate $\mathbb{U}_i f$ directly without canonical preconditioning. Choose a smooth spectral cutoff $\vartheta_\varepsilon(s)$ equal to zero for $s \le \varepsilon$ and one for $s \ge 2\varepsilon$, and apply the bounded-resolvent construction to $u_\varepsilon = \vartheta_\varepsilon u$. If $|u| \le C w^p$, then

$$\|(u - u_\varepsilon)(T) f\|_2 \le C w(2\varepsilon)^{-m} \|w(T)^{p+m} f\|_2 \longrightarrow 0 \qquad (m > 0). \tag{357}$$

The cutoff is removed in a strong limit on the declared domain. It is not a physical ultraviolet regulator. Thus the admitted endpoint evolution itself, and not just the bounded canonical leg, has a transform-free convergent realization.

23.5 All-order logarithmic accuracy, not just a third-order check

The logarithmic statement is made in the usual perturbative sense: it is a theorem about the formal logarithmic expansion and, with explicit remainder hypotheses, a uniform resummed estimate. It is not a proof that QCD perturbation theory converges. Consider the canonical perturbative region $0 < s \le s_0/2$, $\mu_c(s) \le Q$, and set

$$a_Q = a(Q), \qquad L = \ln \frac{Q}{\mu_c(s)}, \qquad a_-(t) = a(Q e^{-t}). \tag{358}$$

At $(\mu, \zeta) = (Q, Q^2)$ the exact retained exponent is

$$E_{i,[4,3,3,4]}(L, a_Q) = \int_0^L \mathrm{d}t \, [\gamma_{i,[3]}(a_-(t)) - 2t\Gamma_{i,[4]}(a_-(t))] - L\, d_{i,[3]}(a_-(L)). \tag{359}$$

The subscripts specify cusp, noncusp, CS boundary, and beta-function loop orders. No expansion of this equation is used to define the numerical operator.

Theorem 23.4 (Retained logarithmic towers)**.** *With Eq. (338), the formal expansion of Eq. (359) has the complete CSS2 exponent towers $a_Q^n L^{n+1}$, $a_Q^n L^n$, $a_Q^n L^{n-1}$, and $a_Q^n L^{n-2}$ wherever the powers exist. A change in perturbative inputs first omitted by this truncation can affect the exponent only through*

$$\Delta E_i \in \sum_{n\geq 4} a_Q^n \, \mathbb{C}[L]_{\leq n-3}. \tag{360}$$

Consequently the evolution is N^3LL in the declared CSS2 convention. Including the physical three-loop hard and canonical TMD matching boundaries, and the consistent three-loop collinear evolution, supplies the primed N^3LL' singular operator. The operator functional calculus adds no perturbative tower: its exact limit is the same scalar canonical RG solution.

Proof. The formal solution of the downward RG equation has $a_-(t) = \sum_{n\geq 1} a_Q^n A_n(t)$ with $A_1 = 1$ and $\deg A_n \leq n-1$. Substitution in $\mathrm{d}a_-/\mathrm{d}t = 2\sum \beta_r a_-^{r+2}$ proves this degree bound by induction. The first omitted cusp term starts at a_-^5; multiplying by t and integrating produces at order a_Q^n at most L^{n-3}. The omitted noncusp term starts at a_-^4 and is integrated once, again yielding at most L^{n-3}. The omitted rapidity boundary is $a_-^4(L)$ multiplied by L, with the same bound. Finally the first omitted beta coefficient multiplies a_-^6: its first effect on a_- is $O(a_Q^6 t)$, with at most degree $n-5$ at order n. Inserting this correction in the leading cusp integral gives at most degree $n-3$; the remaining terms are less enhanced. Products generated by the RG recursion obey the same degree inequalities, completing the all-order induction. The retained towers are those of the parent canonical CSS2 solution because the equations, boundary convention, and integral solution coincide, not because a finite number of expansion coefficients matched.

A first omitted canonical hard or matching boundary starts at fourth order; its running from μ_c generates $a_Q^n L^p$ with $p \leq n-4$. The three-loop finite boundaries are therefore the required primed boundary information. For the matrix-valued collinear evolution, the first omitted splitting function starts at fourth order and contributes a single scale integral; its tower bound is the same as the omitted noncusp term. Ordered products do not increase the degree count. The existing ordered recurrence is used for finite-order matrix checks, not replaced by a commuting approximation. The scalar RG resummation and this convention agree with the precision literature [39, 49, 37]. □

The all-order content can already be seen at LL with one-loop running:

$$E_i^{\mathrm{LL}} = \frac{\Gamma_0^i}{2\beta_0^2 a_Q}[\ln(1-\lambda)+\lambda], \qquad \lambda = 2\beta_0 a_Q L < 1. \tag{361}$$

It generates infinitely many $a_Q^n L^{n+1}$ terms. It is not its Taylor polynomial at $n = 3$.

Proposition 23.5 (Uniform perturbative remainder on a resummation window)**.** *Suppose the comparison RG functions have, on the weak-coupling branch, remainders $\Delta\beta = O(a^6)$, $\Delta\Gamma = O(a^5)$, $\Delta\gamma = O(a^4)$, and $\Delta d = O(a^4)$ with corresponding local derivative bounds. On a family of intervals with $0 \leq a_Q L \leq \lambda_*$ below the first RG singularity and $a_-(t) \leq C a_Q$, the exponent error is $O(a_Q^3)$ uniformly. If the finite hard/matching boundaries are retained through a^3, their finite-boundary error is $O(a_Q^4)$. The usual organization is therefore $g_1(a_Q L)/a_Q + g_2(a_Q L) + a_Q g_3(a_Q L) + a_Q^2 g_4(a_Q L)$ plus an $O(a_Q^3)$ exponent remainder.*

Proof. Variation of the running-coupling ODE and Gronwall give $\Delta a_- = O(a_Q^6 L) = O(a_Q^5)$ uniformly on the interval. The cusp remainder, including the changed running, enters a double-logarithmic integral and is $O(a_Q^5 L^2) = O(a_Q^3)$. The noncusp integral and the rapidity boundary contribute $O(a_Q^4 L) = O(a_Q^3)$. The fourth-order finite boundary is not multiplied by a large logarithm and is

$O(a_Q^4)$. The constants depend on the chosen compact resummation window, not on a_Q. These stated remainder hypotheses are additional to the purely formal tower theorem and are not assumptions of a convergent all-order QCD series. □

This estimate is not asserted uniformly across a Landau pole, an arbitrary nonperturbative region, zeros caused by channel cancellations, or the full large-q_T tail. The infrared completion is exactly the same in the two representations; it is separately modeled physics outside the perturbative window. For finite-order leading-power tails, the completion is harmless in a more precise sense: it is the identity at sufficiently small s, and the difference of any completed and canonical logarithmic coefficient has integrable sufficiently high derivatives in the two-dimensional Fourier variable. Repeated integration by parts therefore suppresses its off-origin momentum image by arbitrarily high inverse powers of k_T/μ_0, for each fixed-order coefficient satisfying these derivative bounds. This does not assign a perturbative error to the unknown infrared TMD itself.

23.6 Primed assembly, universality, and the optional fitted module

The spectral-canonical realization uses $C^{\mathrm{can},[3]}$ and $d_{[3]}$ at $L_b = 0$. It *does not* insert the momentum-disk constants a second time. When a fixed-scale coefficient is later projected onto the $\mathcal{L}_n$ basis or integrated over a disk, its J_m contact shifts are generated automatically. Adding those shifts already at the spectral canonical boundary would double count and would spoil the prime-level finite terms. This branch distinction is essential.

For a sufficient model class, let $S_{i/h}(x, \boldsymbol{k})$ be a localized $L^1 \cap L^2$ radial input with unit integral and finite moments as in Sec. 19. A DNN, analytic model, or spline can parameterize it. Define

$$F^c_{i/h}(x; \mu, \zeta) = M_{i/h}(x, T; \mu, \zeta) S_{i/h}(x), \tag{362}$$

with M from Eq. (351); the perturbative contact limit $S = \delta^{(2)}$ is defined by Theorem 23.3 instead. The same input is used at every endpoint, so Eq. (349) proves universal endpoint evolution. The high-order canonical matching inputs are known[20, 22], but their physical values and source conversion must be supplied in a numerical N^{3}LL$'$ run.

A nonperturbative CS contribution may additionally be specified at a fixed reference ζ_r as the commuting factor $\exp[-\ln\sqrt{\zeta/\zeta_r}\, D_i^{\mathrm{NP}}(T)]$, provided its product domain is admitted. It is μ independent, common to the appropriate partonic universality class, and power suppressed as $s \to 0$. For example $D_i^{\mathrm{NP}}(T) = g_i T$ gives the forward momentum-space heat semigroup for a nonnegative rapidity interval. Backward heat evolution is not declared bounded. An arbitrary hadron- or x-dependent CS kernel is not introduced by this model interface. The admissibility condition is about the module's action, not its choice of parameterization.

For the scalar DY channel the singular operator is

$$W^c_{AB} = \sum_q e_q^2 \widehat{H}^{[3]}_{q\bar{q}}(Q, Q) \left[F^c_{q/A}(Q, Q^2) \otimes_T F^c_{\bar{q}/B}(Q, Q^2) + (q \leftrightarrow \bar{q}) \right]. \tag{363}$$

The hard function stays outside the universal TMD legs. For ordinary L^2 legs their convolution is bounded by Cauchy–Schwarz; finite bins therefore exist. For perturbative contact legs the full two-leg symbol is again constant plus a compactly supported L^2 remainder in spectral space, so Theorem 23.3 can be applied directly to the assembled symbol. This avoids illicit convolutions of arbitrary distributions. For these perturbative contact states the evolution action is defined by the admitted product $u_{i;ba}(s) M_i(s; e_a) = M_i(s; e_b)$, since the canonical matching block is endpoint independent. Both products lie in the constant-plus-L^2 class just described. This is an explicit

extension to these states, not an application of the L^2 theorem to $\delta^{(2)}$, and not an extension to every arbitrary tempered input. The same argument applies to each specified flavor channel; fragmentation factors, tensor structures, and measurements still require their own factorization inputs.

For proof, Fourier transformation of Eq. (363) gives the *complete*, infrared-completed canonical CSS2 product. It is exactly that product, including unexpanded coupling and PDF scale dependence, not a finite L_b dictionary applied to them. Therefore the transform-free spectral realization and this specified parent have identical bin actions. This exact representation plus Theorem 23.4 establishes the N^{3}LL$'$ *formal prescription*. It does not turn the delivered numerical demonstration into an N^{3}LL$'$ data extraction.

Polynomial-approximation, elliptic-grid, collinear-interpolation, and RG quadrature errors must be bounded separately from perturbative uncertainty. The continuum operator bound permits the approximation degree to be chosen so that, for example, $\epsilon_N = O(a_Q^4)$ on a stated parameter range. Relative errors are not promised in bins where the prediction vanishes. The Y term is formed from the perturbative leading-power expansion in the same scheme, with the nonperturbative module and completion-dependent power terms separated as specified in Sec. 32.

23.7 Reproducibility and the status of this construction

The companion v1.5 package supplies a generic retained-RG scalar engine, a momentum-space resolvent evaluator, and two distinct test categories. Symbolic/formal tests verify the canonical three-loop CS expansion, the first omitted logarithmic powers, and the infinite-tower LL expression. A numerical fixed-n_f demonstration uses explicitly listed lower-order QCD coefficients to test endpoint paths, composition, ultraviolet behavior, and the resolvent recurrence against an independently diagonalized *same finite momentum-grid operator*. That reference diagonalization is not used in the production recurrence, and is not a numerical Fourier transform. It validates the finite-matrix algorithm, not its continuum discretization error. Higher-order fixture coefficients, where used in formal-series tests, are marked as mathematical fixtures and never as physical matching data.

Generated reports, script hashes, coefficient choices, and library versions are included in the delivered package. No physical three-loop matching payload, external resummed-spectrum benchmark, or data fit is inferred from these tests. The mathematical completion in this section is the admitted unexpanded operator, its constructive continuum evaluation, and its conventional perturbative logarithmic accuracy for the stated class. Efficiency, full coefficient ingestion, and phenomenological validation remain distinct tasks. The later physical N^{3}LL/strict-NLO realization and its continuum tests are recorded in Sec. 39; they do not turn these original fixtures into physical three-loop matching data.

24 Method revision: fitting a CSS2 TMD directly in momentum space

The objective of the v1.6 revision was a usable inverse problem with the Collins/CSS2 TMD definition and its accuracy convention held fixed. Publication organization was secondary at that stage; the subsequent two-paper route is given in Sec. 39.6. Three layers are kept separate: the renormalized universal TMD operator; its deterministic numerical representation; and the process measurement and matching that turn TMDs into cross sections. The source results in Sec. 23 supply the first layer for their stated fixed-flavor, infrared-completed class. The new results below develop the second layer and test which probabilistic and kinematic properties can consistently be added at the third. The new propositions are developments of this revision; cited parent results and

numerical demonstrations are identified separately.

For a canonical input $S_{i/h,\theta}$ the retained forward map is

$$F^c_{i/h}(x;e) = M_{i/h}(x,T;e)S_{i/h,\theta}(x), \qquad T=-\nabla_k^2, \qquad e=(\mu,\zeta). \tag{364}$$

The exact scalar target M includes the complete RG solution and the longitudinal matching block in Eq. (351). A numerical approximation is made to this target, not to a truncated recoil multiplicity series. Its numerical error is measured separately from the retained QCD logarithmic order. The model could instead specify an admitted reference-scale TMD $A_0(x,\boldsymbol{k})$; then $F(e)=\mathbb{U}(e;e_0)A_0$ on the common domain. Such a boundary must satisfy the scheme, OPE, and domain conditions; an arbitrary normalized A_0 is not an operator definition of a TMD. Canonical inputs avoid the unbounded inverse of a smoothing reference operator.

The motivation from direct-k_T inverse fits [38] is correlated x–k_T flexibility and direct control of localization, regularity, and model normalization. Fourier representation alone adds neither statistical information nor a guaranteed computational advantage. The Parton Branching (PB) method and CASCADE illustrate other useful momentum-space evolution and recoil constructions [51, 52]. A published PB–CSS translation uses an extracted CS kernel [31]; it is relevant evidence for a connection, rather than proof that every PB implementation already supplies all Collins-scheme finite three-loop terms. Positive event weights, exact event kinematics, and a CSS2 accuracy label are distinct properties.

Two common identifications should therefore be avoided. Large-b_T dependence is not a pointwise condition exclusively on small k_T: its Fourier image is nonlocal, with large-k_T suppression only under appropriate regularity/OPE assumptions. Also, $q_T<Q$ characterizes the resummation region, not an exact scattering boundary. The kinematic bound is derived in Sec. 28.

25 Positivity, probability, and exact recoil: compatibility tests

25.1 The correct positivity criterion

The following standard Fourier-probability criterion is stated to fix precisely what a branching interpretation would require [53]. A continuous function $m(\boldsymbol{b})$ is the Fourier transform of a finite positive measure if and only if it is positive definite; its mass is $m(0)$. In particular,

$$\sum_{a,b}\bar{z}_a z_b m(\boldsymbol{b}_a-\boldsymbol{b}_b)\geq 0, \qquad |m(\boldsymbol{b})|\leq m(0). \tag{365}$$

A probability convolution has $m(0)=1$. A positive spectral multiplier $m(\boldsymbol{b})\geq 0$ instead gives positivity of a quadratic form on L^2; this is not positivity preservation of nonnegative momentum-space functions.

For a symmetric, strongly continuous probability convolution semigroup in rapidity interval $\rho\geq 0$, the necessary and sufficient characteristic form is

$$\begin{aligned} m_\rho(\boldsymbol{b}) &= e^{-\rho\psi(\boldsymbol{b})}, \qquad \psi(0)=0, \\ \psi(\boldsymbol{b}) &= \tfrac{1}{2}\boldsymbol{b}^T A\boldsymbol{b}+\int_{\mathbb{R}^2\setminus\{0\}}[1-\cos(\boldsymbol{b}\cdot\boldsymbol{p})]\,\nu(\mathrm{d}\boldsymbol{p}), \quad A\succeq 0, \end{aligned} \tag{366}$$

where ν is positive and satisfies $\int \min(1,p_T^2/\kappa_*^2)\nu(\mathrm{d}\boldsymbol{p})<\infty$ for a reference momentum $\kappa_*>0$. This is the symmetric Lévy–Khintchine criterion; a scale-dependent family can use ordered increments with positive rates. It is a test, not a consequence of two-scale RG integrability. Higher-order matching matrices and process hard functions are not included in this scalar probability statement.

Proposition 25.1 (Obstruction for the present uncapped canonical operator)**.** *Let $m(\boldsymbol{b}) = M(b_T^2)$ be a nonzero continuous canonical symbol satisfying $M(0) = 0$, as in Sec. 23.3. It is not the Fourier transform of any finite positive measure. Thus $M(T)$ is not a positive, finite-mass branching convolution. If the output $F = M(T)S$ is nonzero and integrable and $\int S = 1$, it cannot be nonnegative everywhere.*

Proof. For a point with $m(\boldsymbol{b}) \neq 0$, the two-point matrix in Eq. (365) has zero diagonal and eigenvalues $\pm|m(\boldsymbol{b})|$. It is not positive semidefinite. Equivalently a positive finite measure with zero total mass is zero. For the final assertion, $\widetilde{F}(0) = M(0)\widetilde{S}(0) = 0$, whereas a nonzero nonnegative integrable function has strictly positive integral. For distribution-valued outputs the integrability qualification is replaced by a justified total-cumulative action; a total integral is not assigned to arbitrary tempered distributions. □

This obstruction is intrinsic to the selected canonical completion and its uncapped ultraviolet endpoint, not to the numerical approximation. The same zero-integral issue for the CSS W term is analyzed in Ref. [48]. It does not say where the sign change occurs or whether a matched prediction is negative in the fit region. Clipping the evolved TMD to zero or passing it through a softplus would change its operator definition and evolution. Positivity of fitted cross-section bins, when imposed, is a constraint on the full matched prediction and parameter domain.

The endpoint propagator also lacks a generic probability interpretation. At fixed μ and $\rho > 0$, $u_\rho(s) = e^{-\rho D^c(s;\mu)}$ grows without bound as $s \to 0$ by Eq. (347); multiplying by a finite scalar cannot turn it into the bounded characteristic function of a finite measure. Even in a different ultraviolet completion, probability preservation would require Eq. (366), not just a nonnegative scalar exponent.

Lemma 25.2 (Positive inverse evolution is an additional obstruction)**.** *If a translation-invariant convolution operator and its inverse are both probability convolutions, each is a deterministic translation. For radial kernels the operator is the identity.*

Proof. If independent random variables X, Y with those probability laws satisfy $X + Y = 0$ almost surely, then their characteristic functions have unit modulus at every point, since their product is one and both moduli are at most one. Each law is concentrated at one point. Rotational invariance fixes that point to zero. Hence an invertible two-scale evolution group should not be identified with a diffusion or branching probability law in both directions. □

25.2 A positive proposal with an exact signed CSS2 correction

A limited branching interpretation can be used constructively without changing the TMD: factor a positive auxiliary proposal out of the exact kernel and retain a signed correction. For $0 < \epsilon < K$ and a fixed nonnegative strength A, set

$$\nu(\mathrm{d}^2\boldsymbol{p}) = \frac{A}{\pi p_T^2}\mathbf{1}_{\epsilon<p_T<K}\,\mathrm{d}^2\boldsymbol{p}, \qquad \lambda = 2A\ln(K/\epsilon),$$
$$\psi_{\epsilon K}(b) = 2A\int_\epsilon^K \frac{\mathrm{d}p}{p}[1 - J_0(bp)]. \tag{367}$$

Its exact probability kernel is the compound-Poisson measure

$$P_\rho = e^{-\rho\lambda}\sum_{n=0}^{\infty}\frac{\rho^n}{n!}\nu^{\otimes_T n}, \qquad \nu^{\otimes_T 0} = \delta^{(2)}, \qquad \int P_\rho = 1. \tag{368}$$

The $n = 0$ contact is the no-emission contribution. Its Fourier multiplier is $e^{-\rho\psi_{\epsilon K}}$. In the intermediate region $\epsilon b \ll 1 \ll Kb$,

$$\psi_{\epsilon K}(b) = A\ln(K^2b^2/b_0^2) + \mathcal{O}\Big(A(\epsilon b)^2\Big) + \mathcal{O}\Big(A(Kb)^{-3/2}\Big). \tag{369}$$

For $A = a_s\Gamma_0$ this reproduces the one-loop CS logarithm in that region. At small b it instead vanishes quadratically and thus is a different ultraviolet completion; Eq. (369) is not an all-order CSS2 conversion. The Bessel function only states its characteristic function; positive recoil convolutions or deterministic jump discretizations can realize the proposal.

At fixed virtuality the exact factorization is

$$e^{-\rho D^c(T;\mu)} = P_\rho \otimes_T C_\rho, \qquad \widetilde{C}_\rho(\boldsymbol{b}) = e^{-\rho[D^c(b_T^2;\mu)-\psi_{\epsilon K}(b_T)]}. \tag{370}$$

This follows by multiplication of the scalar symbols. Since $0 \le \psi_{\epsilon K} \le 2\lambda$, the extra factor $e^{\rho\psi_{\epsilon K}}$ is bounded for finite ρ; the admitted logarithmically weighted endpoint domain is preserved. The scalar virtuality factor is restored for general endpoint paths. The correction is retained unexpanded if an exact representation is claimed. Replacing it by one, or by a finite unassessed expansion, loses the exact CSS2 content. Its signed nature is required by Proposition 25.1. Numerical conditioning and variance reduction from this factorization remain to be tested.

Finite multiplicity truncation changes Eq. (368). For a pure proposal its missing mass is the explicit Poisson tail; that is an algorithmic error, not automatically a statement about retained logarithmic towers. Also, bounded individual jumps do not bound the total recoil: arbitrarily many nearly aligned jumps have nonzero probability. Exact event energy and longitudinal momentum constraints require an extended state or a process measurement, not a universal convolution depending on transverse recoil alone. PB and shower kinematics can guide that extended construction; a Collins-scheme matching/conversion is then required before importing their probabilistic advantages.

26 A deterministic shared-resolvent evaluator for direct fitting

Status update. This section retains the distinct-shift rational construction developed for v1.6. The accepted high-resolution benchmarks now use the Chebyshev-resolvent path in Sec. 39.2. The rational results and their error hypotheses remain valid for their own approximation; they are not reassigned to the newer numerical sequence.

26.1 Replacing the production role of Bernstein polynomials

The Bernstein construction of Sec. 23.4 remains an existence and contact-control proof. It is not selected as the production algorithm. For quarks at $n_f = 5$,

$$p_q = \frac{\Gamma_0^q}{2\beta_0} = \frac{8}{23} \simeq 0.3478. \tag{371}$$

Taking the optimistic limiting exponent $\nu \to p_q$ and setting the unknown Hölder constant to one, the sufficient Bernstein bound gives $N \gtrsim 7.6\times10^6$ at absolute symbol error 0.05 and $N \gtrsim 7.9\times10^{10}$ at 0.01. These are illustrative worst-case bounds, not measured costs or relative cross-section accuracies. The operator de Casteljau recurrence as written also has $O(N^2)$ right-hand-side resolvent applications. Alternative polynomial evaluations reduce that count but do not remove the small endpoint-regularity exponent.

We instead construct an endpoint-constrained linear combination of distinct resolvents, motivated by rational approximation and parallel elliptic solution methods [54, 55]:

$$M_J(s;\lambda) = M_\infty(\lambda) + \sum_{j=1}^{J} c_j(\lambda)\frac{\tau_j}{\tau_j + s}, \qquad \tau_j > 0, \tag{372}$$

$$M_J(T;\lambda)S_\theta = M_\infty(\lambda)S_\theta + \sum_j c_j(\lambda) R_{\tau_j} S_\theta. \tag{373}$$

Here λ collects x, hadron/parton labels, μ, ζ, and any fitted kernel parameters; it is not a recoil probability rate. The pole set is fixed across a declared kinematic region. Coefficients change with the target symbol, whereas elliptic matrices and their factorizations are shared. Distinct solves are independent and admit parallel or batched evaluation. Residues are signed. A nonnegative-residue assumption would replace the allowed multiplier class by a much smaller Stieltjes-like class and would fail for the current zero endpoint.

For the constant-infrared symbol of Sec. 23, enforce

$$\sum_j c_j = M(0) - M_\infty = -M_\infty, \qquad \sum_j c_j \tau_j = 0. \tag{374}$$

The first condition preserves the zero spectral endpoint; $M_J(\infty) = M_\infty$ preserves the delta coefficient automatically. The optional second condition eliminates the $1/s$ asymptotic error and gives a regular tail of order s^{-2}. A modified process symbol with a different finite value at zero uses that value on the right-hand side; it is not silently forced to zero. Nonzero approximation error at an endpoint must be included in the error budget rather than interpreted as a change of normalization.

Theorem 26.1 (Weighted rational error and contact preservation)**.** *Let $M - M_\infty$ be continuous, bounded and zero for sufficiently large s, and let M_J have the same contact constant. Define*

$$E_J(s) = M_J(s) - M(s), \qquad \varepsilon_J = \sup_{s\geq 0}(1 + s/\tau_*)|E_J(s)|, \quad \tau_* > 0. \tag{375}$$

Whenever the displayed supremum is finite,

$$\begin{aligned} \|(M_J - M)(T)f\|_2 &\leq \varepsilon_J \|f\|_2, \qquad f \in L^2, \\ \|(M_J - M)(T)\delta^{(2)}\|_2^2 &= \frac{1}{4\pi}\int_0^\infty |E_J(s)|^2 \mathrm{d}s \leq \frac{\tau_*}{4\pi}\varepsilon_J^2. \end{aligned} \tag{376}$$

The finite-bin error for the contact action is bounded by $\sqrt{|\mathcal{B}|\tau_*/(4\pi)}\varepsilon_J$. For two ordinary legs the same Cauchy–Schwarz argument as before propagates their L^2 errors into a finite-bin convolution error. No positivity of the residues or target is assumed.

Proof. The spectral theorem bounds the ordinary action by $\|E_J\|_\infty$. For the contact state, Plancherel with the declared Fourier normalization and $\mathrm{d}^2\boldsymbol{b} = \pi \mathrm{d}s$ gives the equality. The weighted bound and $\int_0^\infty (1 + s/\tau_*)^{-2}\mathrm{d}s = \tau_*$ prove the estimate. Both approximate and exact contact coefficients equal M_∞, so their error has no residual delta term. Integrating its L^2 remainder over a finite bin gives the last assertion. □

Proposition 26.2 (Existence of constrained shifted-resolvent approximations)**.** *For a continuous h in Eq. (352), finite signed sums in Eq. (372) can approximate M arbitrarily well in the weighted norm of Eq. (375), while satisfying both endpoint conditions in Eq. (374).*

Proof. Put $r = \tau_*/(\tau_* + s)$. Dividing a shifted resolvent by r gives

$$\phi_\tau(r) = \frac{\tau}{\tau_* + (\tau - \tau_*)r}. \tag{377}$$

For $\tau = \tau_*(1 + a)$ near $a = 0$, the uniform Taylor series of $(1 + a)/(1 + ar)$ has coefficients that are consecutive differences of powers of r. Uniform limits of finite differences in a therefore put every polynomial in the closed linear span of the ϕ_τ. Polynomial approximation on $[0, 1]$ proves density in $C[0, 1]$. Two fixed basis functions with distinct τ have linearly independent endpoint-value pairs $(\phi_\tau(0), \phi_\tau(1)) = (\tau/\tau_*, 1)$. Small linear corrections using those functions enforce $h_J(0) = 0$ and $h_J(1) = h(1)$, with size tending to zero as the original uniform error tends to zero. Multiplication by r then gives the two endpoint conditions and the weighted convergence. □

This density proof establishes possibility, not a useful rate or numerical conditioning. Pole clustering can be very effective for endpoint powers, but the smooth, nonanalytic infrared splice and PDF dependence of the full RG symbol need their own measured or certified errors. A root-exponential rate for a different fractional-operator class is not asserted for every CSS2 symbol here.

26.2 Error control, stability, and differentiation

For a target constant above s_0 and $s_R \geq s_0$, let $A_1 = |\sum c_j\tau_j|$ and $A_2 = \sum |c_j|\tau_j^2$. The rational tail obeys

$$\sup_{s \geq s_R} (1 + s/\tau_*)|E_J(s)| \leq (1/s_R + 1/\tau_*) \left[A_1 + \frac{A_2}{s_R + \tau_{\min}}\right]. \tag{378}$$

This follows by subtracting $\sum c_j\tau_j/s$ from the resolvent sum and bounding the remaining denominators. It includes any finite arithmetic violation of the tail constraint. Near $s = 0$, an exact endpoint match and a proved target bound $|M(s) - M(0)| \leq Cs^\nu$ give $|E_J(s)| \leq Cs^\nu + s\sum |c_j|/\tau_j$. On a finite interval a rigorous supremum can be bounded with derivative/interval estimates or an enclosure method. A dense sampled residual alone is explicitly called *observed*, not certified.

If an approximate shifted solve has residual $\eta_j = \tau_j f - (\tau_j + T)\widehat{u}_j$, then

$$\left\| \sum_j c_j(\widehat{u}_j - R_{\tau_j} f) \right\|_2 \leq \sum_j \frac{|c_j|}{\tau_j} \|\eta_j\|_2. \tag{379}$$

Thus large signed residues amplify solver and roundoff errors even when the scalar approximation looks excellent. Pole selection, truncated-SVD or other regularization, coefficient cancellation, and residual tolerances are part of the numerical method. Stability is checked on gradients as well as predictions.

With fixed shifts and coefficient parameters λ independent of input parameters θ,

$$\partial_{\theta_a} F_J = M_J(T; \lambda)\partial_{\theta_a} S_\theta. \tag{380}$$

If λ is also fitted, the additional terms are $(\partial_\lambda M_\infty)S + \sum_j (\partial_\lambda c_j)R_{\tau_j} S$. Holding the sampling nodes, shifts, and linear coefficient solver fixed during a training stage makes the residue map differentiable. Adaptive rank or pole changes are outer preparation steps, not differentiated as though they were smooth. Errors in parameter derivatives of M require their own estimates.

For radial inputs the deterministic solves are one-dimensional:

$$\left[\tau_j - \frac{1}{k_T}\partial_{k_T} k_T \partial_{k_T}\right] u_j(k_T) = \tau_j S(k_T), \tag{381}$$

with regularity/no flux at the origin and a controlled outer-domain treatment. The natural norm uses $2\pi k_T \mathrm{d}k_T$, not the unweighted radial-grid norm. A cell-volume finite-volume discretization leads to a positive symmetric matrix after the square-root-volume similarity transform; its adjoint uses the same metric. One finite box gives neither a physical k_T cutoff nor a continuum certificate. Small shifts describe long-range responses and can require a large numerical box. Shift factorizations, preconditioners, and responses for an identical input can be reused; the symbol residues still vary with x, Q, μ, ζ.

26.3 Accuracy connection and acceptance criterion

The exact target remains the Section 23 solution with its physical retained coefficients, canonical boundary convention and infrared completion. Taking a controlled rational/grid/quadrature limit therefore preserves its CSS2 definition and formal logarithmic towers. At finite numerical resolution the prediction carries an additional absolute error estimate; a rational fit to a few nodes is not independently an $\mathrm{N^3LL'}$ proof. For unbounded endpoint ratios use the admitted cutoff/range-space treatment already given, rather than applying the bounded-symbol theorem to them without checking its hypotheses. There is no truncation of a multi-emission series in this evaluation route.

Acceptance is based on measured or bounded error versus computational cost in specified bins and gradient directions. The first comparison holds the scalar symbol, infrared completion, longitudinal inputs, and fitted input fixed and varies only the evaluation algorithm. A second comparison changes the input parameterization while holding the physical theory fixed. Their effects must not be conflated. FFTLog and Bessel-zero quadratures are important additional competitors [56, 57]; a continuum adaptive Hankel reference is included in the first limited study below.

27 Nonperturbative input, prescription changes, and identifiable structure

The spectral fit has the equivalent representations

$$F(x, \boldsymbol{k}; e) = M(x, T; e) S_\theta(x, \boldsymbol{k}), \qquad \widetilde{F}(x, \boldsymbol{b}; e) = M(x, b_T^2; e) \widetilde{S}_\theta(x, \boldsymbol{b}). \tag{382}$$

Thus the convolutional model is the usual perturbative factor times a nonperturbative factor, with a different choice of function coordinates and regularization. A parameter bijection between representations leaves predictions, likelihood, and local information content unchanged. Finite neural architectures, envelopes, positivity constraints and training regularizers can instead select different model classes. Their bias, identifiability, calibration and optimization must be tested, not inferred from the representation. Impact-parameter neural fits are therefore direct comparators [30]. The newer Fernando–Keller impact-parameter fit also explicitly separates the neural factor from retained perturbative inputs [58]; its published setup is a comparator, not input validation performed in this revision.

A sufficient k_T neural class remains

$$S_\theta(x, \boldsymbol{k}) = \frac{\omega(k_T^2) e^{n_\theta(x, k_T^2)}}{2\pi \int_0^\infty k \, \mathrm{d}k \, \omega(k^2) e^{n_\theta(x, k^2)}}. \tag{383}$$

For a fixed smooth compact envelope, smooth bounded network outputs on that support, and normalization bounded away from zero, $S_\theta \in C_c^\infty$. It has finite moments and a controlled L^2 derivative. Then

$$\partial_{\theta_a} S_\theta = S_\theta [\partial_{\theta_a} n_\theta - \langle \partial_{\theta_a} n_\theta \rangle_{S_\theta}]. \tag{384}$$

This is normalization of an input component, not of the renormalized TMD. A finite-support input does not truncate the perturbative output tail.

For $S \in C_c^\infty$ its Fourier image decreases faster than every inverse power along real $\boldsymbol{b}$, by repeated integration by parts. Algebraic oscillatory tails follow from a sharp cutoff or limited smoothness, not from smooth compact support itself. The transform is also an entire function of exponential type; compact support is a substantive functional prior. If $S \geq 0$ and $\int S = 1$, $\widetilde{S}$ is positive definite and $|\widetilde{S}| \leq 1$, but it need not be pointwise nonnegative. A b_T network constrained to be pointwise positive does not represent exactly the same class. Oscillations and zeros, envelope-scale variation, and the universal nonperturbative CS factor all affect the fitted long-distance behavior. Large-b_T changes have a nonlocal k_T image.

27.1 Keep the physical TMD fixed when varying the infrared split

Let completions A and B have the same perturbative cusp and define $\Delta D = D^B_{\rm pert} - D^A_{\rm pert}$. Their RG equations imply $\partial_{\ln\mu}\Delta D = 0$. The same total CS kernel is represented by

$$D^B_{\rm NP} = D^A_{\rm NP} - \Delta D. \tag{385}$$

At a common reference endpoint e_r, impose equality of the full TMD boundaries. For the multiplicative class with the nonperturbative CS factor referenced to ζ_r, this requires, wherever the quotient and domains exist,

$$\widetilde{S}^B(x,s) = \frac{M^A(x,s;e_r)}{M^B(x,s;e_r)}\widetilde{S}^A(x,s). \tag{386}$$

Equality of total CS kernels and reference TMDs then propagates to all endpoints by the common RG equations. This is a split covariance identity, not a guarantee that the quotient preserves positivity, compact support, or a chosen neural architecture. Zeros or an unbounded quotient require a different admitted boundary parameterization. Repeating a fit with the same restricted positive input ansatz under two completions is consequently a model/split sensitivity study, not an exact change of coordinates automatically removed by Fourier duality.

Nonperturbative CS effects can be learned within an admitted common partonic universality class, with their parameter derivatives included in the symbol fit. The heat kernel in Eq. (299) is a valid forward example. Its positivity does not repair the canonical zero endpoint: multiplying by a factor equal to one at $s = 0$ leaves $M(0) = 0$. Backward heat evolution requires its own restricted domain.

28 Exact support and process-level matching without changing the TMD

28.1 The kinematic limit is not $q_T < Q$

Let S denote the squared hadronic center-of-mass energy, distinct from the spectral variable $s = b_T^2$. In the hadronic center-of-mass frame, $q^0 = m_T \cosh y$, $q^z = m_T \sinh y$, $m_T^2 = Q^2 + q_T^2$, so

$$M_X^2 = (P_A + P_B - q)^2 = \mathsf{S} + Q^2 - 2\sqrt{\mathsf{S}}\, m_T \cosh y. \tag{387}$$

The necessary inclusive condition $M_X^2 \geq M_{X,\min}^2$, with positive remnant energy, gives

$$q_T^2 \leq \left[\frac{\mathsf{S} + Q^2 - M_{X,\min}^2}{2\sqrt{\mathsf{S}}\cosh y}\right]^2 - Q^2. \tag{388}$$

Masses, PDFs, cuts, fragmentation variables and recoil definitions can further restrict the available phase space. For a massless minimal remnant at $y = 0$, $q_{T,\max} = (\mathsf{S} - Q^2)/(2\sqrt{\mathsf{S}})$, which can exceed Q. Therefore a universal $k_T < Q$ or $q_T < Q$ cutoff would remove allowed hard recoil and is not a kinematic completion of a TMD. The exact fixed-order calculation uses its actual partonic fractions rather than the low-q_T Born projection.

28.2 A matched measurement with exact support

Keep the universal TMD F^{CSS2} and its endpoint evolution fixed. Denote the process measurement and support by Π_{phys}. Choose a smooth turnoff $X_Q(q_T)$, independent of the coupling, equal to one in the selected TMD region and zero in the hard region. With the subtraction consistently defined on the same measured bins, use

$$\mathrm{d}\sigma_{\mathrm{match}} = \mathrm{d}\sigma_{\mathrm{FO}}^{[m]} + \Pi_{\mathrm{phys}} X_Q \left(\mathrm{d}\sigma_{W,\mathrm{pert}}^{\mathrm{res}} - [\mathrm{d}\sigma_{W,\mathrm{pert}}]_m + \Delta\mathrm{d}\sigma_{W,\mathrm{NP}} \right). \tag{389}$$

The fixed-order term includes the same physical support and cuts. At small q_T this reduces to the usual resummed expression plus its nonsingular remainder; in the turned-off region it reduces to fixed order. Expanding the perturbative bracket through a_s^m gives zero. Thus the support operation can be exact while the dynamics retains its explicitly stated leading-power and fixed-order accuracy. It does not reconstruct all subleading-power terms from the universal TMD. No per-leg kinematic truncation or redefinition is involved. Coupling-dependent profile variations, when used, must also be expanded in the same subtraction.

At finite perturbative order the matched bins are not guaranteed positive. Their signs, cancellation sensitivity and perturbative uncertainty are diagnostics of the chosen domain and matching. A useful binwise quantity is

$$\kappa_{\mathcal{B}} = \frac{|W_{\mathcal{B}}| + |Y_{\mathcal{B}}|}{|W_{\mathcal{B}} + Y_{\mathcal{B}}|}, \tag{390}$$

with absolute-error reporting when the denominator is zero or small. Clipping components would spoil the controlled expansion. A physical rate constraint or unitarity-restoring remainder requires an explicit matched prescription and its perturbative/power counting, rather than identifying $M(0)$ with a collinear PDF.

28.3 An optional process small-distance completion

The uncapped canonical W has zero total integral when that integral is justified, and therefore can require cancellations in the transition region [48]. A remedy can be applied to the *process approximation* without redefining the universal TMD. For example, let $t_Q = c/Q^2$ and choose a smooth positive map $\sigma_Q(s)$ that equals t_Q for $s \le t_Q$, equals s for $s \ge 2t_Q$, and interpolates between. Define

$$\widetilde{W}_{\mathrm{pc}}(s;Q) = \widetilde{W}(\sigma_Q(s);Q). \tag{391}$$

This entire product is a process-level measurement approximation. The functions $F(x,\boldsymbol{k};\mu,\zeta)$ and their CS kernel remain the ones defined earlier; an individual factor evaluated at σ_Q must not be advertised as an unchanged universal TMD. At finite Q the new total endpoint is $\widetilde{W}_{\mathrm{pc}}(0) = \widetilde{W}(t_Q)$, but its positivity-preserving kernel is still not automatic.

Proposition 28.1 (A sufficient leading-power condition for this completion)**.** *Suppose* $\Delta\widetilde{W} = \widetilde{W}_{\mathrm{pc}} - \widetilde{W}$ *vanishes for* $s \ge 2c/Q^2$ *and*

$$\int_0^{2c/Q^2} |\Delta\widetilde{W}(s;Q)|\mathrm{d}s \le C(Q)/Q^2. \tag{392}$$

Then its ordinary transverse image satisfies $|\Delta W(q_T;Q)| \le C(Q)/(4\pi Q^2)$. *If its finite-order coefficient counterparts satisfy the analogous integrable logarithmic bounds, they are analytic in* q_T^2 *near zero and change only nonsingular terms at that order.*

Proof. Use the radial inverse transform $\Delta W = (4\pi)^{-1}\int \mathrm{d}s\, J_0(q_T\sqrt{s})\Delta\widetilde{W}$ and $|J_0| \le 1$. On its compact support the Bessel series and all its moments are integrable under the coefficient assumptions, giving an expansion in nonnegative powers of q_T^2. There is no contact term from a spectral difference that is integrable and supported at finite s. These are sufficient conditions and must be verified for the completed process expression and scale variation. □

Consequently this optional completion can preserve the singular logarithmic content while reorganizing power and nonsingular terms. Its own fixed-order expansion replaces $[W]_m$ in Eq. (389); the same completion is used on both sides of the comparison. The operator/resolvent method also evaluates this completed radial symbol, with its actual value at zero imposed in the coefficient constraints. This is a defined candidate for comparison with standard improved matching prescriptions, not a claim that its phenomenological parameters, inclusive rate or positivity have been validated.

The next physical benchmark should compare baseline and completed W, their consistent Y terms, matched spectra and integrated rates against a common SCETlib or DYTurbo setup. It must use the same fixed-order channels, PDFs, cuts, scales and perturbative order. At low Q the required overlap region may be narrow or absent; adding exact support alone cannot cure the power expansion.

29 Executed method study and revised completion criteria

The preserved v1.6 `verification/run_study.py` study evaluates the retained lower-order massless $n_f = 5$ quark RG symbol from the earlier code bundle, with $\mu_0 = 2\,\mathrm{GeV}$, $a(\mu_0) = 0.02$, and canonical matching block $B = 1$. The coefficient fixture contains $\beta_{0,1}$, $\Gamma_{0,1}$, γ_0 and d_2; it is explicitly not a physical N^{3}LL$'$ payload. The existing scalar module is retained except for an endpoint-only input bug: a call containing only $s = 0$ now returns its known zero limit without asking the ODE interpolation routine to evaluate an empty array. The original and revised module hashes and the small patch are supplied with the new tests.

The study uses $S(k) = e^{-k^2/v}/(\pi v)$ with $v = 0.9\,\mathrm{GeV}^2$. The production path is the one-dimensional weighted finite-volume rational evaluator. For an independent continuum reference, the exact Gaussian transform and analytic bin integration give

$$F_{[a,b]} = M_\infty(e^{-a^2/v} - e^{-b^2/v}) + \int_0^{\sqrt{s_0}} \mathrm{d}\beta\, [M(\beta^2) - M_\infty] e^{-v\beta^2/4}[bJ_1(\beta b) - aJ_1(\beta a)]. \tag{393}$$

Here a, b are momentum bin edges and β is the radial dual coordinate. The formula is used only as a validation reference. It uses exactly the same symbol and infrared completion as the direct evaluator, and keeps the contact contribution analytic. The reference gradient follows by differentiating both the Gaussian bin term and the exponential. It is adaptive continuum Bessel quadrature, not a comparison to FFTLog or Ogata and not a physical DY benchmark.

The tested edges are $(0, 0.5, 1, 2, 4, 8)\,\mathrm{GeV}$. Shifts are log-spaced from $10^{-12}s_0$ to $10s_0$. A null-space constrained linear solve fits the weighted symbol, enforcing Eq. (374). The default relative singular-value cutoff is 10^{-8}. Independent dense check nodes measure, rather than rigorously enclose, the symbol residual. Table 7 shows selected actual outputs; the full CSV includes additional spatial refinement.

Q [GeV]	J	R [GeV]	Δk [GeV]	max bin error	max gradient error
5	24	80	0.05	0.000943	0.000291
5	48	80	0.05	0.000498	0.000318
5	72	80	0.05	0.000357	0.000319
20	24	80	0.05	0.00581	0.000976
20	48	80	0.05	0.00417	0.000778
20	72	80	0.05	0.00221	0.000425

Table 7: Direct-evaluator differences from continuum single-leg bin actions. Values are absolute errors in the declared fixture; gradients are with respect to v in GeV^2.

The results reveal, rather than hide, stability issues. An initial near-unregularized fit with relative SVD cutoff 2×10^{-13} produced residues with an absolute sum near 1.6×10^7 at $Q = 20$ GeV, $J = 72$, and failed the declared finite-difference gradient check. The 10^{-8} cutoff reduced the residue sum to approximately 3.15×10^5 and passed that same check. The earlier failed run is retained in a separately labeled pilot directory. Significant cancellation remains, and the observed symbol residual and bin error are not monotonic in every combination of degree and box size. The v1.6 prototype is a deterministic, differentiable method study, not an optimized or globally certified evaluator. The missing certification and efficiency tests are specific numerical tasks.

29.1 A direct-input two-leg closure check

A further test uses identical inputs on the two legs, $S_\theta = \theta G_{v_1} + (1-\theta) G_{v_2}$, with $v_1 = 0.5\,\mathrm{GeV}^2$, $v_2 = 1.4\,\mathrm{GeV}^2$ and true $\theta = 0.37$. The scalar target is $M(s)^2$ at $Q = 5$ GeV, with unit luminosity and unit hard function. The input convolution is the exact combination of Gaussians of variances $2v_1$, $v_1 + v_2$, $2v_2$ and weights $(\theta^2, 2\theta(1-\theta), (1-\theta)^2)$. Noiseless reference bin moments are generated by Eq. (393) applied to that product. Only the input mixture coefficient is fitted, using precomputed shifted responses.

J	recovered θ	bias	max bin residual
24	0.371569904	0.00157	0.0005488
48	0.369494121	−0.0005059	0.0004117
72	0.369712439	−0.0002876	$5.462e-05$

This is a mathematical closure experiment for a shared-input scalar convolution, not experimental data or a physical DY likelihood. The reference moments in $[3, 4]$ and $[4, 6]$ GeV are negative. Their signs are retained, exposing the uncapped singular operator's limitation rather than turning them into positive pseudo-cross-sections. Physical fitting requires the measured-domain and matching analysis in Sec. 28. The test demonstrates that the input can be optimized in k_T through a deterministic fixed operator and that numerical approximation shifts its recovered parameters. A full DNN fit and its identifiability are separate tasks.

The final executable study reports 34/34 checks of endpoint constraints, finite-matrix bounds, input gradients, closure optimization, the positivity counterexamples, the compound-Poisson proposal, its exact signed correction and kinematic support identities. Finite-dimensional Gram tests are examples, not proof of positive definiteness for every set of points; the proposal's proof is Eq. (368). The suite does not convert sampled errors or successful optimization into a physical-accuracy certificate. Source hashes, coefficient choices, dependencies, all measured residuals and machine-dependent timings are recorded in JSON/CSV. Reference quadrature errors are algorithmic estimates; the sampled rational residuals are not global enclosures.

29.2 Priority sequence for completing a usable extraction

The immediate numerical target at the v1.6 stage was a stable shared-pole or hybrid evaluator with controlled endpoint, contact, gradient and continuum/bin errors over a declared kinematic range. Pole adaptation, a smoother split or alternative rational bases must be compared at fixed physical symbol before changing the theory. Next, compare against optimized FFTLog/Ogata evaluations of that same symbol and input, including setup cost, repeated batches, gradient accuracy and memory. In parallel, implement the actual perturbative matched process and measure transition-region cancellation and rate stability. The optional small-distance process completion requires the bound and subtraction checks stated above. No benchmark should quietly change the PDF, infrared prescription or perturbative order.

At that stage the remaining physical tasks were convention-checked coefficient ingestion, independent perturbative spectrum comparisons, controlled nonperturbative split variation, and data-level closure. A positive auxiliary branching implementation with signed correction is an additional research route, not a prerequisite for fitting in k_T and not a replacement for the CSS2 operator. The present uncapped operator has a proved obstruction to an everywhere positive finite branching interpretation. Exact observable support can nevertheless be imposed at the process layer. This is the corrected separation of what is preserved, what has been constructed, and what still requires a quantitative test. The accepted canonical N^{3}LL/strict-NLO comparisons in Sec. 39 now resolve part of this earlier list; full matching and global extraction remain separate obligations.

30 Physical high-order coefficient import gate

Section 22 defined the abstract coefficient-table contract and used a synthetic payload only to validate the map

$$L_b^0 \mapsto \delta^{(2)}(\boldsymbol{k}), \qquad L_b^m \mapsto \mathfrak{L}_{m-1}(\boldsymbol{k};\mu), \quad m \geq 1. \tag{394}$$

That test is necessary, but it is not physics input. The physical-import layer introduces the hard gate that separates a mock interface test from a source-stamped perturbative coefficient package. A run may not advertise an $N^3\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$ boundary unless the imported three-loop matching payload carries provenance, convention metadata, a scheme map to CSS2, and a successful fixed-order expansion check.

Working decision. Synthetic coefficient tables are allowed only for importer regression tests. They are not allowed to satisfy the hard, matching, or expansion requirements of an $N^3\mathrm{LL}'$ physics claim. If the raw ancillary files or a verified normalized export are absent, the physical import gate must remain blocked.

30.1 Registered source families

The first source registry is deliberately conservative. It contains the NNLO analytic reference used for lower-order checks and the two public N^{3}LO source families needed for the unpolarized quark, antiquark, and gluon matching interface.

Source key	Order and legs	Local payload status	Role in the $\mathcal{K}_{\mathrm{CSS2}}$ importer
ESV2016	NNLO PDF/FF	paper-level reference	Analytic NNLO reference for unpolarized TMDPDF and TMDFF matching in all flavor combinations; used for the $\mathcal{O}(a_s^2)$ gate and lower-order checks of any N^{3}LO import.
EMV2020	N^{3}LO PDF	ancillary files expected	Independent N^{3}LO TMDPDF/beam-function source for quark and gluon initial-state channels. Registered files include `TMDPDF.m`, `PTBF.m`, `PTBF_ZExpansion.m`, `PTBF_ZbExpansion.m`, and soft-function files.
LYZZ2021	N^{3}LO PDF/FF	ancillary files expected	Primary all-leg unpolarized matching source for PDFs and FFs. Registered files include `TMDPDF.m`, `TMDPDFN.m`, `BeamFunction.m`, `BeamfunctionN.m`, `TMDFF.m`, `TMDFFN.m`, `FFMatchingKernels.m`, `FFMatchingKernelsN.m`, `ancillary_readme.pdf`, and `softfunction.m`.

This registry is not itself a coefficient table. It is a provenance and convention layer that decides which external payloads are eligible to be converted into canonical coefficient records. The public arXiv records for EMV2020 and LYZZ2021 list the relevant ancillary files, while the publications describe the N^{3}LO unpolarized TMDPDF/TMDFF matching calculations [20, 22].

30.2 Source stamp and hash protocol

Each physical coefficient payload must carry a source stamp

$$S_C = \{\text{source}, \text{arXiv version}, \text{journal record}, \text{file name}, \text{SHA256}, \text{parser version}, \text{scheme map}\}. \tag{395}$$

The importer must reject a payload if

$$\text{source} \in \{\text{mock}, \text{synthetic}, \text{placeholder}, \text{toy}\} \tag{396}$$

or if the source is not convertible to the CSS2 parent convention. In other words, the v0.14 mock table validates the map in Eq. (394), but only a source-stamped physical table validates the perturbative content.

The accepted coefficient package should use a local read-only layout such as

```
coefficients/
  2012.03256/
    TMDPDF.m
    TMDPDFN.m
    BeamFunction.m
    BeamfunctionN.m
```

```
    ancillary_readme.pdf
    ...
  2006.05329/
    TMDPDF.m
    PTBF.m
    PTBF_ZExpansion.m
    PTBF_ZbExpansion.m
    info.txt
    ...
```

A released analysis should archive the exact source package or a normalized export with hashes. Live network access should not be required to reproduce a fit.

30.3 Canonical record schema

Most public high-order matching tables are distributed as Mathematica ancillary files. The extraction code should not evaluate arbitrary Mathematica programs during a fit. The preferred contract is a two-stage conversion,

$$\texttt{*.m source} \longrightarrow \texttt{canonical JSON/YAML coefficient table} \longrightarrow \mathcal{K}_{\mathrm{CSS2}} \text{ operator.} \tag{397}$$

The canonical table exposes records

$$\mathcal{R}_C = \{n, \text{object}, i \leftarrow j, m, r, c_{i\leftarrow j}^{(n,m,r)}(z), S_C\}, \tag{398}$$

where n is the loop order, m is the power of L_b^m, r labels rapidity-logarithm structure such as powers of ℓ_ζ, and S_C is the source stamp. The z-space object may remain symbolic, provided the importer can apply it to test functions and can check DGLAP and CS logarithmic recursions.

30.4 Acceptance predicate

The accepted physical payload for an $N^3\mathrm{LL}'$ claim must satisfy

$$\begin{aligned}\mathrm{Accept}_{N^3\mathrm{LL}'} = \mathrm{Source} \wedge \mathrm{Hash} \wedge \mathrm{NotMock} \wedge \mathrm{Scheme} \wedge \mathrm{Power} \wedge \mathrm{Flavor} \\ \wedge\ \mathrm{RG} \wedge \mathrm{Expansion}.\end{aligned} \tag{399}$$

The individual gates are:

1. **Source/hash:** the raw or normalized file has a documented source and SHA256 hash;
2. **NotMock:** the source is not synthetic or placeholder data;
3. **Scheme:** the payload is already CSS2 or has an explicit finite conversion to CSS2;
4. **Power:** an n-loop coefficient has no power higher than L_b^{2n};
5. **Flavor:** the flavor basis covers the process class being claimed;
6. **RG:** the logarithmic terms satisfy the CS, μ-RG, and DGLAP reconstruction equations;
7. **Expansion:** the fixed-order expansion of the imported W operator agrees with the reference singular spectrum through the claimed order.

30.5 NNLO then N3LO staged replacement

The physical replacement of the mock payload should be staged:

1. **NNLO gate:** import the vetted two-loop coefficient package and verify the $\mathcal{O}(a_s^2)$ singular W expansion, including all one-loop products, $H^{(1)}C^{(1)}$, $C^{(1)} \otimes_T C^{(1)}$, $P^{(1)}$, $D^{(2)}$, and $H^{(2)}$ terms.

2. **N^{3}LO gate:** import the vetted three-loop coefficient package and verify the $\mathcal{O}(a_s^3)$ singular expansion, including $H^{(3)}$, $C^{(3)}$, three-loop DGLAP, three-loop CS evolution, and all lower-order products generated by the convolution exponential.

The expansion check is a binned or test-function equality,

$$\int_{\text{bin}} \mathrm{d}^2\boldsymbol{q} \left[W^{\text{imported}}_{\mathcal{K}_{\text{CSS2}}} - W^{\text{singular}}_{\text{CSS2}} \right]_{\mathcal{O}(a_s^n)} = 0, \qquad n = 2, 3, \tag{400}$$

for every channel, flavor basis element, and chosen test luminosity.

30.6 Executable physical-payload audit status

The companion script `kt_kcss_physical_coeff_import_validator.py` implements the source-level audit. It does not load the synthetic coefficient payload. It checks whether the expected EMV2020 and LYZZ2021 ancillary files are present in the local coefficient staging directory and records whether the higher-order expansion gate can be enabled.

The audited package does not contain a parsed, convention-checked physical NNLO/N^{3}LO hard/matching boundary payload. Its recorded physics status is

> Source registry: defined. Mock payload: disabled for physics. Validated physical NNLO/N^{3}LO boundary payload: absent from the audited package. Physical NNLO/N^{3}LO expansion gates: specified but not yet executed.

This is a deliberate hard stop. It prevents the note from overclaiming that physical three-loop matching coefficients have been imported when only the interface algebra has been tested. The sourced four-loop beta/cusp, three-loop noncusp/CS, and strict-NLO hard/OPE realization added in Sec. 39 is a separate completed input layer and does not unlock this primed-order gate.

31 N3LL-prime label certificate

The previous sections define the ingredients and the physical import gate. We now make explicit what it means for a calculation, a code path, or an extraction to carry the label $\mathrm{N^3LL}'_{\mathcal{K}_{\text{CSS2}}}$. The goal is to make the eventual paper's claim precise: the paper may present a generic N^{3}LL$'$ prescription in k_T space, while any numerical implementation or data extraction must satisfy the concrete certificate below.

31.1 Representation, resummed prescription, and implementation

Three statements must be separated. A fixed-order CSS2 representation is proved by the distribution dictionary. A transform-free N^{3}LL$'$ *resummed prescription* additionally requires a defined ordered evolution solution and an all-order logarithmic accuracy argument. A numerical implementation then requires physical inputs and independent numerical validation. The second statement is not established merely by writing an ingredient manifest for the third.

In particular, a process cross section contains two universal TMD legs,

$$W = \sum_{ij} H_{ij\to F}\, \mathcal{M}_F[\mathbb{F}_{i/A}, \mathbb{F}_{j/B}], \qquad \mathbb{F}_{i/h} = \mathcal{U}_i \otimes_T (\mathcal{C}_{i\leftarrow a} \otimes_x f_{a/h}), \tag{401}$$

with the process measurement $\mathcal{M}_F$ defined as in Sec. 8. The process-dependent hard function H is *not* an argument or constituent of an individual universal TMD boundary. For the convolutional nonperturbative interface, for example,

$$F^{\text{full}}_{i/h} = \mathcal{P}_{i/h}[f_h; D, \mathcal{C}, P, \beta, \Gamma, \gamma_F] \otimes_T \mathcal{M}^{\text{NP}}_{i/h} + \text{declared overlap or power corrections.} \tag{402}$$

The symbol $\mathcal{P}$ here excludes H. Other nonperturbative interfaces are alternatives with their own consistency conditions, not interchangeable products. The high-accuracy label is attached to a complete perturbative prediction with a specified process, not to the fitted neural network.

31.2 Minimal N3LL-prime certificate

Define a certificate

$$\begin{aligned}\mathfrak{C}_{N^3\text{LL}'} = \{&\text{scheme}, \text{ingredients}, \text{coefficient provenance}, \text{basis}, \text{scales},\\ &\text{RG}, \text{expansion}, \text{process}, \text{NP separation}\}.\end{aligned} \tag{403}$$

The certificate is accepted only if

$$\begin{aligned}\text{ClaimAllowed}_{N^3\text{LL}'} = \text{ParentScheme}_{\mathcal{K}_{\text{CSS2}}} &\wedge \text{IngredientComplete}_{N^3\text{LL}'} \wedge \text{PhysicalPayload}\\ &\wedge\ \text{NoMockCoefficients} \wedge \text{NoNumericalBTransform} \wedge \text{RGConsistent}\\ &\wedge\ \text{ResummedKernelExists} \wedge \text{AllOrderLogAccuracy}\\ &\wedge\ \text{ScaleRealizationConsistent} \wedge \text{ExpandedTo}\{a_s^3\} \wedge \text{NPSeparated}.\end{aligned} \tag{404}$$

For the spectral-canonical route, ScaleRealizationConsistent means using the completion, operator ordering, and parent canonical boundaries of Sec. 23, with the endpoint domain of Theorem 23.1. It does not require a second cumulative profile or a momentum-disk shift of those boundaries. For the alternative distributional route, the same condition requires the cumulative scale operator of Definition 11.1, coupled-(μ, ζ) RG integrability, the process-specific rapidity constraint, and an explicit declaration of whether the profile is applied leg by leg or to the process-level convolved spectrum. Because the profile operation is not generally a convolution homomorphism [Remark 11.4], the selected placement must pass the profiled-cumulant expansion check and the separate all-order logarithmic argument. ResummedKernelExists rules out the bare cubic exponential of Proposition 22.1; AllOrderLogAccuracy is not inferred from ExpandedTo$\{a_s^3\}$.

The ingredients required by IngredientComplete$_{N^3\text{LL}'}$ are the manifest of Section 21: four-loop cusp anomalous dimension, four-loop QCD beta function, three-loop noncusp TMD anomalous dimension, three-loop Collins-Soper/rapidity kernel, three-loop hard function, three-loop TMD matching coefficients, and DGLAP splitting functions through $P^{(2)}$. For full-spectrum predictions outside the singular region, the separate Y-term also requires the chosen fixed-order accuracy.

The claim ledger distinguishes the following modes:

Mode	What may be claimed	What may not be claimed
Representation framework	Exact fixed-order CSS2 distribution identities and a target manifest.	A completed numerical extraction.
Spectral-canonical prescription	Admitted unexpanded evolution and $\mathrm{N^3LL'}$ logarithmic prescription under the stated fixed-flavor and input hypotheses.	A numerical high-order prediction, arbitrary profiles, or all process classes.
Implementation dry run	Importer, generator algebra, RG, and expansion machinery using mock or partial payloads.	A physical $\mathrm{N^3LL'}$ prediction.
Certified implementation	A physical $\mathrm{N^3LL'}_{\mathcal{K}_{\mathrm{CSS2}}}$ perturbative W operator for the declared process class.	Universality beyond the declared process, flavor basis, or nonperturbative module assumptions.

Working decision. Do not declare the resummed formal prescription complete on the strength of a manifest or a fixed-order map. The all-order kernel and logarithmic proof are theoretical requirements, followed by physical-payload and numerical checks.

31.3 Physical payloads required to unlock the label

For a Drell–Yan TMDPDF-only implementation, the minimal physical payload is

$$\mathcal{P}_{\mathrm{DY}}^{\mathrm{min}} = \left\{ H_{q\bar{q}}^{(0,1,2,3)}, C_{i\leftarrow j}^{(0,1,2,3)}, D_i^{(1,2,3)}, \Gamma_i^{(0,1,2,3)}, \gamma_{F,i}^{(0,1,2)}, \beta_{0,1,2,3}, P_{ij}^{(0,1,2)} \right\}. \tag{405}$$

For SIDIS or back-to-back hadron production in e^+e^-, the payload must additionally include TMDFF matching through three loops and the appropriate timelike DGLAP kernels. For gluon-initiated color-singlet processes, the gluon hard function and gluon matching channels must pass the same source and expansion checks.

The public high-order source records justify the availability of these matching inputs: EMV2020 states that the $\mathrm{N^3LO}$ quark and gluon TMDPDF matching kernels are the last missing ingredient for $\mathrm{N^3LL'}$ q_T spectra, and LYZZ2021 provides $\mathrm{N^3LO}$ unpolarized quark/gluon TMDPDF and TMDFF matching coefficients with arXiv ancillary files [20, 22]. The present note does not claim that those files have been parsed unless their hashes appear in the local coefficient ledger.

31.4 Reproducible coefficient staging

Acquire the versioned ancillary files from the cited primary sources, retain source and normalization metadata, and hash their unmodified bytes. A portable implementation should use a repository-relative coefficient directory, not an absolute path belonging to an earlier working environment. File presence and hash agreement establish provenance, not correctness: parsing, CSS2 conversion, finite momentum-boundary shifts, RG checks, and observable comparisons are separate steps. No new physical high-order coefficient import is claimed in this revision. For the spectral-canonical branch use the parent canonical constants, not an additional momentum-disk shift; Sec. 23.6 explains the distinction.

31.5 Current certificate status

Fixed-order representation and contact-complete profile calculus: **derived**.
Ordered logarithmic reconstruction through three loops: **derived and tested**.
Finite momentum-canonical boundary conversion through three loops: **derived**.
Admitted unexpanded spectral-canonical evolution: **constructed and proved** on $\mathscr{D}_{\log}$; bounded canonical-leg evaluation is constructive.
$\mathrm{N^3LL}$ logarithmic evolution and the $\mathrm{N^3LL'}$ boundary prescription: **proved for the stated fixed-flavor, infrared-completed CSS2 class**.
Physical $\mathrm{N^3LL}$ evolution with strict-NLO boundaries: **implemented and validated for the declared inputs and UV class**; see Sec. 39.
Alternative cumulative-profile production implementation: **not claimed**.
Physical NNLO/$\mathrm{N^3LO}$ payload and independent expansion benchmarks: **not validated**.
Numerical $\mathrm{N^3LL'}$ implementation or extraction: **not claimed**.

These are separate mathematical and physical obligations. They cannot be reduced to whether an aggregate script reports a pass.

32 Full-spectrum matching and perturbative uncertainties

The singular operator $W^{\mathcal{K}_{\mathrm{CSS2}}}$ is the object that carries the TMD resummation accuracy. For a full transverse-momentum spectrum it must be combined with a nonsingular fixed-order remainder,

$$\mathrm{d}\sigma = \mathrm{d}\sigma_W^{\mathcal{K}_{\mathrm{CSS2}},N^k\mathrm{LL'}} + \mathrm{d}\sigma_Y + \mathrm{d}\sigma_{\mathrm{pow}}, \tag{406}$$

with

$$\mathrm{d}\sigma_Y = \mathrm{d}\sigma_{\mathrm{FO}}^{[m]} - \left[\mathrm{d}\sigma_{W,\mathrm{pert}}^{\mathcal{K}_{\mathrm{CSS2}},\mathrm{LP}}\right]_{\text{expanded to } \mathcal{O}(\mathrm{a_s^m})}. \tag{407}$$

The subtraction in Eq. (407) must use the same PDFs, hard normalization, flavor basis, scale choices, electroweak inputs, recoil prescription, and bin definitions as the fixed-order calculation. The subtraction is the expansion of the *perturbative leading-power* W, not of a fitted nonperturbative smearing. The remainder $\mathrm{d}\sigma_{\mathrm{pow}}$ contains only power contributions not already included in the fixed-order remainder or in a specified recoil improvement; adding those terms again would double count. Its first power can differ for inclusive and fiducial observables [47]. External programs such as DYTurbo, SCETlib, CuTe-MCFM/MCFM, or other fixed-order/resummed implementations are benchmark and provider tools for $\mathrm{d}\sigma_{\mathrm{FO}}$ and matched spectra [40, 41, 42, 37]; they do not define the universal TMD legs.

Writing $W_{\mathrm{full}} = W_{\mathrm{pert}} + \Delta W_{\mathrm{NP}}$ makes the bookkeeping explicit:

$$\mathrm{d}\sigma_{\mathrm{matched}} = \mathrm{d}\sigma_{\mathrm{FO}}^{[m]} + \left(\mathrm{d}\sigma_{W,\mathrm{pert}}^{\mathrm{resum}} - [\mathrm{d}\sigma_{W,\mathrm{pert}}]_m\right) + \Delta\mathrm{d}\sigma_{W,\mathrm{NP}}, \tag{408}$$

up to the explicitly unaccounted power terms. The nonperturbative module is not Taylor-expanded as though it were a finite QCD loop coefficient. Resummation turnoff must be implemented consistently in both the retained term and its subtraction; the formula alone does not prove the transition-region accuracy.

The two tails remain distinct:

$$\boxed{\text{TMD large-}k_T\text{ tail}} \iff \mathcal{C}_{i\leftarrow j}\otimes_x f_j, \qquad \boxed{\text{matched large-}q_T\text{ spectrum}} \iff W+Y. \tag{409}$$

Conflating these two objects would turn a universal TMD extraction into a process-specific spectrum fit.

For the formalism paper the perturbative uncertainty sources are separated from the nonperturbative model uncertainty:

$$\{\Delta_{\rm pert}\} = \{\Delta_{\rm scale}, \Delta_{\rm match}, \Delta_{\rm scheme}, \Delta_{\rm numerical}\}. \tag{410}$$

This notation is a classification, not an instruction to add the sources in quadrature; correlations and envelope prescriptions must be specified by the phenomenological implementation. Here $\Delta_{\rm scale}$ comes from variations of μ_H, boundary scales, final scales, and rapidity scales subject to constraints such as $\zeta_A\zeta_B = Q^4$ in Drell-Yan. Variations in the cumulative-profile route use the contact-complete weak operation of Sec. 11. The spectral route instead varies the complete scalar RG/matching symbol and reevaluates its operator approximation; these placements are not assumed interchangeable. In either route, $\Delta_{\rm match}$ comes from the W-Y transition and fixed-order truncation; $\Delta_{\rm scheme}$ comes from any finite conversion between $\mathcal{K}_{\rm CSS2}$ and a future child scheme; and $\Delta_{\rm numerical}$ comes from binned distribution evaluation, quadrature, interpolation, and convolution-grid errors. Experimental errors, PDF-replica errors, and nonperturbative model/DNN uncertainties belong to the later extraction paper.

33 Implementation status and extraction workflow

The first paper presents the full formal prescription and the scoped physical validation, without claiming a completed precision extraction. The accepted $\mathrm{N^3LL}$/strict-NLO implementation is detailed in Sec. 39. The stronger claim hierarchy remains:

Claim	Allowed when	Not implied
Formal $N^3\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$ prescription	In addition to the ingredient map and contact-complete distributions, a defined resummed evolution solution and a proof of its retained logarithmic towers are supplied. Section 23 supplies these under its fixed-flavor domain hypotheses. Section 29 separately records practical validation still required.	A numerical extraction has been performed.
Numerical $N^3\mathrm{LL}'_{\mathcal{K}_{\mathrm{CSS2}}}$ implementation	Physical three-loop coefficient payloads are staged, hashed, parsed, and converted; the profile-operator algebra check passes; a declared leg-wise or process-level profile is integrated into the production kernel and its cumulants are validated; and the $\mathcal{O}(a_s^3)$ singular gate passes.	Data have been fitted.
k_T-space $N^3\mathrm{LL}'$ extraction	The numerical operator is combined with a specified nonperturbative module, data set, fit protocol, and uncertainty model.	The nonperturbative module is unique.
DNN-based extraction	The chosen $\mathcal{M}_{\mathrm{NP}}$ is a neural parameterization satisfying the admissibility conditions of Sec. 19.	The perturbative label is produced by the DNN.

The earlier method-development sequence is given in Sec. 29: first separate numerical representation error from the physical input, then resolve matched-spectrum stability, then perform controlled input-parameterization and data-closure studies. A document outline or a count of passing tests is not an acceptance criterion for a physical extraction. Current methods-paper completion and future comparison work are separated in Sec. 39.6.

34 Numerical representation

Because perturbative kernels are distributions, the numerical implementation should avoid unbinned pointwise evaluation near $q_T = 0$. Preferred representations are:

1. bin-integrated kernels matching experimental binning;
2. cumulants $\Sigma(q_T^{\max}) = \int^{q_T^{\max}} \mathrm{d}^2\boldsymbol{q}\, W(\boldsymbol{q})$;
3. smooth test-function projections;
4. radial basis expansions with analytically known integrals over bins.

The transverse convolution of radial functions can be evaluated through the angular integral

$$(A \otimes_T B)(q) = \int_0^\infty \mathrm{d}k\, k\, A(k) \int_0^{2\pi} \mathrm{d}\phi\, B\Big(\sqrt{q^2 + k^2 - 2qk\cos\phi}\Big)\,, \tag{411}$$

for ordinary functions. Distributional pieces should be handled analytically before numerical quadrature. When a transverse profile is used, the numerical object is the profiled cumulative and its derivative in Eq. (60), not a pointwise differential kernel with a substituted scale.

Working decision. The paper-level recommendation is the hybrid approach: singular kernels are handled analytically through bin or test-function actions, while ordinary nonperturbative modules are represented on radial grids or flexible bases and combined with analytic subtraction terms.

35 Validation sequence

The formal prescription becomes a numerical implementation only after a hierarchy of regression and physics-input checks has been completed. These checks test consequences of the parent theory and the correctness of the code; they do not replace the parent factorization proof. The required checks are:

1. **Distribution algebra.** Check $\delta^{(2)}$, $\mathcal{L}_n$, $\mathfrak{L}_n$, generator derivatives, bin actions, and transverse convolution identities.
2. **One-loop singular spectrum.** Reproduce the $A_q^{(1)} = C_F$ and $B_q^{(1)} = -3C_F/2$ Drell-Yan singular spectrum in bins, including the correct $d^2\boldsymbol{q}$ versus dq_T^2 normalization.
3. **Flavor-vector and x-convolution tests.** Check the longitudinal plus-prescription, DGLAP kernels, and quark/gluon mixing independently of the transverse plus prescription.
4. **Evolution tests.** Check path independence in (μ, ζ), semigroup closure, fixed-coupling limits, and running-coupling endpoint identities.
5. **Distributional profile tests.** Check constant-profile reduction, cumulative consistency, canonical removal of boundary logarithmic distributions, profile-derivative terms, and additivity under bin subdivision.
6. **All-order resummation gate.** Establish a weak resummed kernel on the declared domain, correct canonical boundary shifts, and the retained logarithmic towers. No finite-order test count substitutes for this theoretical step.
7. **NNLO singular gate.** Import the physical $\mathcal{C}^{(2)}$, $H^{(2)}$, and $P^{(1)}$ ingredients; expand $W^{\mathcal{K}_{\mathrm{CSS2}}}$ through $\mathcal{O}(a_s^2)$ and compare bin by bin with the parent CSS2 singular expansion.
8. **N^{3}LO singular gate.** Import the physical $\mathcal{C}^{(3)}$, $H^{(3)}$, $D^{(3)}$, and $P^{(2)}$ ingredients; expand through $\mathcal{O}(a_s^3)$ and compare bin by bin with the parent CSS2 singular expansion.
9. **Y-term consistency.** Add Eq. (407); verify that the matched result tends to the resummed result at small q_T and to fixed order at large q_T.
10. **Nonperturbative closure.** Generate pseudo-data from a known $\mathcal{M}_{\mathrm{NP}}$ and recover it without allowing the model to absorb missing perturbative ingredients.

Only after this ladder is passed should a data fit advertise an implementation-level $N^3\mathrm{LL}'$ extraction.

36 Relation to Fernando-Keller pipeline

Fernando-Keller can be summarized schematically as

$$\text{data} \longrightarrow \text{learned structure kernel} \longrightarrow s(x,k;Q) \longrightarrow F_{q/h}(x,k;Q) \simeq f_{q/h}(x,Q)\, s(x,k;Q). \quad (412)$$

The perturbative upgrade is instead

$$\text{data} \longrightarrow N^k\text{LL}_{k_T} \text{ forward operator} \longrightarrow F_{i/h}^{\text{core}}(x,k;\theta) \longrightarrow F_{i/h}^{\text{pert+NP}-\text{sub}}(x,k;\mu,\zeta). \quad (413)$$

The neural network is moved from "the full transverse profile" to "the nonperturbative boundary condition."

37 Audit reproducibility and interpretation of test results

The v1.4/v1.5 execution descriptions below are retained historical records from the supplied note. They were not rerun as part of v1.6. The v1.6 execution results, including a failed pilot and its revised linear solve, are separately generated by `verification/run_study.py` and recorded in Sec. 29. The retained scalar source came from the locally supplied code archive, not a fresh GitHub clone; the source ledger records that distinction. The accepted later scientific snapshot and its public immutable revision are described in Sec. 39.6. Earlier archive observations in this section are not assertions about that subsequent public revision.

The historical v1.4 audit described in the input note contains `validators/validate_v1p4.py` with generated `audit_results.json`, `audit_tests.csv`, and contact-complete bin outputs. It records the exact input-source SHA256, script SHA256, Python/library versions, arithmetic precision, and test-specific tolerances. It performs 102 checks spanning polynomial dictionaries, canonical contact constants, weak profiles, ordered matrix RG through three loops, product partitions, and illustrative nonperturbative heat evolution. Negative controls reject the missing-contact formula, an invalid global multiplier closure, reversed flavor ordering, and the inference from finite-order agreement to all-order logarithmic accuracy.

The development repository is Ref. [3]. An archived local repository snapshot was also rerun and returned its historical 14/14 aggregate pass. This is *not* a claim that the current mutable GitHub branch was cloned or that it has that test count. More importantly, the archived profile script omitted the first-bin contact contribution, and its text-consistency checker merely compared the note with that output. Its successful aggregate result therefore did not verify the mathematical statement at issue. The new test of $C(t) = c$ rejects the old zero-bin formula immediately. The supplied provenance record identifies the archive and files used, so these statements can be checked.

A separate eight-check demonstration of the convergent resolvent evaluator is provided by `validators/resolvent_operator_demo.py`. It uses a bounded analytic test multiplier and a finite momentum grid. Its same-grid approximation errors satisfy the theorem's bound; continuum/grid errors and a QCD multiplier are not certified by that demonstration.

The other historical one-/two-loop numerical figures quoted in their respective sections are regression results of the archived scripts, not physical predictions or certificates of resummed accuracy. Their raw rerun outputs are included for traceability. The v1.4 summary remains a historical record, including its then-open all-order construction. The v1.5 construction and its new test reports are given in Sec. 23. The physical high-order coefficient payload and an externally benchmarked numerical $\text{N}^3\text{LL}'$ prediction remain unvalidated. No script is allowed to infer a theorem's hypotheses merely from the presence of labels, equation strings, or an ingredient count.

38 Method ledger: established constructions and remaining validation

The continuing goal is an accurate, stable k_T fit of the same renormalized TMD. The current accepted implementation and publication route are summarized in Sec. 39. The following ledger preserves the formal and developmental dependencies through v1.6. The v1.6 method fixes, proofs, and measured limitations are summarized in Sec. 29. Positivity and exact event kinematics are not inferred from linear evolution; positive proposal kernels are auxiliary and process support belongs to matching. The rational evaluator is a deterministic candidate with contact constraints and explicit error accounting, not a claim of a completed physical N^{3}LL$'$ prediction or an efficiency advantage.

The v1.4 corrections remain in force. Fixed-order distribution identities, contact-preserving cumulative profiles, canonical-disk conversions, and the ordered flavor recurrence do not by themselves prove resummation. Version 1.5 adds a distinct constructive result with the following dependency chain:

1. The explicit scale in Eq. (340) fixes the infrared completion without an ultraviolet cap or a process-dependent CS kernel.

2. Equation (341) resums the CS logarithms from their canonical boundary before exponentiation. The endpoint kernel exists on $\mathscr{D}_{\log}$ and obeys exact retained-RG path and composition identities (Theorem 23.1).

3. The complete canonical leg is bounded (Theorem 23.2). Momentum-resolvent polynomials and a controlled cutoff limit evaluate both the canonical action and arbitrary admitted endpoint actions without numerical Fourier transformations.

4. Theorem 23.3 covers perturbative contact kernels and finite-bin errors, not only L^2 test inputs.

5. Theorem 23.4 proves the retained all-order exponent towers. Proposition 23.5 states the additional hypotheses for a uniform resummed error estimate. Three-loop physical hard and matching boundaries supply the prime under the declared convention.

This completes *one specified formal prescription* for a transform-free, fixed-flavor N^{3}LL$'$ singular operator. It is a spectral implementation of the canonical CSS2 solution, not a new factorization theorem or proof that all possible momentum-profile constructions are equivalent. The spectral nodes encode the same transverse resolution as the parent variable; what is eliminated is the numerical transformation of the fitted functions and cross sections. No claim of computational speedup follows from the existence theorem.

At the physical primed-order level, remaining work is explicitly separated. A numerical N^{3}LL$'$ prediction needs physical coefficient packages with scheme conversion and provenance, an efficient controlled elliptic or resolvent discretization, numerical RG/PDF accuracy, independent fixed-order and resummed benchmarks, and the specified process hard and measurement functions. Heavy-flavor thresholds, tensor harmonics, fiducial power corrections, and other processes require their own inputs and checks. A full-spectrum result also needs consistent $W + Y$ matching. A data extraction then adds a chosen nonperturbative module, closure tests, and uncertainty analysis. The alternative cumulative-profile production route remains available for later development but is not a prerequisite for the spectral realization proved here.

The delivered v1.5 numerical demonstrations use disclosed lower-order QCD inputs and mathematical fixtures. They do not stand in for physical three-loop matching or an external phenomenological benchmark. Earlier historical pass counts remain historical; the v1.6 reports were generated

by the scripts accompanying that revision. The formalism is model agnostic, and no absolute priority claim is made.

39 Physical canonical realization and validation update (1 October 2026)

This section adds the results obtained after the v1.6 method study. It updates implementation and publication status while retaining the preceding derivations, conventions, historical tests, and higher-order import requirements. The scientific evidence is pinned to revision `04cf2ff` of the public validation repository [59]. Later prose or formatting changes do not change that numerical record.

39.1 Current accuracy and the connection to the proved prescription

The implemented and validated physical claim is **$\mathrm{N^3LL}$ evolution with strict-NLO hard/OPE boundaries**. The evolution uses four-loop beta and cusp functions and three-loop noncusp and canonical Collins–Soper inputs. Its scale is the endpoint-independent, ultraviolet-uncapped canonical scale of Eq. (340), with the declared infrared completion. Thus the physical benchmark instantiates the spectral construction of Sec. 23. This closes the earlier gap between a formal retained-RG engine and an actual-PDF evaluation of that same prescription. It does not complete the separate physical three-loop hard/matching import gate.

The general matching formula in Eq. (351) was written with a third-order boundary to specify a primed-order extension. For the present calculation its finite boundary order is instead $n_B = 1$:

$$B^{[1]}_{i/h}(x;s) = \sum_j \big[C^{\mathrm{can},[1]}_{i\leftarrow j}(a_c) \otimes_x f_{j/h}(\mu_c)\big](x), \qquad M^{[1]}_{i/h}(x,s;e) = v_i(s;e) B^{[1]}_{i/h}(x;s), \tag{414}$$

where $a_c = \alpha_s(\mu_c(s))/(4\pi)$. The boundedness and input-gradient arguments use the same stated continuity and logarithmic-growth assumptions; they do not require that the finite boundary be truncated specifically at third order. The error $O(a_Q^4)$ for a finite boundary in Proposition 23.5 is conditional on retaining that boundary through a^3. It is not the finite-boundary remainder of the strict-NLO calculation in this update.

Table 8: Accuracy ingredients and current implementation.

Ingredient	Formal primed-order extension	Validated physical realization
Cusp and running	Four loops	Sourced four-loop quark cusp and fixed-n_f beta function
Noncusp and CS boundary	Three loops	Three-loop noncusp and $d^{(2)}, d^{(3)}$, with $d^{(1)} = 0$
Hard and OPE boundaries	Through a^3 in a common scheme	Strict NLO; six linear two-leg components
Collinear input	Order- and convention-consistent PDFs/evolution	Pinned NNPDF4.0 NNLO member 0, with separately evolved fixed-$n_f = 5$ coupling
Full-spectrum matching	A separately specified nonsingular remainder	The $W + Y$ interface is defined; numerical full-spectrum completion remains open
Extraction	Admitted model, data and inference protocol	Conditional E866 and restricted mixture closure only

The coefficient manifest records source versions, locations, normalization conversions and hashes. In particular, the four-loop quark cusp is compared with a separately written primary-source expression [33, 60]; the largest absolute disagreement at $n_f = 3, 4, 5, 6$ is below 2.4×10^{-11}. Quartic color invariants are retained. Source agreement verifies the ingestion and convention check; the two numerical paths do not independently compute those QCD coefficients.

39.2 Accepted evaluator and strict-NLO physical assembly

The distinct-shift rational evaluator in Sec. 26 remains an available construction and a preserved development study. The accepted high-resolution calculation uses an endpoint-preserving Chebyshev polynomial of one resolvent. Write

$$r = \frac{\tau}{\tau + s}, \qquad m(r) = M\left(\tau \frac{1-r}{r}\right), \qquad X = 2R_\tau - I, \tag{415}$$

and interpolate m on Chebyshev–Lobatto nodes, including $r = 0, 1$:

$$p_N(r) = \sum_{j=0}^{N} c_j T_j(2r-1), \qquad p_N(0) = M_\infty, \qquad p_N(1) = 0, \tag{416}$$

$$y_0 = S, \quad y_1 = XS, \quad y_{j+1} = 2Xy_j - y_{j-1}, \qquad p_N(R_\tau)S = \sum_{j=0}^{N} c_j y_j. \tag{417}$$

Here T_j denotes a Chebyshev polynomial, not the Laplacian T. The endpoint identities hold in exact arithmetic; their floating-point realization is checked. Each recurrence step applies the same momentum-space elliptic solver. A Clenshaw implementation supplies a separate recurrence check. For repeated physical components, cached bin actions on y_j can be contracted with different coefficient vectors. The fitted input is never numerically Fourier transformed by this production path.

The radial finite-volume operator uses the $2\pi k\,\mathrm{d}k$ measure, a regular zero-flux condition at the origin, and a homogeneous Dirichlet condition at the outer radius. Similarity by the square roots of the cell weights gives a symmetric discrete operator. Degree, radial spacing and outer radius must be refined separately. Increasing degree on a fixed grid cannot certify the continuum limit. The Bernstein theorem in Sec. 23.4 continues to supply an existence and contact-error result for its own sequence; it does not certify a global Chebyshev error from sampled residuals. The current Chebyshev budgets are observed refinement and independent-reference comparisons.

For one flavor orientation, abbreviate a leg's canonical longitudinal boundary as $B_A^{(0)} + a_c C_{qq,A}^{(1)} + a_c C_{qg,A}^{(1)}$, with the longitudinal PDF convolutions included in each C insertion. At symmetric final scales, $a_Q = \alpha_s(Q)/(4\pi)$, the physical boundary product is

$$\begin{aligned}\mathcal{P}_{AB}^{[1]} = {} & B_A^{(0)} B_B^{(0)} + a_Q h_q^{(1)} B_A^{(0)} B_B^{(0)} \\ & + a_c (C_{qq,A}^{(1)} + C_{qg,A}^{(1)}) B_B^{(0)} + a_c B_A^{(0)} (C_{qq,B}^{(1)} + C_{qg,B}^{(1)}).\end{aligned} \tag{418}$$

The complete evolution factors remain unexpanded. The charge sum and charge-conjugate orientation are then included. This gives the Born term, hard insertion, qq insertion on each leg, and qg insertion on each leg. Hard–OPE and OPE–OPE products would be incomplete higher-order boundary terms and are excluded. The one-loop hard function and longitudinal kernels are those already given in the note; their scale derivatives and normalization are checked.

39.3 Continuum, physical-PDF, contact and gradient comparisons

The ordinary-input continuum benchmark uses

$$S(k;\lambda,p)=\frac{p-1}{\pi\lambda^2}\left(1+\frac{k^2}{\lambda^2}\right)^{-p},\qquad p>1, \tag{419}$$

with $p=3$ and $\lambda=1.2$ GeV. The adaptive continuum reference uses its analytic Bessel-K transform and analytic λ derivative, retaining the constant-contact contribution exactly. At degree 1024, spacing 0.0125 GeV and radius 1280 GeV, seven bins from 0 to 8 GeV have maximum peak-normalized bin error 4.51×10^{-5} and derivative error below 1.13×10^{-4}. The last degree, spacing and radius changes are respectively 2.90×10^{-5}, 1.32×10^{-4} and 2.17×10^{-5}. Their conservative observed bin budget is 2.28×10^{-4}, below the 4×10^{-4} criterion. Reference quadrature error is below 3×10^{-13} in the same normalization.

The physical two-leg study uses NNPDF4.0 NNLO member 0 through LHAPDF6 [61, 62], photon charges, five massless quark flavors, both beam orientations and every term of Eq. (418). The four-loop coupling is evolved independently with $n_f=5$ from $\alpha_s(91.1876\,\mathrm{GeV})=0.1180024$; it is not the PDF provider's lower-order tabulated coupling. The benchmark is not a massive-threshold calculation. The three (Q,x_A,x_B) configurations are $(8,0.08,0.20)$, $(20,0.15,0.30)$ and $(91.1876,0.02,0.07)$, with Q in GeV. The highest value does not mean that a Z channel was included.

The low- and high-scale cases use Student legs of powers 3 and 4 with $\lambda_A=0.9+0.8x_A$ GeV and $\lambda_B=1.2+0.6x_B$ GeV; the middle case uses a Gaussian pair. The native two-Student convolution is evaluated directly:

$$\begin{aligned}(S_{p,\lambda_A}\otimes_T S_{q,\lambda_B})(k)&=\mathcal{N}_{pq}\int_0^1\frac{u^{p-1}(1-u)^{q-1}\,\mathrm{d}u}{[u\lambda_A^2+(1-u)\lambda_B^2+u(1-u)k^2]^{p+q-1}},\\ \mathcal{N}_{pq}&=\frac{(p-1)(q-1)\lambda_A^{2p-2}\lambda_B^{2q-2}}{\pi}\frac{\Gamma(p+q-1)}{\Gamma(p)\Gamma(q)}.\end{aligned} \tag{420}$$

Combining the two denominators with a Feynman parameter, shifting the momentum integration variable, and integrating its radial square yields this identity. It requires no transform of the input. Normalization, bins and the analytic logarithmic scale derivative are checked separately.

The independent reference shares the source coefficient values and official PDF provider, but imports none of the production RG, matching, flavor-sum or physical-symbol routines. It integrates the endpoint anomalous dimension by RK45, performs longitudinal convolutions in $\ln\xi$ rather than z, and uses adaptive continuum Bessel quadrature with the product of analytic input transforms. An adaptive longitudinal calculation checks its cache. Separate RG tolerance refinement changes log evolution by less than 10^{-9}, while production agrees with the tight reference to better than 4.8×10^{-12} at the sampled points. Contact inputs and a derivative with respect to $\ln\lambda_A$ accompany ordinary-bin comparisons.

All primary criteria in Table 9 pass. The low- and middle-scale calculations use degree 2048 and radius 1280 GeV; the high-scale case uses degree 4096 and radius 5120 GeV. All use spacing 0.0125 GeV, with separate degree, spacing and box refinements. These large radii are numerical domains, not physical momentum support or proposed production-fit settings. The budget adds the final discrepancy, last refinement changes, reference quadrature and UV class contribution. It is not a rigorous supremum over all inputs or a physical perturbative uncertainty. Signed and cancellation-sensitive bins, including the region $q_T>0.2Q$, remain in the tests as operator stress checks. The zero spectral endpoint discussed in Sec. 25 is unchanged; positivity of the complete singular output is not imposed.

Table 9: Physical errors including the broader conditional PDF continuation bound. Entries are rounded upward and normalized to the peak reference total in each case; the gradient is with respect to $\ln \lambda_A$.

Q [GeV]	Bin	Component	Gradient	Contact	Budget
8	$2.25\,10^{-5}$	$2.00\,10^{-5}$	$2.56\,10^{-5}$	$2.39\,10^{-5}$	$2.63\,10^{-4}$
20	$4.27\,10^{-5}$	$3.91\,10^{-5}$	$6.12\,10^{-6}$	$4.08\,10^{-5}$	$3.47\,10^{-4}$
91.1876	$2.50\,10^{-5}$	$2.31\,10^{-5}$	$2.64\,10^{-5}$	$2.14\,10^{-5}$	$1.85\,10^{-4}$
Limits	$5\,10^{-4}$	$5\,10^{-4}$	10^{-3}	$5\,10^{-4}$	10^{-3}

39.4 Finite PDF support and the conditional ultraviolet bound

The uncapped canonical scale tends to infinity as $s \to 0$, whereas the PDF table stops at $\mu_{\max} = 10^5$ GeV. Production nodes stay inside the table. The unresolved interval is bounded for the explicit admitted class

$$|f_j(x,\mu)| \le |f_j(x,\mu_{\max})|[1+\ln(\mu/\mu_{\max})]^4, \qquad \mu > \mu_{\max}, \tag{421}$$

including every argument used inside longitudinal convolutions. This is an input assumption; a finite PDF table does not establish it. The same condition must accompany any quoted class-wide numerical bound.

Let $b_{\rm cut} = b_0/\mu_{\max}$ and $u = \ln(\mu_c/\mu_{\max})$. Each strict-NLO two-leg component has an absolute longitudinal envelope $A_{\rm comp}(1+u)^8$, formed before cancellations and including bounded coupling factors and flavor/charge sums. Monotone ultraviolet decay of the canonical evolution follows on the audited weak-coupling interval from

$$\frac{\mathrm{d}\ln v}{\mathrm{d}\ln \mu_c} = d(a_c) - \gamma(a_c) + 2[\Gamma(a_c) + \mathcal{B}_\beta(a_c) d'(a_c)] \ln \frac{Q}{\mu_c}, \quad \mathcal{B}_\beta(a) = \sum_{r=0}^{3} \beta_r a^{r+2}. \tag{422}$$

The signs of $(d-\gamma)/a$ and $(\Gamma + \mathcal{B}_\beta d')/a$ are checked using polynomial endpoints and stationary points. They are respectively nonpositive and positive on $0 < a \le a(\mu_{\max})$, and $\mu_{\max} > Q$, so v decreases above the table endpoint.

For a bin $q_a < q_T < q_b$, the identity $q_b J_1(q_b b) - q_a J_1(q_a b) = b \int_{q_a}^{q_b} q J_0(qb)\,\mathrm{d}q$, the bound $|J_0| \le 1$, and the unit bound on transforms of normalized positive inputs give the sufficient estimate

$$|\Delta \mathcal{B}_{\rm comp}| \le \frac{q_b^2 - q_a^2}{2}\, b_{\rm cut}^2 v(\mu_{\max})^2 A_{\rm comp} \int_0^\infty e^{-2u}(1+u)^8\,\mathrm{d}u. \tag{423}$$

The actual bound partitions $u \in [0, 24]$, uses the decreasing evolution at each interval's left endpoint, integrates the polynomial moments analytically, and bounds the final infinite tail. An RG integration margin is included; this is a numerical bound evaluation, not interval-arithmetic certification. The absolute L^1 norm of the logarithmic input-scale derivative is at most four for the audited $p = 3$ Student and Gaussian inputs, giving the corresponding derivative envelope. Contact inputs use their unit transform directly.

The largest achieved class contribution is 1.6683×10^{-6} of the peak bin, passing the declared 2×10^{-6} numerical criterion. That tolerance has no identified special physical significance. The class contribution remains in the original primary errors and budgets in Table 9. No negative bin is removed and no physical prediction is changed to meet a limit.

Two explicitly completed PDF inputs provide a separate sensitivity check: constant endpoint PDFs above the table, or growth by $(1+u)^4$. Both retain uncapped coupling and evolution. Their unresolved integral estimates are below 10^{-17} of the peak, and their largest bin spread is 1.532×10^{-6}.

Neither continuation is asserted to solve DGLAP outside the table. Their spread is input dependence, while their integration residuals are numerical errors; neither replaces the broader-class contribution in the accepted budget.

39.5 Qualified Drell–Yan and neural inverse demonstrations

The process demonstrator now uses 41 proton–proton E866 bins transcribed from Ref. [63], with $\sqrt{s} = 38.8$ GeV, $Q_{\text{low}} \geq 6.2$ GeV and $q_{T,\text{high}}/Q_{\text{low}} \leq 0.2$. Proton–proton data avoid introducing a copper nuclear model for this example. Its separate capped process profile keeps perturbative scales above 4.99 GeV. It tests physical measurement integration under that declared prescription; it does not inherit the uncapped theorem by sharing implementation components.

The native forward benchmark agrees with independently evaluated PDFs, longitudinal matching, RG integration and continuum bins. Its worst relative bin error is 2.26×10^{-6}, component error is 2.12×10^{-6} of the row total, and scaled gradient error is 3.78×10^{-5}. A two-parameter Gaussian pseudodata closure recovers its injections with biases 1.21×10^{-6} and -9.7×10^{-8}, and maximum held-out prediction error 8.2×10^{-6}.

A restricted neural pilot uses six fixed transverse widths and a 1–12–12–12–6 network for x-dependent softmax mixture weights. Precomputed native operator responses and analytic derivatives train the forward model. In a non-Gaussian correlated-input closure, an entire ten-bin mass window is withheld. The accepted seed has maximum training error 7.6×10^{-5}, held-out error 6.7×10^{-4} and native/refined forward discrepancy 6.9×10^{-6}. The generating input lies in the same six-width family. This establishes a restricted inverse workflow, not arbitrary joint-function recovery or a flavor-resolved extraction.

The actual-data diagnostic retains a measurement qualification: 253 of 279 transcribed thesis rows have a mean-energy entry inconsistent with the stated bin kinematics, and the exact finite-bin invariant conversion is not completely documented. Under the preserved tabulated-mean estimator, the neural diagnostic gives point plus normalization-prior $\chi^2 = 21.68$ for 41 bins with normalization $N = 1.024$. The nonperturbative CS coefficient is fixed from a conditional Gaussian diagnostic. No parameter-counting χ^2/dof, precision extraction, or agreement with the thesis's full analysis is claimed.

The intended general model remains a network in both x and k_T, subject to the input, overlap and evolution conditions already specified in this note. The six-width pilot is a limited numerical closure, not a decision to restrict the ultimate DNN to x dependence. The separate comparison with the b_T-space extraction of Fernando and Keller [58] will use identical data, a comparable joint-input architecture, most likely FiLM, and matched cuts, likelihood, nuisance treatment and uncertainty procedure. The controlled difference is the native-k_T perturbative/matching realization within the established CSS2 prescription. That project begins after the methods paper, rather than serving as an unfinished claim of the present update.

39.6 Evidence, reproducibility, and remaining work

At the cited scientific revision, 121 software tests and all 14 registered scientific suites pass. The suites include 25 physical-canonical checks and 16 derivation checks, as well as the algebraic and interface layers; overlapping counts are not added into a physics certificate. The public snapshot contains 32 numerical records, with source, coefficient, PDF and figure provenance. The accepted physical reports pass directly under the stated criteria. Older v1.4–v1.6 counts and development attempts retained in this note are historical records, not a competing current acceptance status.

The portable offline entry point is

```
python scripts/run_method_validation.py
```

The full physical run adds `--physical` and explicit `--library` and `--pdf-data` paths after executing the documented local LHAPDF setup. The reproduction guide is `docs/REPRODUCE_METHOD_PAPER.md`; the accepted snapshot is `validation/reference/2026-10-01`. Source and PDF bytes are hash checked. The reported physical acceptance was run on macOS/Apple silicon; a provided Linux recipe is not itself evidence of a completed Linux acceptance run. The optional manual full-physical GitHub workflow is not claimed as executed.

The numerical reference paths were separately coded within the same AI-assisted development effort. They are not independent human review or independent determinations of the shared perturbative coefficients. The companion PRD manuscript discloses computational assistance and cites the same immutable scientific revision. Its full two-column exposition and separate BibTeX reference file present the method for author review. This technical note retains its broader developmental derivations and historical record.

Table 10: Current completion ledger, superseding older implementation-status language while preserving the earlier mathematical statements.

Item	Status after the recent work
Formal spectral prescription	Retained domain, contact, endpoint and $\mathrm{N^3LL}$ tower results apply under their stated hypotheses.
Physical perturbative realization	Source-qualified 4/4/3/3 evolution and strict-NLO hard/OPE assembly implemented; physical NNLO/$\mathrm{N^3LO}$ boundary import remains open.
Native numerical action	Non-Gaussian continuum, three-scale physical bins/components/contacts/derivatives and separated refinements pass for the stated inputs and UV class.
Inverse interface	Restricted closure and conditional E866 workflow demonstrated; no unrestricted DNN recovery or global uncertainty extraction.
Method publication	Public validation revision and expanded PRD draft prepared; author scientific review, final metadata and declarations remain. No journal submission is implied.
Future comparison	Identical-data joint-(x, k_T) versus b_T extraction is a separate subsequent project.
Further physical extensions	Numerical full $W + Y$, higher-order boundaries, massive thresholds and additional tensor/fragmentation channels require their own calculations.

The new evidence therefore resolves the earlier practical question of evaluating one physically instantiated uncapped canonical prescription accurately in native momentum space. It does not remove the note's hypotheses, promote NLO boundaries to $\mathrm{N^3LL'}$, certify a global operator-norm error for the practical approximation, or establish a computational speed advantage. The earlier alternatives, including positive proposals with signed corrections and cumulative-profile constructions, remain optional research routes rather than prerequisites for the accepted canonical method.

A Minimal notation dictionary

Symbol	Meaning

$F_{i/h}(x, \boldsymbol{k}; \mu, \zeta)$	Renormalized TMDPDF for parton i in hadron h in transverse-momentum space.
$H_{ij}(Q, \mu)$	Hard function for the partonic channel ij.
$\mathcal{D}_i(\boldsymbol{k}; \mu)$	Momentum-space Collins-Soper kernel.
$\mathcal{U}_i$	Evolution Green's function acting by transverse convolution.
$\mathcal{C}_{i\leftarrow j}$	Perturbative matching coefficient from collinear PDF j to TMD parton i.
W	Resummed/singular TMD contribution to the cross section.
Y	Fixed-order remainder: full fixed order minus the fixed-order expansion of W.
F^{core}	Neural or parametric nonperturbative core at the boundary scale.
F^{sub}	Overlap subtraction between the perturbative tail and nonperturbative core.

B Reference anchors

- Foundations of the baseline b_T framework and scheme translation: Collins-Soper, CSS, Collins 2011, Collins-Rogers, and the TMD Handbook [4, 5, 6, 39, 7].
- Early direct-q_T and transverse-momentum-space resummation: Ellis-Veseli, Frixione-Nason-Ridolfi, and Kulesza-Stirling [8, 9, 10, 11].
- Modern direct-space and distribution-space resummation: Monni-Re-Torrielli, Bizon et al., Ebert-Tackmann, and Simonelli [12, 13, 14, 15].
- Perturbative ingredients for a high-order k_T implementation: RRG soft/beam functions, N^3LO TMD matching, high-order rapidity anomalous dimensions, four-loop cusp information, three-loop hard functions, and modern N^{3}LL$'$ Drell–Yan benchmarks [19, 20, 22, 43, 18, 23, 24, 33, 34, 37].
- Integral constraints and large-k_T consistency: Ebert-Michel-Stewart-Sun, Gonzalez-Hernandez-Rainaldi-Rogers, and HSO phenomenology [25, 26, 27].
- Existing high-accuracy and neural TMD extractions: Moos-Scimemi-Vladimirov-Zurita and MAP neural-network fits [28, 29, 30].
- Direct k_T inverse reconstruction template: Fernando-Keller [38].
- Cross-section fixed-order and matched benchmarks for the Y term: DYTurbo and CuTe-MCFM/MCFM [40, 41, 42].
- Momentum-space TMD evolution in a neighboring framework: Parton Branching and PB-to-CSS comparisons [32, 31].

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